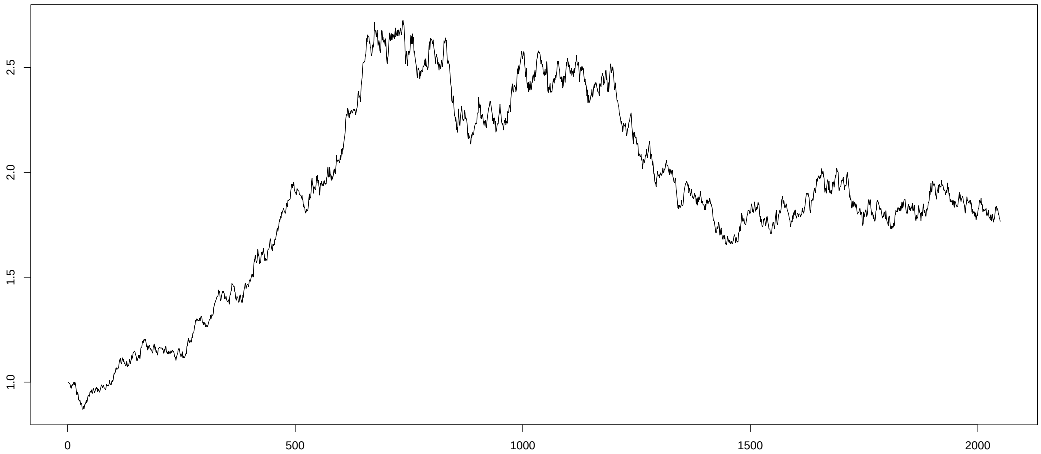
# Financial Mathematics

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## Course outline

- **Two versions:** Level 3 and Level M
- **Lectures:** Two weekly in-person lectures (Tuesdays, 12-1; Wednesdays, 11-12) and some online videos
- **Problem class:** Fridays, 2-3
- **Support session:** Fridays, 5-6
- **Assessment:** 90% exam in May, 10% coursework (best 3 out of 4), details announced on Blackboard

# Textbooks

Particularly useful books:

- S.R. Pliska, *Introduction to Mathematical Finance: Discrete Time Models*, Blackwell Publishers (1997) [main resource for the first half of the course]
- M. Baxter and A. Rennie, *Financial Calculus*, Cambridge University Press (1996).
- N.H. Bingham and R. Kiesel, *Risk-Neutral Valuation: Pricing and Hedging of Financial Derivatives*, Springer (1998).

Further books mentioned in unit description.



# Some myths and facts

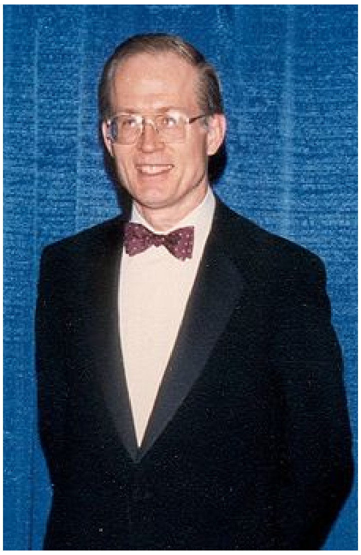
Financial Mathematics. . .

- . . . explains how to earn money. No, better do Time Series Analysis?
- . . . is an applied course. Actually rather theoretical, much in the style of Definition-Lemma-Theorem-Proof.
- . . . is an easy course. We make use of fairly sophisticated concepts like stochastic integration.

In fact . . .

- Hedging, not forecasting.
- Derivation of Black-Scholes formula
- Nobel prize for Merton and Scholes in 1997

# Black–Scholes–Merton



Fischer Black
(1938-95)

Myron Scholes
(1941-)

Robert Merton
(1944-)



# Structure of course

1. Financial Terminology and Single-Period models (Weeks 13, 14)
2. Stochastic Processes in Discrete Time (Weeks 14, 15)
3. Multi-Period models (Weeks 16, 17, 19, 20)
4. Stochastic Processes in Continuous Time (Weeks 20, 21)
5. Financial Market Models in Continuous Time (Weeks 22, 23)

# 1. Financial Terminology and Single-Period models

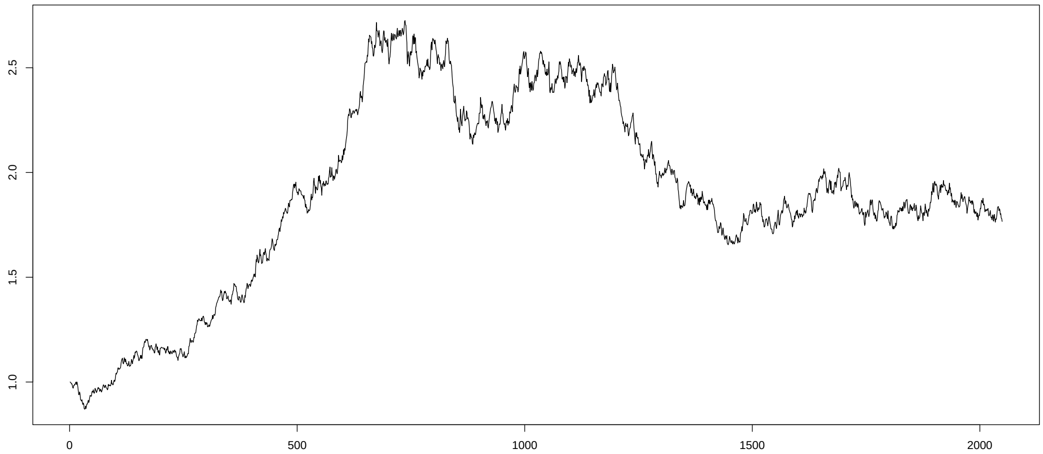




## 1.1: Motivation

- Course distinguishes between two types of financial objects:
  1. 'Underlying' traded assets: e.g. oil, interest rates, exchange rates.
  2. Derivative security: contracts based on these assets.
- For example: a piece of paper which says 'I will pay you \$1 million for every dollar the price of oil is over \$80 on 1st December 2025.'
- Such contracts act as insurance – for example, an airline might find such a contract useful.
- Key question: what is such a piece of paper worth?
- Answer comes in two parts: modelling and arbitrage.

# 1.1: Motivation




Modelling:

- The value of the contract depends on a (plausible) model for oil price.
- Classical models for price processes involve Brownian motion - will appear in the second half of the course.
- How to choose a model is an important practical issue - beyond the scope of this course (that's why you should take courses in statistics!).



# 1.1: Motivation

Arbitrage and some key concepts:

- Given a model, need to decide a sensible price for the contract
- Key idea is that of arbitrage – free money.
- Specifically . . . we will arrive at prices for contracts by assuming that there isn't any free money (markets move to remove it) 'no arbitrage principle'.
- Hence we can decide the price of something because 'if it was worth something else', there would be arbitrage.
- Assume that assets are divisible into arbitrary fractions, no transactions costs, and short-selling is possible.
- Short selling: Borrowing and selling an asset with the obligation to buy it back later. [hoping the price of asset will fall]

# 1.1: Motivation

## Example 1.1.

- Consider a share which trades on 1st January at £10. Suppose we know that on 1st July with probability $p = 2/3$ it will be worth £25, and with probability $1/3$ it will be worth £5.
- On 1st January, the following contract is freely traded (bought and sold) on the market:
  1. 'If the share goes up, I will pay you £4. If the share goes down, I pay you nothing.'
  2. You also have a bank account that pays no interest, which you can pay into or borrow from.
- What is the contract worth on 1st January?
- It might appear that it depends on the probability that the share goes up. E.g., you might argue it should be $2/3 \times 4 + 1/3 \times 0 =$ £$8/3$. (Expected value)
- Surprisingly there is a good argument why the contract is worth £1 – whatever the value of $p$!!



## Example 1.1.

- Create a replicating portfolio as follows: buy $1/5$ unit of the stock, and borrow £1 from the bank (put £$-1$ in the bank).
- If the stock goes up, the portfolio is now worth $1/5 \times 25 - 1 =$£4 – exactly the same as the contract.
- If the stock goes down, the portfolio is now worth $1/5 \times 5 - 1 =$£0 – exactly the same as the contract.
- **Key insight**: whatever happens, this replicating portfolio pays exactly the same as the contract, so it must be worth exactly the same.
- The portfolio costs $1/5 \times 10 - 1 =$£1, so must the contract.
- If contract trades for more – say £1.05, then sell contract for that much, and buy portfolio for £1. On 1st July, whatever happens, the portfolio pays out same as contract, so we can cover our promise. However, we have £0.05 left – guaranteed profit!
- If contract trades for less – say £0.95, then buy contract for that amount, and sell portfolio for £1. Again, contract and portfolio match, we make £0.05 guaranteed.

## 1.1: Motivation

**Example 1.1.**

- Will learn how to create replicating portfolios in general.
- The 'true probability' of going up doesn't affect the argument above.
- However, there exists a 'phantom probability' $q = 1/4$ that does make things interesting.
- The price of the contract is the expected value as if the probability of going up is $1/4$ – that expectation is $(1/4) \times 4 + (3/4) \times 0 = £1$.
- Interestingly, under this phantom probability, the expected value of the stock on 1st July is $(1/4) \times 25 + (3/4) \times 5 = £10$ – exactly what it was on 1st January.
- In fact, this is how to find $q$. This $q$ is chosen to make the average value of the stock constant.
- Probabilities with respect to this $q$ called equivalent martingale measures.



## 1.1: Motivation

More interesting behaviour is possible when we consider that money does not have constant value. (Interest rates/inflation).

**Definition 1.2 (Bank Process).**

We assume the existence of a risk-free bank account with known interest rate $r$ which is often assumed constant. Then an initial deposit of one unit becomes

1. $B_t = e^{rt}$ for each time point $t \in [0, T]$ (continuous time models)
2. $B_t = (1 + r)^t$ for each time point $t \in \{0, \dots, T\}$ (multi-period models).

- We must consider the performance of the portfolio relative to the bank process.
- We will discount our profits by factor of $B_t$.
- Some textbooks call the bank account process $B_t$ a numeraire or refer to the bank process as a bond.

# 1.2: Derivative securities

**Definition 1.3.**

Derivative security: is a contract whose value at expiration date $T$ ('expiry') is a function of the values of the assets within the time interval $[0, T]$.

Often, the value is just a function of the value at time $T$.

**Example 1.4.**

A piece of paper which says 'I will pay you \$1 million for every dollar the price of oil is over \$80 on 1st December 2025' is a derivative.



# 1.2: Derivative securities

**Definition 1.5.**

A forward contract: is an agreement to buy or sell asset $S$ at future date $T$ for price $K$.

- The agent who agrees to buy underlying asset is said to have a long position, the other agent assumes a short position.
- $T$ is called the delivery date or settlement date
- $K$ is called the delivery price
- Forward price $f(t; T)$: the delivery price which would make the contract have zero value at time $t$.

## 1.2: Derivative securities

**Example 1.6.**

A piece of paper which says 'I will buy 1000 litres of petrol for £1,000 next January 1st' is a forward contract.

We will find below the fair delivery price (see Proposition 1.12). The forward price equals at the time where the contract is set up, $t = 0$, the delivery price, hence $f(0; T) = K$. The forward prices $f(t; T)$ need not (and will not) necessarily be equal to the delivery price $K$ during the life-time of the contract.



## 1.2: Derivative securities

Very common form of derivative studied in this course is an 'option'.

**Definition 1.7.**

An option is a financial instrument giving one the right, but not the obligation to buy or sell an asset $S$ at (or by) a specified date $T$ at a specified strike price $K$.

- Call options give one the right to buy, put options give one the right to sell.
- European options give one the right to buy/sell on the specified date, the expiry date, on which the option expires or matures.
- American options give one the right to buy/sell at any time prior to or at expiry.

## 1.2: Derivative securities

**Example 1.8.**

- A piece of paper which says 'I have the right to buy 1000 litres of petrol for £1,000 next January 1st' is a call option.
- If the market price $S_T$ on 1st January is more than £1,000, it is worth exercising the option. You will make a profit of $S_T - 1000$. If the market price $S_T$ is less than £1,000, it is not worth exercising the option, and you will make no profit.
- The value on January 1st of the option will be $(S_T - 1000)_+$, a function of the (random) price $S_T$.
- Here the positive part $x_+ = x$ if $x \geq 0$ and $x_+ = 0$ otherwise.
- Similarly a put option says 'I have the right to sell 1000 litres of petrol for £1,000'. If $S_T \geq 1000$, there is no point exercising the option. If $S_T \leq 1000$, it is worth exercising, to make a profit of $1000 - S_T$. Overall, we receive $(1000 - S_T)_+$.
- Put and call options are exercised in complementary cases.



## 1.3: Arbitrage

**Definition 1.9.**

An arbitrage opportunity is the possibility of being able to make a profit on a transaction without being exposed to the risk of incurring a loss. Traders who try to lock in riskless profit are called arbitrageurs.

## 1.3: Arbitrage

**Example 1.10.**

Consider stock that is traded on both the New York Stock exchange and the London Stock Exchange. Suppose stock price is \$189 in New York and £100 in London at a time when the exchange rate is \$1.8700 per pound.
Arbitrage opportunity:

- Buy 100 shares in London
- Sell all shares in New York
- Change Dollars to Pounds

Riskless profit is £106.95 (in the absence of transaction costs).



## 1.3: Arbitrage

One might hypothesize that this arbitrage opportunity cannot last for long—the forces of supply and demand will increase the sterling price of the stock in London and decrease the dollar price in New York.

**Definition 1.11 (No-arbitrage principle).**

When determining prices we assume an idealized market without arbitrage opportunities.

Remarkably, we can determine prices of many derivatives, only by assuming the no-arbitrage principle.

## 1.4: The forward price

In this section we derive the price of a forward contract by just assuming no-arbitrage. This can be thought off as a one-off payment at a certain time to the holder of the asset. We also make use of short selling which reverses the order of buying and selling. Short selling allows the seller to sell an asset that he does not own and buy it a later point of time.

**Proposition 1.12.**

*Consider a forward contract on an asset with current price $S_0$ The delivery date of the forward contract is $T$. We assume the existence of a risk-free bank account with constant interest rate $r$ during the interval $[0, T]$, so that an initial deposit of one unit grows to $B_t = e^{rt}$ up to time point $t$. Then the delivery price of the forward contract $K$ is*

$$K = S_0 e^{rT}.$$



**Proof.**

We use the no-arbitrage principle to prove this statement and assume first that

$$K > S_0 e^{rT}.$$

Then an arbitrageur can adopt the following strategy:

- Borrow $S_0$ at an interest rate of $r$
- Buy the underlying asset.
- Take short position in forward contract – agree to sell the asset at time $T$.

At time point $T$ we sell the asset for $K$ and repay the loan. The riskless profit is

$$K - S_0 e^{rT} > 0$$

whatever the price of the underlying asset at time point $T$ is.

**Proof.**

If, on the other hand,

$$K < S_0 e^{rT},$$

an arbitrageur could adopt the following strategy:

- Sell short the underlying asset.
- Invest the proceeds of $S_0$ at the risk-free interest rate of $r$.
- Enter a long position in the forward contract to buy the underlying asset at time $T$.

On the delivery date the arbitrageur buys the asset for $K$, returns it to the short-seller and makes a riskless profit of

$$S_0 e^{rT} - K > 0.$$

□



## 1.5: Put-call-parity and arbitrage bounds

**Proposition 1.13.**

*Consider two combinations of financial derivatives that both have the same value $V = W$ at time point $T$. Then their prices $V_t$ and $W_t$ at any time point $t < T$ must also coincide.*

**Proof.**

We use no-arbitrage and assume wlog that $V_t > W_t$. At time $t$ we

- sell (or short) the first combination,
- buy the second combination and
- invest the difference $(V_t - W_t)$.

At time point $T$ we

- sell the second combination for $V = W$ and
- rebuy the first combination for $W$.

Profit is $(V_t - W_t)e^{r(T-t)}$ assuming a risk-free interest rate of $r$. □

## 1.5: Put-call-parity and arbitrage bounds

We apply this result to European calls and puts with the same underlying asset, strike price $K$ and expiry date $T$. Assume that $S_T$ is the price of the underlying asset at time point $T$, then the payoff of the call at that time is

$$C = (S_T - K)^+$$

and payoff of the put is

$$P = (K - S_T)^+.$$

Now choose for the first combination the underlying asset and the European put. The value at time point $T$ is

$$V = S_T + (K - S_T)^+ = \max\{S_T, K\}.$$

For the second combination we choose the European call option and a bond that matures at time point $T$ with a price of $K$. The value of the second combination is

$$W = K + (S_T - K)^+ = \max\{S_T, K\}.$$



## 1.5: Put-call-parity and arbitrage bounds

Using the result above we obtain:

**Proposition 1.14 (Put-call-parity).**

*Let $S_t$ be the price of the asset at time point $t$, $Ke^{-r(T-t)}$ the discounted value of the bond and $C_t$ and $P_t$ the prices of European call and put options at time $t$. Then*

$$S_t + P_t - C_t = Ke^{-r(T-t)}.$$

As a consequence from the put-call-parity we obtain the following lower bound for a European call option:

**Corollary 1.15.**

*With the same notation as in the put-call-parity Proposition 1.14*

$$C_t \geq \max\{0, S_t - Ke^{-r(T-t)}\}.$$

## 1.5: Put-call-parity and arbitrage bounds

Finally we can obtain another interesting relationship without any specification of the underlying mathematical model.

**Proposition 1.16.**

*For a non-dividend paying stock we have*

$$C_A = C_E$$

*where $C_A$ and $C_E$ are the prices of the American and European call options on the same underlying stock, with the same strike price and expiry date.*

Proof.

To see that note that exercising the American call early at time $t < T$ generates the cash-flow $S_t - K$. On the other hand from the inequality above we know that selling the call options yields a cash-flow of at least $S_t - Ke^{-r(T-t)}$. Thus early exercising the call early is suboptimal. □



## 1.6: The single-period model

**Definition 1.17.**

- A *sample space* $\Omega$ is a set consisting of all elementary outcomes.

**Definition 1.18.**

- A *random variable* $X$ is a function $X : \Omega \to \mathbb{R}$

## 1.6: The single-period model

Single-period models illustrate many of the important economic principles associated even with the most complex, continuous time models.

**Definition 1.19.**

A single period-model is a model for a financial market made up of the following components:

- Initial date $t = 0$ and terminal date $t = 1$ with trading and consumption possible at these two dates.
- A finite sample space $\Omega$ with $K < \infty$ elements:

$$\Omega = \{\omega_1, \ldots, \omega_K\}$$

each corresponding to a certain state of the world.
- A probability measure $\mathbb{P}$ on $\Omega$ with $\mathbb{P}(\{\omega\}) > 0$ for all $\omega \in \Omega$.



## 1.6: The single-period model

**Definition 1.19.**

- A bank account process $B = \{B_t : t = 0, 1\}$ where $B_0 = 1$ and $B_1 \geq 1$ should be thought of as the time $t = 1$ value of the bank account when 1 unit of a currency is deposited at time $t = 0$ and $r = B_t - 1 \geq 0$ should be thought of as the interest rate. In this course, $B_1$ and $r$ are considered deterministic scalars, but one could allow $B_1$ (and $r$) to be random variables.
- A price process $S = \{S_t : t = 0, 1\}$ where $S = (S_1(t), S_2(t), \ldots, S_N(t)), N < \infty$ and $S_n(t)$ is the time $t$ price of the $n$-th security. For many applications these $N$ risky securities are stocks. The time $t = 0$ prices are positive scalars that are known to the investors whereas the time $t = 1$ prices are non-negative random variables whose values become known to the investors only at time $t = 1$.

## 1.6: The single-period model

**Definition 1.20.**

A trading strategy $H = (H_0, \dots, H_N)$ describes an investor's portfolio as carried from time $t = 0$ to time $t = 1$. In particular $H_0$ is the number of units of the currency invested in the savings account and for $n \geq 1$ the scalar $H_n$ is the number of units of the $n$-th security. In general, $H_n$ can be positive or negative where negative means borrowing or selling short.



## 1.6: The single-period model

**Example 1.21.**

> *Consider a share which trades on 1st January at £10. Suppose we know that with probability $p = 2/3$ it will be worth £25, and with probability 1/3 it will be worth £5 on 1st July. . . . You also have a bank account that pays no interest, which you can pay into or borrow from. . . . Create a replicating portfolio as follows: buy $1/5$ unit of the stock, and borrow £1 from the bank.*

We can rephrase this in terms of a single-period ($T = 1$) model.

- Bank account process $B_0 = B_1 = 1$.
- Sample space $\Omega$: has two elements ($K = 2$), with $\omega_1$ ('up') and $\omega_2$ ('down').
- There are $N = 1$ stocks. The stock process $S_1(0)(\omega_i) = 10$ for $i = 1, 2$, $S_1(1)(\omega_1) = 25$, $S_1(1)(\omega_2) = 5$.
- Can write replicating portfolio as a vector $(H_0, H_1) = (-1, 1/5)$.

## 1.6: The single-period model

**Definition 1.22.**

The value process $V = \{V_t : t = 0, 1\}$ describes the total value of the portfolio at each point in time:

$$V_t = H_0 B_t + \sum_{n=1}^{N} H_n S_n(t).$$

**Definition 1.23.**

The gains process $G$ is a random variable that describes the total profit or loss generated by the portfolio between times 0 and 1:

$$G = H_0 r + \sum_{n=1}^{N} H_n \Delta S_n$$

where $\Delta S_n = S_n(1) - S_n(0)$.



## 1.6: The single-period model

It is convenient to normalize the prices, so that the bank account becomes constant by defining discounted versions of the processes defined above:

**Definition 1.24.**

We define the discounted price process $S^* = \{S_t^* : t = 0, 1\}$ by

$$S_n^*(t) = S_n(t)/B_t, \quad n = 1, \ldots, N; \quad t = 0, 1,$$

the discounted value process $V^* = \{V_t^* : t = 0, 1\}$ by

$$V_t^* = H_0 + \sum_{n=1}^{N} H_n S_n^*(t), \quad t = 0, 1$$

and the discounted gains process $G^*$ by the random variable

$$G^* = \sum_{n=1}^{N} H_n \Delta S_n^* \quad \text{where} \quad \Delta S_n^* = S_n^*(1) - S_n^*(0).$$

# 1.6: The single-period model

Simple calculations show:

**Proposition 1.25.**

*Let H be a trading strategy in a single-period model, then*

$$V_t^* = V_t/B_t, \quad t = 0, 1$$

*and*

$$V_1^* = V_0^* + G^*.$$



# 1.6: The single-period model

We return to the setting of Example 1.1, and introduce a non-zero interest rate for the sake of generality.

**Example 1.26.**

- Bank account process $B_0 = 1$, $B_1 = 1 + r$.
- Probability space $\Omega$ has two elements ($K = 2$), with $\omega_1$ ('up') and $\omega_2$ ('down').
- There are $N = 1$ stocks. The stock process $S_1(0)(\omega_i) = 10$ for $i = 1, 2$, $S_1(1)(\omega_1) = 25$, $S_1(1)(\omega_2) = 5$.
- The discounted stock process

$$\begin{aligned} S_1^*(0)(\omega_i) &= 10 \text{ for } i = 1, 2, \\ S_1^*(1)(\omega_1) &= 25/(1+r), \\ S_1^*(1)(\omega_2) &= 5/(1+r). \end{aligned}$$

# 1.6: The single-period model

**Example 1.26.**

- For an arbitrary trading strategy $H$ we have $V_0 = V_0^* = H_0 + 10H_1$.
- Hence in state $\omega_1$

$$\begin{aligned}
V_1(\omega_1) &= (1+r)H_0 + 25H_1 \\
V_1^*(\omega_1) &= H_0 + 25H_1/(1+r) \\
G(\omega_1) &= rH_0 + 15H_1 \\
G^*(\omega_1) &= (25/(1+r) - 10)H_1.
\end{aligned}$$

- Whereas in state $\omega_2$

$$\begin{aligned}
V_1(\omega_2) &= (1+r)H_0 + 5H_1 \\
V_1^*(\omega_2) &= H_0 + 5H_1/(1+r) \\
G(\omega_2) &= rH_0 - 5H_1 \\
G^*(\omega_2) &= (5/(1+r) - 10)H_1
\end{aligned}$$



# 1.7: Arbitrage in the single-period model

**Definition 1.27.**

An arbitrage opportunity is some trading strategy $H$ such that

(a) $V_0 = 0$,

(b) $V_1 \geq 0$,

(c) $\mathbb{P}(V_1 > 0) > 0$.

- (b) means that $V_1(\omega) \geq 0$ for all $\omega$.
- Note that (b) and (c) together imply $\mathbb{E}V_1 > 0$ and $\mathbb{E}V_1^* > 0$.
- An equivalent condition for arbitrage in the single-period model is given by the following Proposition.

## 1.7: Arbitrage in the single-period model

**Proposition 1.28.**

*There exists some arbitrage opportunity if and only if there is some trading strategy $H$ such that $G^* \geq 0$ and $\mathbb{E}(G^*) > 0$.*

Proof.

- To see this, suppose $H$ is an arbitrage opportunity. Since $G^* = V_1^* - V_0^*$ and $B_t > 0$ for all $t$ and $\omega$ we find that $G^* \geq 0$ and thus $\mathbb{E}(G^*) = \mathbb{E}(V_1^*) > 0$.
- Conversely suppose that $H$ satisfies $G^* \geq 0$ and $\mathbb{E}(G^*) > 0$. Then define the strategy $\hat{H} = (\hat{H}_0, H_1, \ldots, H_N)$ where $\hat{H}_0 = -\sum_{n=1}^{N} H_n S_n^*(0)$.
- Under $\hat{H}$ one has $V_0^* = 0$ and $V_1^* = V_0^* + G^* = G^*$.
- Hence $V_1^* \geq 0$ and $\mathbb{E}(V_1^*) > 0$, in which case $\hat{H}$ is an arbitrage opportunity.

□



## 1.8: Risk neutral probability measures

**Definition 1.29.**

A probability measure $\mathbb{Q}$ on $\Omega$ is said to be a risk neutral probability measure if

(a) $\mathbb{Q}(\{\omega\}) > 0$ for all $\omega \in \Omega$ and

(b) $\mathbb{E}_{\mathbb{Q}}(S_n^*(1)) = S_n^*(0)$ for $n = 1, 2, \ldots, N$.

A very important result is the no-arbitrage theorem:

**Theorem 1.30 (No-arbitrage theorem).**

*There are no arbitrage opportunities if and only if there exists a risk neutral probability measure $\mathbb{Q}$.*

## 1.8: Risk neutral probability measures

The proof of the no-arbitrage theorem relies on the separating hyperplane theorem, a consequence of the Hahn-Banach theorem in functional analysis. We will not prove this theorem in our lecture and just give below one of its many versions.

**Theorem 1.31 (Separating hyperplane theorem).**

*Let $\mathbb{W}$ be a linear subspace of $\mathbb{R}^K$ and let $\mathbb{K}$ be a compact convex subset in $\mathbb{R}^K$ disjoint from $\mathbb{W}$. Then we can separate $\mathbb{W}$ and $\mathbb{K}$ strictly by a hyperplane containing $\mathbb{W}$, ie there exists some vector $v \in \mathbb{R}^K$ orthogonal to $\mathbb{W}$*

$$u^T v = 0 \text{ for all } u \in \mathbb{W}$$

*such that*

$$u^T v > 0 \text{ for all } u \in \mathbb{K}$$



## 1.8: Risk neutral probability measures

Proof.

We consider three sets:

- $\mathbb{W} = \{X \in \mathbb{R}^K : X = G^* \text{ for some trading strategy } H\}$.

  Think of it as a set of random variables each representing a possible $t = 1$ discounted wealth when the initial value of the investment is zero. Actually, $\mathbb{W}$ is a linear subspace of $\mathbb{R}^K$.

- $\mathbb{A} = \{X \in \mathbb{R}^K : X \geq 0, X \neq 0\}$.

  In view of Proposition 1.28 above there exists an arbitrage opportunity if and only if $\mathbb{W} \cap \mathbb{A} \neq \emptyset$.

- $\mathbb{A}^+ = \{X \in \mathbb{R}^K : X \geq 0, X \neq 0, \sum_{i=1}^K X_i = 1\}$.

  This is a compact convex subset of $\mathbb{R}^K$.

## Proof.

Let us now assume that there is no arbitrage opportunity, then $\mathbb{W}$ and $\mathbb{A}^+$ are disjoint.
By the separating hyperplane theorem there exists some vector $Y \in \mathbb{R}^K$ orthogonal to $\mathbb{W}$

$$X^T Y = 0 \text{ for all } X \in \mathbb{W}$$

such that

$$X^T Y > 0 \text{ for all } X \in \mathbb{A}^+.$$

For each $k = 1, \ldots, K$ the $k$-th unit vector $e_k$ is an element of $\mathbb{A}^+$.
Therefore for all $k = 1, \ldots, K$

$$Y_k = e_k^T Y > 0,$$

ie all entries of $Y$ are strictly positive.



## Proof.

We define a probability measure $\mathbb{Q}$ by setting

$$\mathbb{Q}(\{\omega_k\}) = Y(\omega_k)/(Y(\omega_1) + \cdots + Y(\omega_K)).$$

Furthermore $\Delta S_n^* \in \mathbb{W}$ for all $n$ because $\Delta S_n^*$ is the discounted wealth for the portfolio $H = e_n$ which consists of one unit of the $n$th asset only.
Since $Y$ is orthogonal to $\mathbb{W}$ we conclude that

$$\mathbb{E}_Q(\Delta S_n^*) = \sum_{k=1}^{K} \Delta S_n^*(\omega_k) \cdot \mathbb{Q}(\{\omega_k\}) = 0 \text{ for all } n.$$

In other words

$$\mathbb{E}_{\mathbb{Q}}(S_n^*(1)) = S_n^*(0); \text{for all } n.$$

Thus $\mathbb{Q}$ is a risk neutral probability measure.

Proof.

For the converse assume that $\mathbb{Q}$ is a risk neutral probability measure. Then for an arbitrary trading strategy $H$ we have

$$\mathbb{E}_{\mathbb{Q}}(G^*) = \mathbb{E}_{\mathbb{Q}}\left(\sum_{n=1}^{N} H_n \Delta S_n^*\right) = \sum_{n=1}^{N} H_n \mathbb{E}_{\mathbb{Q}}(\Delta S_n^*) = 0$$

and, in particular,

$$\sum_{k=1}^{K} G^*(\omega_k)\mathbb{Q}(\omega_k) = 0$$

which shows that either $G^*(\omega_k) < 0$ for some $k$ or $G^* = 0$, but then also $\mathbb{E}_{\mathbb{P}}(G^*) = 0$. Hence by Proposition 1.28 there cannot be any arbitrage opportunities. □



## 1.9: Valuation of contingent claims

**Definition 1.32.**

A contingent claim in the single period model is a random variable $X$ representing a payoff at time $t = 1$. A contingent claim $X$ is said to be attainable if there exists some trading strategy $H$ such that $V_1 = X$. In this case one says that $H$ generates $X$ and $H$ is called the replicating portfolio.

## 1.9: Valuation of contingent claims

What is the fair price $p$ that the buyer should pay at time $t = 0$ for an attainable contingent claim? Let $H$ be a replicating portfolio and $V_0$ its value at time point 0. Then $V_0$ must be the fair price for the contingent claim because otherwise:

- If $p > V_0$ then an arbitrage opportunity arises because an arbitrageur would sell the contingent claim for $p$ at time 0, follow the trading strategy $H$ at a time $t = 0$ cost of $V_0$ and pocket the difference $p - V_0$. At time $t = 1$ the value $V_1$ of the portfolio matches the obligation $X$ of the contingent claim in every state of the world.
- If $p < V_0$ an arbitrageur would follow the trading strategy $-H$ and purchase the contingent claim, again making riskless profit.

(could also be seen using Proposition 1.13.)



## 1.9: Valuation of contingent claims

Could there be a second trading strategy $\hat{H}$ such that $\hat{V}_1 = X$, but $\hat{V}_0 \neq V_0$? Let $\mathbb{Q}$ be any risk-neutral probability measure, then for any trading strategy $H$ we have as seen in the proof of no-arbitrage theorem

$$\mathbb{E}_{\mathbb{Q}}(G^*) = 0$$

and consequently

$$V_0 = V_0^* = \mathbb{E}_{\mathbb{Q}}(V_0^*) = \mathbb{E}_{\mathbb{Q}}(V_1^* - G^*) = \mathbb{E}_{\mathbb{Q}}(V_1^*) - 0 = \mathbb{E}_{\mathbb{Q}}(V_1/B_1). \quad (1.1)$$

In particular any replicating strategy with $V_1 = X$ has time 0 value

$$V_0 = \mathbb{E}_{\mathbb{Q}}(X/B_1).$$

This calculation does not depend on the choice of $\mathbb{Q}$ and therefore $\mathbb{E}_{\mathbb{Q}}(V_1^*)$ is constant even if there are two or more risk neutral probability measures.

## 1.9: Valuation of contingent claims

Summarizing we have the following result on the valuation of contingent claims:

**Proposition 1.33 (Risk neutral valuation principle:).**

*If the single period model is free of arbitrage opportunities, then the time $t = 0$ value of an attainable contingent claim $X$ is $\mathbb{E}_{\mathbb{Q}}(X/B_1)$ where $\mathbb{Q}$ is any risk neutral probability measure.*



## 1.9: Valuation of contingent claims

**Example 1.34.**

Consider the setting of Example 1.1.

- In order to find a risk neutral probability measure we need to find strictly positive numbers $\mathbb{Q}(\{\omega_1\})$ and $\mathbb{Q}(\{\omega_2\})$ so that $\mathbb{E}_{\mathbb{Q}}(S_n^*(1)) = S_n^*(0)$ is satisfied, that is

$$10 = 25/(1+r)\mathbb{Q}(\{\omega_1\}) + 5/(1+r)\mathbb{Q}(\{\omega_2\}).$$

- Also $\mathbb{Q}$ must be a probability measure, so it must satisfy

$$1 = \mathbb{Q}(\{\omega_1\}) + \mathbb{Q}(\{\omega_2\}).$$

- $\mathbb{Q}(\{\omega_1\}) = (1+2r)/4$ and $\mathbb{Q}(\{\omega_2\}) = (3-2r)/4$ satisfy both equations, so this is a risk neutral probability measure and by Theorem 1.30 there cannot be any arbitrage opportunities.
- In particular for $r = 0$ we recover $\mathbb{Q}(\{\omega_1\}) = 1/4$, as in Example 1.1.

## 1.9: Valuation of contingent claims

**Example 1.34.**

- Suppose $X$ is a contingent claim with $X(\omega_1) = 4$ and $X(\omega_2) = 0$.
- If $X$ is attainable, then the time $t = 0$ value of $X$ is

$$\mathbb{E}_{\mathbb{Q}}(X/B_1) = \frac{(1+2r)}{4} \times \frac{4}{1+r} = \frac{1+2r}{1+r}.$$

- Check whether $X$ can be generated by solving for $i = 1, 2$:

$$\begin{aligned} 4 &= H_0(1+r) + H_1(25) \\ 0 &= H_0(1+r) + H_1(5) \end{aligned}$$

- Solution $H_0 = -1/(1+r)$ and $H_1 = 1/5$, so $X$ is indeed attainable.
- Can check time 0 value:

$$V_0 = H_0 + H_1 S_1 = -\frac{1}{1+r} + 1/5 \times 10 = \frac{1+2r}{1+r}.$$

- Note we find $(H_0, H_1)$ by solving $AH = X$, for certain $A$, $X$.



## 1.10: Complete and incomplete markets

**Definition 1.35.**

A model is said to be complete if every contingent claim $X$ can be generated by some trading strategy. Otherwise, the model is said to be incomplete.

One way to check whether a model is complete is to define the $K \times (N+1)$-matrix $A$ by

$$A = \begin{pmatrix} B_1(\omega_1) & S_1(1)(\omega_1) & \dots & S_N(1)(\omega_1) \\ B_1(\omega_2) & S_1(1)(\omega_2) & \dots & S_N(1)(\omega_2) \\ \vdots & \vdots & & \vdots \\ B_1(\omega_K) & S_1(1)(\omega_K) & \dots & S_N(1)(\omega_K) \end{pmatrix}$$

The vector $AH$ gives the value process at time 1, ie $(V_1(\omega_1), \dots, V_K(\omega_k))$.

**Proposition 1.36.**

*A model is complete if and only if, for any $X \in \mathbb{R}^K$, there exists $H$ such that $AH = X$.*

# 1.10: Complete and incomplete markets

Usually the choice of a risk neutral probability measure is not unique. However for complete markets we have the following theorem:

**Theorem 1.37.**

*Suppose there are no arbitrage opportunities. Then the model is complete if and only if there is a unique risk neutral probability measure.*



Proof.

We denote by $\mathbb{M}$ the set of all risk neutral probability measures. Since there are no arbitrage opportunities we know that $\mathbb{M} \neq \emptyset$.

Let us first assume that the model is complete, but $\mathbb{M}$ contains two distinct risk neutral probability measures $\mathbb{Q}$ and $\hat{\mathbb{Q}}$. In this case there must exist some state $\omega_k$ with $\mathbb{Q}(\{\omega_k\}) \neq \hat{\mathbb{Q}}(\{\omega_k\})$, so take the contingent claim $X$ defined by

$$X(\omega) = \begin{cases} B_1(\omega_k) & \text{if } \omega = \omega_k \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}_{\mathbb{Q}}(X/B_1) = \mathbb{Q}(\{\omega_k\}) \neq \hat{\mathbb{Q}}(\{\omega_k\}) = \mathbb{E}_{\hat{\mathbb{Q}}}(X/B_1).$$

But this contradicts the calculation (1.1) where we saw that if $X$ is attainable, then $\mathbb{E}_{\mathbb{Q}}(X/B_1)$ takes the same value for all $\mathbb{Q} \in \mathbb{M}$.

## Proof.

To show the converse let us assume that there is only one risk neutral probability measure $\hat{\mathbb{Q}}$, but there exists a contingent claim $X$ which is not attainable. Then there is no solution $H$ to the system $AH = X$. By the separating hyperplane theorem it follows that there must exist a row vector $\pi \in \mathbb{R}^K$ satisfying

$$\pi^T A = 0, \quad \pi^T X > 0.$$

Let $\lambda > 0$ be small enough so that

$$\mathbb{Q}(\{\omega_j\}) = \hat{\mathbb{Q}}(\{\omega_j\}) + \lambda \pi_j B_1(\omega_j) > 0, \quad j = 1, \ldots, K.$$

Since $B_1$ is the first column of $A$ and $\pi^T A = 0$ the quantity $\mathbb{Q}$ just defined is actually a probability measure.



## Proof.

Moreover, for any discounted price process $S^* = (S_1^*, \ldots, S_N^*)$ and for every $n = 1, \ldots, N$ we have

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}(S_n^*(1)) &= \sum_{j=1}^K \mathbb{Q}(\{\omega_j\}) S_n(1, \omega_j)/B_1(\omega_j) \\
&= \sum_{j=1}^K \hat{\mathbb{Q}}(\{\omega_j\}) S_n(1, \omega_j)/B_1(\omega_j) + \lambda \sum_{j=1}^K \pi_j S_n(1, \omega_j) \\
&= \sum_{j=1}^K \hat{\mathbb{Q}}(\{\omega_j\}) S_n^*(1, \omega_j) \\
&= \mathbb{E}_{\hat{\mathbb{Q}}}(S_n^*(1)) = S_n^*(0).
\end{aligned}$$

Thus $\mathbb{Q} \in \mathbb{M}$ which is a contradiction to the uniqueness of $\hat{\mathbb{Q}}$. □

# 1.10: Complete and incomplete markets

**Example 1.38.**

In the context of Example 1.34, another way to check that $X$ is attainable would be to check if the model is complete.

- Consider the matrix
$$A = \begin{pmatrix} 1+r & 25 \\ 1+r & 5 \end{pmatrix}.$$
- Since $A$ has two linearly independent columns $AH$ can take all values in $\mathbb{R}^2$.
- Hence the model is complete and hence all contingent claims are attainable.



# 1.10: Complete and incomplete markets

# 2. Stochastic Processes in Discrete Time





## 2.1: Multi-period models

Multi-period models of securities markets are much more realistic than single-period models.

**Definition 2.1.**

A multi-period model is a model for a financial market made up of the following elements:

- $T+1$ trading dates: $t = 0, 1, \ldots, T$
- A finite sample space $\Omega$ with $K < \infty$ elements:

$$\Omega = \{\omega_1, \ldots, \omega_K\}$$

  each corresponding to a certain state of the world.
- A probability measure $\mathbb{P}$ on $\Omega$ with $\mathbb{P}(\omega) > 0$ for all $\omega \in \Omega$.

## 2.1: Multi-period models

**Definition 2.1.**

- A filtration $\mathcal{F} = \{\mathcal{F}_t : t = 0, 1, \dots, T\}$ which is a submodel describing how the information about the security prices is revealed to the investors.
- A bank account process $B = \{B_t : t = 0, 1, \dots, T\}$ where $B_t$ should be thought of as the time $t$ value of a savings account when 1 unit of a currency is deposited at time $t = 0$. In general, $B$ is a stochastic process, but in this course we always assume that there is a constant and deterministic interest rate $r$ in which case $B$ becomes deterministic and is
$$B_t = (1 + r)^t.$$
- A price process $S = \{S(t) : t = 0, 1, \dots, T\}$ where $S(t) = (S_1(t), \dots, S_N(t))$ and $S_n$ is a non-negative stochastic process for each $n = 1, 2, \dots, N$. Here $S_n(t)$ should be thought of as the time $t$ price of the $n$th risky security.



## 2.2: Information structures

**Example 2.2.**

- Consider a multi-period model with $T = 3$, where the stock jumps up or down at each stage. The sample space has $K = 2^3 = 8$ different outcomes

| $\omega_1$ | $\omega_2$ | $\omega_3$ | $\omega_4$ | $\omega_5$ | $\omega_6$ | $\omega_7$ | $\omega_8$ |
|---|---|---|---|---|---|---|---|
| UUU | UUD | UDU | UDD | DUU | DUD | DDU | DDD |

- Initially, all of these outcomes are possible, ie $A_0 = \Omega = \{\omega_1, \dots, \omega_8\}$.
-
  1. When first move occurs, there are now $4 = 2^2$ options, whatever happens. Say we move up; possibilities are $A_1 = \{\omega_1, \dots, \omega_4\}$.
  2. When second move occurs, there are 2 options. Say we move down, possibilities are $A_2 = \{\omega_3, \omega_4\}$..
  3. When third move occurs, 1. Say we move up, possibility is $A_3 = \{\omega_3\}$.
- Possible subsets $A_t$ at time $t$ form a partition $\mathcal{P}_t$. (disjoint, union is everything)
- For example, for $t = 2$, $\mathcal{P}_2 = \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \{\omega_5, \omega_6\}, \{\omega_7, \omega_8\}\}$.

## 2.2: Information structures

- At time $t = 0$ every state $\omega \in \Omega$ can happen, although usually not with the same probability.
- At time $t = T$ the true state $\omega$ of the world is known.
- Each new piece of information rules out certain states.
- One can view the evolution of information as a random sequence $\{A_t\}$ of subsets of $\Omega$ where

$$A_0 = \Omega, A_T = \{\omega\} \text{ and } A_0 \supseteq A_1 \supseteq \cdots \supseteq A_T.$$

- There exist $K$ possible information sequences $\{A_t\}$ of subsets, corresponding to the $K$ elements of $\Omega$.
- At time $t = 0$ the investors are aware of all these sequences, but they do not know which one is going to occur.



## 2.2: Information structures

**Definition 2.3.**

The Information structure is fully described by a sequence $\mathcal{P}_0, \mathcal{P}_1, \ldots, \mathcal{P}_T$ of partitions of $\Omega$ with

- $\mathcal{P}_0 = \{\Omega\}$,
- $\mathcal{P}_T = \{\{\omega_1\}, \{\omega_2\}, \ldots, \{\omega_K\}\}$, and
- each $A \in \mathcal{P}_t$ is equal to the union of some elements in $\mathcal{P}_{t+1}$ for every $t < T$.

# 2.2: Information structures

**Example 2.4.**

- Consider one arbitrary sequence $\{\widehat{A}_t\}$ and some time $s < T$.
- For each $\omega \in \hat{A}_s$ the sequence $\{A_t\}$ with $A_T = \{\omega\}$ will coincide with $\{\widehat{A}_t\}$ at least up to time $t = s$.
- The collection of subsets $\{A_{s+1}\}$ that can follow $\widehat{A}_s$ forms a partition of $\widehat{A}_s$, that is, a collection of disjoint subsets whose union equals $\widehat{A}_s$.
- In particular, taking $s = 0$, we see that the collection $\{A_1\}$ of all possible time $t = 1$ subsets forms a partition of $\Omega$. This partition is denoted $\mathcal{P}_1$.
- Moreover, the collection $\{A_2\}$ of all possible time $t = 2$ subsets also forms a partition of $\Omega$, denoted $\mathcal{P}_2$.



# 2.2: Information structures

**Example 2.5.**

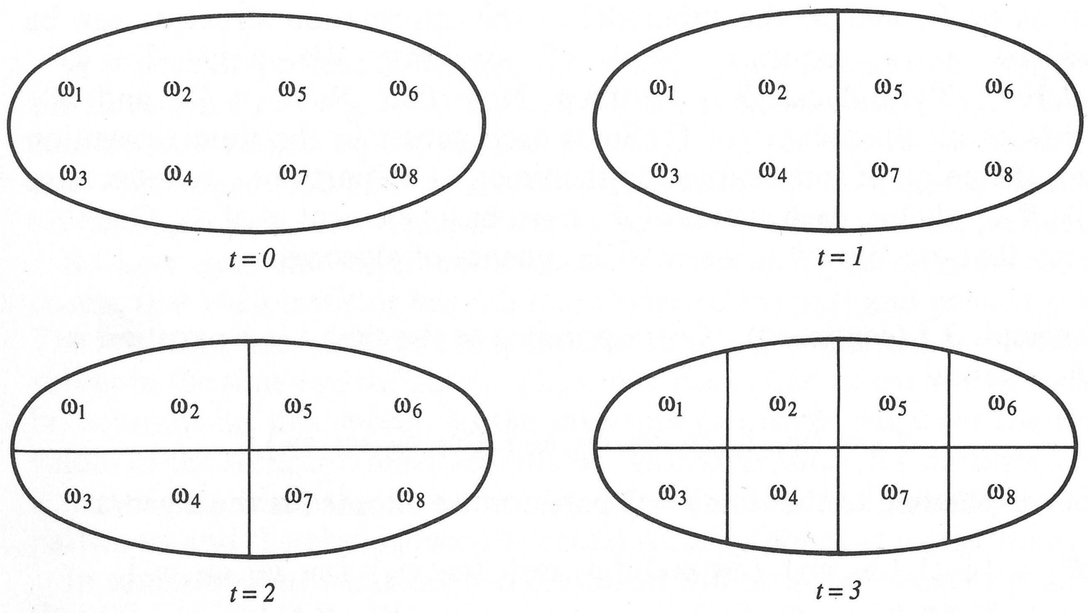
Relabelling motivating Example 2.2, with $K = 8$ and $T = 3$, the partitions are

$$\begin{aligned}
\mathcal{P}_0 &= \{\{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7, \omega_8\}\} \\
\mathcal{P}_1 &= \{\{\omega_1, \omega_2, \omega_3, \omega_4\}, \{\omega_5, \omega_6, \omega_7, \omega_8\}\} \\
\mathcal{P}_2 &= \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \{\omega_5, \omega_6\}, \{\omega_7, \omega_8\}\} \\
\mathcal{P}_3 &= \{\{\omega_1\}, \{\omega_2\}, \{\omega_3\}, \{\omega_4\}, \{\omega_5\}, \{\omega_6\}, \{\omega_7\}, \{\omega_8\}\}.
\end{aligned}$$

We can visualize the information structure with

- a tree diagram where each node corresponds to one element $A_t$ of the time $t$ partition and where there is one arc going from this node to each node corresponding to some $A_{t+1} \subseteq A_t$
- or a sequence of pictures of the sample space.

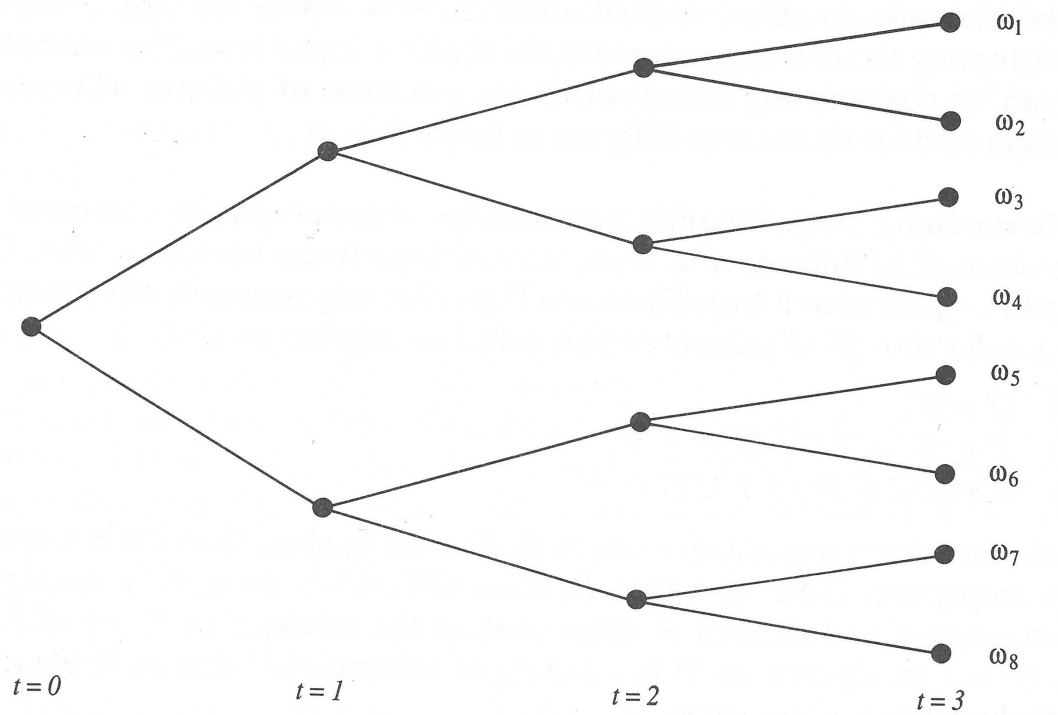
# 2.2: Information structures




(Pliska, 1997, p. 75)



# 2.2: Information structures




(Pliska, 1997, p. 76)

# 2.2: Information structures

**Definition 2.6.**

A collection $\mathcal{F}$ of subsets of $\Omega$ is called a $\sigma$-algebra on $\Omega$ if

(a) $\Omega \in \mathcal{F}$

(b) For each $F \in \mathcal{F}$ the $F^C \in \mathcal{F}$.

(c) For all $F, G \in \mathcal{F}$ the $F \cup G \in \mathcal{F}$.

To any partition $\mathcal{P}$ of $\Omega$ we can associate a $\sigma$-algebra $\mathcal{F}$ as follows:

- Let $\mathcal{F}$ be the collection of all unions of elements of $\mathcal{P}$ together with the complements of all such unions.
- This $\mathcal{F}$ is called the $\sigma$-algebra generated by $\mathcal{P}$.

Hence the submodels of the information structure can be organized as a sequence $\{\mathcal{F}_t\}$ of $\sigma$-algebras.



# 2.2: Information structures

**Definition 2.7.**

A family of $\sigma$-algebras $\mathcal{F} = \{\mathcal{F}_t : t = 0, 1, \ldots, T\}$ is called a filtration if

- $\mathcal{F}_0 = \{\emptyset, \Omega\}$
- $\mathcal{F}_T$ consists of all subsets of $\Omega$.
- $\mathcal{F}_n \subset \mathcal{F}_{n+1}$ for all $n < T$, by which we mean that each subset in $\mathcal{F}_n$ must also be a subset in $\mathcal{F}_{n+1}$.

## 2.2: Information structures

**Example 2.8.**

In the context of Example 2.5, the corresponding filtration is given by

$$\begin{aligned}
\mathcal{F}_0 =&\{\emptyset, \Omega\} \\
\mathcal{F}_1 =&\{\emptyset, \Omega, \{\omega_1, \omega_2, \omega_3, \omega_4\}, \{\omega_5, \omega_6, \omega_7, \omega_8\}\} \\
\mathcal{F}_2 =&\{\emptyset, \Omega, \{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \{\omega_5, \omega_6\}, \{\omega_7, \omega_8\}, \{\omega_1, \omega_2, \omega_3, \omega_4\}, \\
&\{\omega_5, \omega_6, \omega_7, \omega_8\}, \{\omega_1, \omega_2, \omega_5, \omega_6\}, \{\omega_1, \omega_2, \omega_7, \omega_8\}, \\
&\{\omega_3, \omega_4, \omega_5, \omega_6\}, \{\omega_3, \omega_4, \omega_7, \omega_8\}, \\
&\{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6\}, \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_7, \omega_8\}, \\
&\{\omega_1, \omega_2, \omega_5, \omega_6, \omega_7, \omega_8\}, \{\omega_3, \omega_4, \omega_5, \omega_6, \omega_7, \omega_8\}\}
\end{aligned}$$

and $\mathcal{F}_3$ containing all the different subsets of $\Omega$.



## 2.3: Stochastic Processes in Discrete Time

**Definition 2.9.**

A stochastic process $S$ is a real-valued function $S(t)(\omega)$ of both $t$ and $\omega$.
For each fixed $\omega \in \Omega$ the function $t \to S(t)(\omega)$ is called the sample path.
For each fixed $t$ the function $\omega \to S(t)(\omega)$ is a random variable.

For simplicity, we shall assume that $S(0)$ is a constant.

**Definition 2.10.**

A function $\omega \to W(\omega)$ is said to be measurable with respect to the $\sigma$-algebra $\mathcal{F}$ if for every real number $x$ the subset

$$W^{-1}(x) = \{\omega \in \Omega : W(\omega) = x\}$$

is an element of the $\sigma$-algebra $\mathcal{F}$.

In other words if we know which set of the $\sigma$-algebra $\omega$ is in, then we know the value of $W(\omega)$.

**Example 2.11.**

Consider $X$ and $Y$ defined by

$$X(\omega) = \begin{cases} 5, & \omega = \omega_1, \omega_2, \omega_3, \omega_4 \\ 7, & \omega = \omega_5, \omega_6, \omega_7, \omega_8 \end{cases}$$

and

$$Y(\omega) = \begin{cases} 8, & \omega = \omega_1, \omega_3, \omega_5, \omega_7 \\ 6, & \omega = \omega_2, \omega_4, \omega_6, \omega_8. \end{cases}$$

Then, using the notation of Example 2.5,

- $X$ is measurable with respect to $\mathcal{F}_1$
- $Y$ is not measurable with respect to $\mathcal{F}_1$





**Definition 2.12.**

A stochastic process $S = \{S(t) : t = 0, 1, \dots, T\}$ is said to be adapted to the filtration $\mathcal{F} = \{\mathcal{F}_t : t = 0, 1, \dots, T\}$ if for every $t = 0, \dots, T$ the random variable $S(t)$ is measurable with respect to $\mathcal{F}_t$.

In practice we often define the stochastic process $S$ first and use the so-called 'natural' filtration $\mathcal{F}$ – defined as follows.

- For each $t = 0, 1, \dots, T$, let $\mathcal{P}_t$ be the partition of $\Omega$ such that the stochastic process $\{S(0), \dots, S(t)\}$ takes the same value for each $\omega \in A$, for each subset $A \in \mathcal{P}_t$.
- Let $\mathcal{F}_t$ be the $\sigma$-algebra generated by $\mathcal{P}_t$
- Then $\mathcal{F} = \{\mathcal{F}_t; t = 0, 1, \dots, T\}$ is called the filtration generated by the process $S$

### Example 2.13.

- In Example 2.2, let $S(t)$ be the number of 'up' moves by time $t$.
- Then $S(t)$ generates the filtration above, ie given a sequence of $(S(t))$ values, can find exact value of $\omega$.
- Can summarise in table

| $\omega_k$ | $t=0$ | $t=1$ | $t=2$ | $t=3$ |
|---|---|---|---|---|
| $\omega_1$ | $S(t)=0$ | $S(t)=1$ | $S(t)=2$ | $S(t)=3$ |
| $\omega_2$ | $S(t)=0$ | $S(t)=1$ | $S(t)=2$ | $S(t)=2$ |
| $\omega_3$ | $S(t)=0$ | $S(t)=1$ | $S(t)=1$ | $S(t)=2$ |
| $\omega_4$ | $S(t)=0$ | $S(t)=1$ | $S(t)=1$ | $S(t)=1$ |
| $\omega_5$ | $S(t)=0$ | $S(t)=0$ | $S(t)=1$ | $S(t)=2$ |
| $\omega_6$ | $S(t)=0$ | $S(t)=0$ | $S(t)=1$ | $S(t)=1$ |
| $\omega_7$ | $S(t)=0$ | $S(t)=0$ | $S(t)=0$ | $S(t)=1$ |
| $\omega_8$ | $S(t)=0$ | $S(t)=0$ | $S(t)=0$ | $S(t)=0$ |



## 2.3: Stochastic Processes in Discrete Time

### Definition 2.14.

Let $X_t = X_0 + Y_1 + \cdots + Y_t$ where $Y_1, Y_2, \ldots$ are i.i.d. random variables with finite variance $\sigma^2$ and $\mathbb{E}(Y_i) = \mu$. Then $\{X_t, t \geq 0\}$ is called a random walk. We say that $\{X_t, t \geq 0\}$ is a simple random walk if $Y_i$ takes only the values 1 with probability $p$ and $-1$ with probability $q = 1 - p$.

The simple random walk takes the value $y$ at time point $t$ if and only if exactly $(t+y)/2$ of $Y_1, \ldots, Y_t$ are equal to 1 and the remaining variables equal to $-1$. This shows that:

### Proposition 2.15.

*Let $\{X_t, t \geq 0\}$ be a simple random walk with parameter $p$ and $X_0 = 0$. Then for all $t$ and $y \in \{-t, -t+2, -t+4, \ldots, t-2, t\}$*

$$\mathbb{P}(X_t = y) = \binom{t}{\frac{t+y}{2}} p^{\frac{t+y}{2}} (1-p)^{\frac{t-y}{2}}$$

# 2.4: Conditional expectations

**Definition 2.16.**

For a finite sample space $\Omega$ the conditional expectation $\mathbb{E}(Y|A)$ of the discrete random variable $Y$ given the event $A$ is defined by

$$\mathbb{E}(Y|A) = \sum_y y\mathbb{P}(Y = y|A).$$

$\mathbb{E}(Y|A)$ is a map from events $A$ to real numbers.

We can write

$$\mathbb{E}(Y|A) = \sum_y y\mathbb{P}(Y(\omega) = y, A)/\mathbb{P}(A) = \sum_{\omega\in A} Y(\omega)\mathbb{P}(\{\omega\})/\mathbb{P}(A).$$



# 2.4: Conditional expectations

**Example 2.17.**

- Consider Example 2.13, with all $\omega_i$ having probability $1/8$.
- If $A = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ then

$$\mathbb{E}(S_3|A) = \frac{(3+2+2+1)(1/8)}{1/2} = 2.$$

- If $A = \{\omega_5, \omega_6, \omega_7, \omega_8\}$ then

$$\mathbb{E}(S_3|A) = \frac{(2+1+1+0)(1/8)}{1/2} = 1.$$

## 2.4: Conditional expectations

For an event $A \subset \Omega$ its indicator function is a random variable that takes the values 1 and 0 depending on whether $A$ happens or not. It is denoted by $1_A$ and defined as

$$1_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A. \end{cases}$$

**Definition 2.18.**

Let $\mathcal{F}$ be a $\sigma$-algebra and $\mathcal{P}$ be the corresponding partition of $\Omega$. Then we define the conditional expectation $\mathbb{E}(Y|\mathcal{F})$ by

$$\mathbb{E}(Y|\mathcal{F}) = \sum_{A \in \mathcal{P}} \mathbb{E}(Y|A) 1_A.$$

This defines a random variable which is measurable with respect to $\mathcal{F}$ and for all $\omega \in A$

$$\mathbb{E}(Y|\mathcal{F})(\omega) = \mathbb{E}(Y|A).$$



## 2.4: Conditional expectations

**Example 2.19.**

Consider Example 2.13, with all $\omega_i$ having probability $1/8$.

- Recall $\mathcal{F}_1 = \{\emptyset, \Omega, \{\omega_1, \omega_2, \omega_3, \omega_4\}, \{\omega_5, \omega_6, \omega_7, \omega_8\}\}$.
- Then
$$\mathbb{E}(S_3|A) = \begin{cases} 2, & \text{if } A = \{\omega_1, \omega_2, \omega_3, \omega_4\}, \\ 1, & \text{if } A = \{\omega_5, \omega_6, \omega_7, \omega_8\}. \end{cases}$$
- Hence $\mathbb{E}(S_3|\mathcal{F}_1)$ is a random variable with
$$\mathbb{E}(S_3|\mathcal{F}_1)(\omega_i) = \begin{cases} 2, & \text{for } i = 1, 2, 3, 4, \\ 1, & \text{for } i = 5, 6, 7, 8. \end{cases}$$
- This random variable is $\mathcal{F}_1$-measurable.

## 2.4: Conditional expectations

**Example 2.19.**

- Observe that

$$\mathbb{E}(\mathbb{E}(S_3|\mathcal{F}_1)) = \sum_i \mathbb{P}(\omega_i)\mathbb{E}(S_3|\mathcal{F}_1)(\omega_i)$$
$$= 4/8(2) + 4/8(1) = 3/2 = \mathbb{E}(S_3).$$

- Note that if

$$Y(\omega_i) = \begin{cases} 7, & \text{for } i \leq 4 \\ 1, & \text{for } i \geq 5 \end{cases}$$

then $\mathbb{E}(Y|\mathcal{F}_1) = Y$.

- Moreover, if $Z = 1_{(\text{3rd move is up})}$ then $Z$ is independent of $\mathcal{F}_1$, and

$$\mathbb{E}(Z|\mathcal{F}_1) = 1/2 = \mathbb{E}(Z).$$



## 2.4: Conditional expectations

The conditional expectation satisfies the following important properties :

**Proposition 2.20.**

*Let $Y$ be a random variable and $\mathcal{F}, \mathcal{F}_1, \mathcal{F}_2$ be $\sigma$-algebras with $\mathcal{F}_1 \subset \mathcal{F}_2$. Then*

(a) $\mathbb{E}(\mathbb{E}(Y|\mathcal{F})) = \mathbb{E}(Y)$

(b) *Slightly more general:* $\mathbb{E}(\mathbb{E}(Y|\mathcal{F}_2)|\mathcal{F}_1) = \mathbb{E}(Y|\mathcal{F}_1) = \mathbb{E}(\mathbb{E}(Y|\mathcal{F}_1)|\mathcal{F}_2)$.

(c) *If the random variable $X$ is measurable with respect to $\mathcal{F}$, then* $\mathbb{E}(XY|\mathcal{F}) = X\mathbb{E}(Y|\mathcal{F})$. *In particular* $\mathbb{E}(X|\mathcal{F}) = X$.

(d) *If $Y$ is independent of $\mathcal{F}$, so for all $A \in \mathcal{P}$ the distribution of $Y$ is independent of $A$, then* $\mathbb{E}(Y|\mathcal{F}) = \mathbb{E}(Y)$.

# 2.4: Conditional expectations

**Proof.**

The first property is satisfied because

$$\begin{aligned}\mathbb{E}\left(\mathbb{E}(Y|\mathcal{F})\right) = \mathbb{E}\left(\sum_{A\in\mathcal{P}} \mathbb{E}(Y|A)1_A\right) &= \sum_{A\in\mathcal{P}} \mathbb{E}(1_A)\mathbb{E}(Y|A)\\ &= \sum_{A\in\mathcal{P}} \mathbb{P}(A) \sum_{\omega\in A} \frac{Y(\omega)\mathbb{P}(\{\omega\})}{\mathbb{P}(A)}\\ &= \sum_{A\in\mathcal{P}}\sum_{\omega\in A} Y(\omega)\mathbb{P}(\{\omega\})\\ &= \mathbb{E}(Y).\end{aligned}$$

Here we used that

$$\mathbb{E}(1_A) = \sum_{\omega\in\Omega} 1_A(\omega)\mathbb{P}(\{\omega\}) = \sum_{\omega\in A} \mathbb{P}(\{\omega\}) = \mathbb{P}(A).$$



# 2.4: Conditional expectations

**Proof.**

Property (b) is a generalization of this result. We only show the first equality and see using the definition of the conditional expectation that

$$\begin{aligned}\mathbb{E}(\mathbb{E}(Y|\mathcal{F}_2)|\mathcal{F}_1) = \mathbb{E}\left(\sum_{B\in\mathcal{P}_2} \mathbb{E}(Y|B)1_B|\mathcal{F}_1)\right) &= \sum_{B\in\mathcal{P}_2} \mathbb{E}(Y|B)\mathbb{E}(1_B|\mathcal{F}_1)\\ &= \sum_{B\in\mathcal{P}_2} \mathbb{E}(Y|B) \sum_{A\in\mathcal{P}_1} \mathbb{E}(1_B|A)1_A\\ &= \sum_{A\in\mathcal{P}_1}\sum_{B\in\mathcal{P}_2} \mathbb{E}(Y|B)\mathbb{E}(1_B|A)1_A\\ &= \sum_{A\in\mathcal{P}_1}\sum_{B\in\mathcal{P}_2} \mathbb{E}(Y|B)\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A)}1_A.\end{aligned}$$

Since the partition $\mathcal{P}_2$ is finer than $\mathcal{P}_1$ for all $B \in \mathcal{P}_2$ and $A \in \mathcal{P}_1$ either $B \subset A$ or $A \cap B = \emptyset$.

## 2.4: Conditional expectations

**Proof.**

Thus

$$
\begin{aligned}
\mathbb{E}(\mathbb{E}(Y|\mathcal{F}_2)|\mathcal{F}_1) &= \sum_{A\in\mathcal{P}_1}\sum_{B\in\mathcal{P}_2, B\subset A} \mathbb{E}(Y|B)\frac{\mathbb{P}(B)}{\mathbb{P}(A)}1_A \\
&= \sum_{A\in\mathcal{P}_1}\sum_{B\in\mathcal{P}_2, B\subset A}\sum_{\omega\in B} Y(\omega)\frac{\mathbb{P}(\{\omega\})}{\mathbb{P}(B)}\frac{\mathbb{P}(B)}{\mathbb{P}(A)}1_A \\
&= \sum_{A\in\mathcal{P}_1}\sum_{\omega\in A} Y(\omega)\frac{\mathbb{P}(\{\omega\})}{\mathbb{P}(A)}1_A \\
&= \sum_{A\in\mathcal{P}_1} \mathbb{E}(Y|A)1_A \\
&= \mathbb{E}(Y|\mathcal{F}_1).
\end{aligned}
$$



## 2.4: Conditional expectations

**Proof.**

In order to prove property (c), since $X$ is measurable it is constant on sets of $\mathcal{P}$ so we write

$$X = \sum_{A\in\mathcal{P}} x_A 1_A \tag{2.2}$$

with suitable scalars $x_A$. Then

$$
\begin{aligned}
\mathbb{E}(XY|\mathcal{F}) = \sum_{A\in\mathcal{P}} \mathbb{E}(XY|A)1_A = \sum_{A\in\mathcal{P}} \mathbb{E}(x_A Y|A)1_A &= \sum_{A\in\mathcal{P}} x_A\mathbb{E}(Y|A)1_A \\
&= \sum_{A\in\mathcal{P}} X\mathbb{E}(Y|A)1_A \\
&= X\mathbb{E}(Y|\mathcal{F}).
\end{aligned}
$$

Taking $Y = 1$ it follows that $\mathbb{E}(X|\mathcal{F}) = X$.

**Proof.**

Finally, to prove (d), if $Y$ is independent of $\mathcal{F}$, then for all $A \in \mathcal{F}$

$$\mathbb{E}(Y|A) = \sum_y y\mathbb{P}(Y = y|A) = \sum_y y\mathbb{P}(Y = y) = \mathbb{E}(Y)$$

and thus $\mathbb{E}(Y|\mathcal{F}) = \mathbb{E}(Y)$. $\square$





The following result is usually used to define $\mathbb{E}(Y|\mathcal{F})$ in more general probability spaces where in particular the number of possible states is not finite.

**Proposition 2.21.**

*Let $Y$ be a random variable, then the conditional expectation $\mathbb{E}(Y|\mathcal{F})$ is the unique random variable such that*

(a) $\mathbb{E}(Y|\mathcal{F})$ *is $\mathcal{F}$-measurable*

(b) *For all $A \in \mathcal{F}$: $\mathbb{E}(\mathbb{E}(Y|\mathcal{F})1_A) = \mathbb{E}(Y1_A)$.*

**Proof.**

For arbitrary $A \in \mathcal{F}$ the indicator function $1_A$ is $\mathcal{F}$-measurable and thus $\mathbb{E}(1_A Y|\mathcal{F}) = 1_A\mathbb{E}(Y|\mathcal{F})$ by Proposition 2.20. Hence for all $A \in \mathcal{F}$

$$\mathbb{E}(\mathbb{E}(Y|\mathcal{F})1_A) = \mathbb{E}(\mathbb{E}(1_A Y|\mathcal{F})) = \mathbb{E}(Y1_A).$$

## 2.4: Conditional expectations

**Proof.**

On the other hand suppose $X$ is $\mathcal{F}$-measurable and satisfies

$$\mathbb{E}(X 1_A) = \mathbb{E}(Y 1_A).$$

for all $A \in \mathcal{F}$. Then write $X$ as in (2.2) and it follows that for all $A \in \mathcal{P}$. $\mathbb{E}(X 1_A) = x_A \mathbb{P}(A)$ and thus

$$\begin{aligned}\mathbb{E}(Y 1_A) = \sum_{\omega \in A} Y(\omega)\mathbb{P}(\{\omega\}) &= \mathbb{P}(A) \sum_{\omega \in A} Y(\omega)\mathbb{P}(\{\omega\})/\mathbb{P}(A) \\ &= \mathbb{P}(A)\mathbb{E}(Y|A).\end{aligned}$$

Therefore we find that

$$x_A \mathbb{P}(A) = \mathbb{E}(X 1_A) = \mathbb{E}(Y 1_A) = \mathbb{P}(A)\mathbb{E}(Y|A).$$

and consequently $x_A = \mathbb{E}(Y|A)$ for all $A \in \mathcal{P}$ and thus $X = \mathbb{E}(Y|\mathcal{F})$. □



## 2.5: Martingales

**Definition 2.22.**

Let $Z = \{Z_t : t = 0, 1, \ldots, T\}$ be an adapted stochastic process defined on a probability space $\Omega$ with a filtration $\{\mathcal{F}_t\}$. The process $Z$ is said to be a martingale if

$$\mathbb{E}(Z_t|\mathcal{F}_{t-1}) = Z_{t-1}, \text{ for all } t \geq 1,$$

a supermartingale if

$$\mathbb{E}(Z_t|\mathcal{F}_{t-1}) \leq Z_{t-1}, \text{ for all } t \geq 1,$$

and a submartingale if

$$\mathbb{E}(Z_t|\mathcal{F}_{t-1}) \geq Z_{t-1}, \text{ for all } t \geq 1,$$

Notice this might seem the wrong way round!

**Proposition 2.23.**

*An adapted stochastic process $Z$ is a martingale if and only if*

$$\mathbb{E}(Z_t|\mathcal{F}_s) = Z_s, \text{ for all } t \geq s. \tag{2.3}$$

*The corresponding results hold for sub- and supermartingales.*

Proof.

- Clearly, if Equation (2.3) holds then Definition 2.22 part 1 holds.
- Hence, assume Definition 2.22 part 1 holds. Proposition 2.20 means

$$\begin{aligned}\mathbb{E}(Z_t|\mathcal{F}_s) = \mathbb{E}(\mathbb{E}(Z_t|\mathcal{F}_{t-1})|\mathcal{F}_s) = \mathbb{E}(Z_{t-1}|\mathcal{F}_s) = \cdots &= \mathbb{E}(Z_s|\mathcal{F}_s)\\ &= Z_s,\end{aligned}$$

  so (2.3) holds.
- Similarly for sub- and supermartingales.

$\square$





**Example 2.24.**

Let $\{X_t, t \geq 0\}$ be a simple random walk with parameter $p$ and $\mathcal{F}_t$ the $\sigma$-algebra generated by $(X_t)$. Then:

(a) $\{X_t, t \geq 0\}$ is a martingale if $p = 1/2$, a supermartingale if $p \leq 1/2$ and a submartingale if $p \geq 1/2$.

(b) If $p = 1/2$ the process $\{Z_t, t \geq 0\}$ defined by $Z_t = X_t^2 - t, t = 0, 1, 2, \ldots$ is a martingale.

(c) If $p \neq 1/2$ the processes $\{L_t, t \geq 0\}$ defined by $L_0 = 1$ and

$$L_t = \left(\frac{1-p}{p}\right)^{X_t}$$

and $\{M_t, t \geq 0\}$ defined by $M_t = X_t - t(2p-1)$ are both martingales.

## 2.5: Martingales

**Proof.**

- **Part (a).** Since $\mathcal{F}_t$ is the natural filtration, $(X_t)$ is $\mathcal{F}_t$-measurable and $Y_t$ is independent of $\mathcal{F}_{t-1}$. Therefore by Proposition 2.20(c) and (d):

$$\begin{aligned}\mathbb{E}(X_t|\mathcal{F}_{t-1}) = \mathbb{E}(X_{t-1}+Y_t|\mathcal{F}_{t-1}) &= \mathbb{E}(X_{t-1}|\mathcal{F}_{t-1}) + \mathbb{E}(Y_t|\mathcal{F}_{t-1}) \\ &= X_{t-1} + \mathbb{E}(Y_t).\end{aligned}$$

- If $p = 1/2$, then $\mathbb{E}(Y_t) = 0$ and thus $\mathbb{E}(X_t|\mathcal{F}_{t-1}) = X_{t-1}$ which means that $(X_t)$ is a martingale.
- If $p \leq 1/2$ then $\mathbb{E}(Y_t) = 2p-1 \leq 0$, so $\mathbb{E}(X_t|\mathcal{F}_{t-1}) \leq X_{t-1}$ and the process is a supermartingale.
- If $p \geq 1/2$ then $\mathbb{E}(Y_t) = 2p-1 \geq 0$, so $\mathbb{E}(X_t|\mathcal{F}_{t-1}) \geq X_{t-1}$ and the process is a submartingale.



## 2.5: Martingales

**Proof.**

In order to prove (b) note that

$$\begin{aligned}\mathbb{E}(Z_t|\mathcal{F}_{t-1}) &= \mathbb{E}(X_t^2 - t|\mathcal{F}_{t-1}) \\ &= \mathbb{E}((X_{t-1}+Y_t)^2|\mathcal{F}_{t-1}) - t \\ &= \mathbb{E}(X_{t-1}^2|\mathcal{F}_{t-1}) + 2\mathbb{E}(X_{t-1}Y_t|\mathcal{F}_{t-1}) + \mathbb{E}(Y_t^2|\mathcal{F}_{t-1}) - t \\ &= X_{t-1}^2 + 2X_{t-1}\mathbb{E}(Y_t|\mathcal{F}_{t-1}) + \mathbb{E}(Y_t^2) - t \\ &= X_{t-1}^2 + 0 + 1 - t \\ &= Z_{t-1}\end{aligned}$$

The proofs for $(L_t)$ and $(M_t)$ are similar and a problem on a problem sheet. □

# 2.6: Stopping times

**Definition 2.25.**

Let $\Omega$ be a probability space with a filtration $\{\mathcal{F}_t : t = 0, 1, 2, \dots\}$. A stopping time is a random variable $\tau$ taking values in the set $\{0, 1, \dots, \infty\}$ such that each event of the form $\{\tau \leq t\}$ for arbitrary $t$ is an element of the $\sigma$-algebra $\mathcal{F}_t$. A stopping time is called bounded if $\mathbb{P}(\tau < K) = 1$.

- Definition allows to decide whether event $\{\tau \leq t\}$ occurs or not by examining the information available at time $t$.
- Value of $\infty$ allows for events that never happen.

**Example 2.26.**

- 'When RBS shares hit 100p' is a stopping time.
- 'When RBS shares hit their maximum' is not a stopping time.



# 2.6: Stopping times

An equivalent characterization is

**Proposition 2.27.**

*A random variable $\tau$ is a stopping time if and only if each event of the form $\{\tau = t\}$ for arbitrary $t$ is an element of the $\sigma$-algebra $\mathcal{F}_t$.*

Proof.

This can be seen from noticing that

$$\{\tau = t\} = \{\tau \leq t\} \setminus \{\tau \leq t - 1\}$$

and

$$\{\tau \leq t\} = \bigcup_{k \leq t} \{\tau = k\}.$$

$\square$

## 2.6: Stopping times

An important example is the first time where a process hits a certain bound.

**Proposition 2.28.**

*Let $(X_t)$ be an adapted process and $c$ an arbitrary number, then a stopping time $\tau_c$ is defined by*

$$\tau_c = \inf\{t \geq 0 : X_t \geq c\}.$$

Proof.

We find that $\tau_c \leq t$ if and only if there is some $k \leq t$ such that $X_k \geq c$. Therefore

$$\{\tau_c \leq t\} = \bigcup_{k \leq t} \{X_k \geq c\}$$

and is thus an element of $\mathcal{F}_t$. Therefore $\tau_c$ is a stopping time. □



## 2.6: Stopping times

**Theorem 2.29 (Optional sampling theorem).**

- *Let $\tau$ be a bounded stopping time, and $X = \{X_t : t = 0, 1, \ldots\}$ be a martingale. Then*
$$\mathbb{E}(X_\tau) = \mathbb{E}(X_0).$$
- *If $\tau$ is a bounded stopping time and $X$ is a supermartingale, then*
$$\mathbb{E}(X_\tau) \leq \mathbb{E}(X_0).$$

- Also known as the optional stopping theorem.
- Weaker alternative conditions that suffice to prove the theorem are
  - $\mathbb{P}(\tau < \infty) = 1$ and $X_\tau$ is bounded: $|X(\omega)_{\tau(\omega)}| < L$ for some $L$ and all $\omega$
  - $\mathbb{E}(\tau) < \infty$ and $(X_t - X_{t-1})$ is bounded

## 2.6: Stopping times

**Proof.**

To see that assume that $\tau \le K$ and write

$$X_{\tau(\omega)}(\omega) = \sum_{t=0}^{K} X_t(\omega) 1_{\{\tau(\omega)=t\}}.$$

Then

$$\mathbb{E}(X_\tau) = \mathbb{E}(\sum_{t=0}^{K} X_t 1_{\{\tau=t\}}) = \sum_{t=0}^{K} \mathbb{E}(X_t 1_{\{\tau=t\}}) = \sum_{t=0}^{K} \mathbb{E}(\mathbb{E}(X_K|\mathcal{F}_t) 1_{\{\tau=t\}})$$

$$= \sum_{t=0}^{K} \mathbb{E}(X_K 1_{\{\tau=t\}}) = \mathbb{E}(X_K \sum_{t=0}^{K} 1_{\{\tau=t\}}) = \mathbb{E}(X_K \cdot 1) = \mathbb{E}(X_0)$$

using the fact that since $\tau$ is a stopping time, $\{\tau = t\}$ is measurable with respect to $\mathcal{F}_t$, so $\mathbb{E}(X_K|\mathcal{F}_t) 1_{\{\tau=t\}} = \mathbb{E}(X_K 1_{\{\tau=t\}}|\mathcal{F}_t)$. □



## 2.6: Stopping times

**Example 2.30 (Gambler's ruin).**

- A gambler has initial wealth $C > 0$ Pounds.
- He or she gambles by guessing heads or tails on each flip of a coin.
- If guess is correct, receives £1, if not loses £1.
- The game ends as soon as the gambler becomes bankrupt, or chooses to walk away when wealth reaches $C + G$ Pounds, where $G > 0$.

## 2.6: Stopping times

**Proposition 2.31 (Gambler's ruin).**

*Let $\{X_t, t \geq 0\}$ be a simple random walk with parameter $p$ and $X_0 = 0$ and $C, G > 0$. Let $\tau = \inf\{t : X_t = G \text{ or } X_t = -C\}$.*
*Then, if $p = 1/2$,*

$$\mathbb{P}(X_\tau = G) = \frac{C}{C+G}, \ \mathbb{P}(X_\tau = -C) = \frac{G}{C+G} \text{ and } \mathbb{E}(\tau) = CG$$

*and, if $p \neq 1/2$,*

$$\mathbb{P}(X_\tau = G) = \frac{1 - (\frac{p}{1-p})^C}{(\frac{1-p}{p})^G - (\frac{p}{1-p})^C}, \quad \mathbb{P}(X_\tau = -C) = \frac{(\frac{1-p}{p})^G - 1}{(\frac{1-p}{p})^G - (\frac{p}{1-p})^C}$$

*and*

$$\mathbb{E}(\tau) = \frac{G\mathbb{P}(X_\tau = G) - C\mathbb{P}(X_\tau = -C)}{2p-1}.$$



## 2.6: Stopping times

Proof.

We want to apply the stopping theorem. The stopping time $\tau$ is not bounded, but $X_\tau$ is bounded and therefore we can use one of the alternative version of the optional stopping theorem provided that $\mathbb{P}(\tau < \infty) = 1$. To see that note first that whenever there is a run of at least $k = C + G$ successive 1's in the process $Y$ which defines $X$ the process will stop and $\tau < \infty$. Thus for every $m$

$$\begin{aligned}\mathbb{P}(\tau > km) &\leq \mathbb{P}(\text{no run of } k \text{ 1's in } Y_1, \dots, Y_{mk}) \\ &\leq \prod_{j=0}^{m-1} \mathbb{P}(\text{no run of } k \text{ 1's in } Y_{jk+1}, \dots, Y_{(j+1)k}) \\ &= (1-p^k)^m.\end{aligned}$$

Therefore $\mathbb{P}(\tau < \infty) = 1$.

## 2.6: Stopping times

**Proof.**

Let us now first consider the case where $p = 1/2$. Then using the optional stopping theorem

$$0 = \mathbb{E}(X_\tau) = G\mathbb{P}(X_\tau = G) + (-C)\mathbb{P}(X_\tau = -C)$$
$$= G\mathbb{P}(X_\tau = G) + (-C)(1 - \mathbb{P}(X_\tau = G)).$$

Therefore $C = (G + C)\mathbb{P}(X_\tau = G)$ and from that we get the required probabilities.

The expectation of $\tau$ can be determined by an application of the optional stopping theorem to the process $Z_t = X_t^2 - t$ which is also a martingale as seen in Example 2.24.

Since $0 = \mathbb{E}(Z_0) = \mathbb{E}(Z_\tau) = \mathbb{E}(X_\tau^2 - \tau)$ we obtain

$$\mathbb{E}(\tau) = \mathbb{E}(X_\tau^2) = G^2C/(C + G) + C^2G/(C + G) = CG.$$



## 2.6: Stopping times

**Proof.**

We consider now the case where $p \neq 1/2$ and apply the optional stopping theorem to the process $\{L_t\}$ defined as in Example 2.24 and obtain

$$1 = \mathbb{E}(L_0) = \mathbb{E}(L_\tau) = \left(\frac{1-p}{p}\right)^G \mathbb{P}(X_\tau = G) + \left(\frac{p}{1-p}\right)^C \mathbb{P}(X_\tau = -C).$$

Using again that $\mathbb{P}(X_\tau = G) + \mathbb{P}(X_\tau = -C) = 1$ we can easily derive the desired expressions for both probabilities.

Finally in order to obtain the expectation of $\tau$ we apply the optional stopping theorem to the process $M_t$ as defined in Example 2.24 and obtain

$$0 = \mathbb{E}(M_\tau) = G\mathbb{P}(X_\tau = G) + (-C)\mathbb{P}(X_\tau = -C) - \mathbb{E}(\tau)(2p - 1).$$

Solving for $\mathbb{E}(\tau)$ we obtain the desired identity. □

# 3. Multi-Period models





## 3.1: Trading strategies

We now formally define a trading strategy for multi-period models.

**Definition 3.1.**

(a) A stochastic process $X_t$ is called *predictable*, if it is measurable with respect to $\mathcal{F}_{t-1}$.

(b) A vector of stochastic processes $H = (H_0, H_1, \dots, H_N)$ is called a *trading strategy* if for each $n = 0, 1, \dots, N$ the process $H_n = \{H_n(t) : t = 1, 2, \dots, T\}$, is predictable.

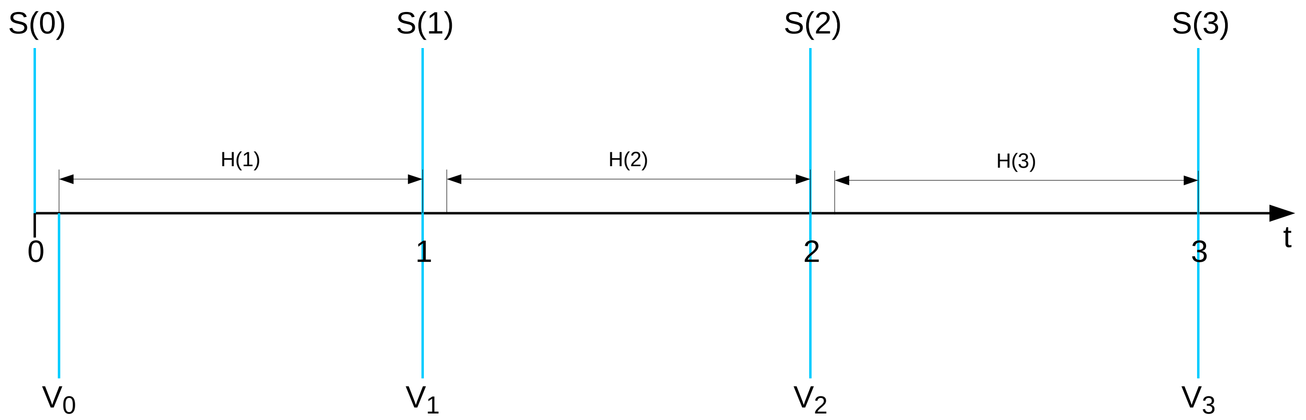
- $H_n(t)$ $(n \geq 1)$): number of units that the investor carries forward from time $t-1$ to time $t$
- $H_0(t)B_{t-1}$: amount of money in the bank account at time $t-1$.
- Negative values of $H_n(t)$ correspond to borrowing money from the bank or selling short.

# 3.1: Trading strategies

## Definition 3.2.

The value process $V = \{V_t : t = 0, 1, \dots, T\}$ is a stochastic process defined by

$$V_t = \begin{cases} H_0(1)B_0 + \sum_{n=1}^N H_n(1)S_n(0), & t = 0 \\ H_0(t)B_t + \sum_{n=1}^N H_n(t)S_n(t), & t \geq 1 \end{cases}$$






# 3.1: Trading strategies

Denote

$$\Delta S_n(t) = S_n(t) - S_n(t-1)$$

for the change in the value of the stochastic process $S_n$ between times $t-1$ and $t$.

## Definition 3.3.

The gains process of a trading strategy $H$ is given by

$$G_t = \sum_{u=1}^{t} H_0(u)\Delta B_u + \sum_{n=1}^{N}\sum_{u=1}^{t} H_n(u)\Delta S_n(u), t \geq 1$$

This process is a first example of a stochastic integral of the trading strategy with respect to the price process.

## 3.1: Trading strategies

We will only consider special trading strategies where the portfolio is adjusted from $t$ to $t+1$ without bringing in or consuming any wealth.

**Definition 3.4.**

The trading strategy $H$ is self-financing if

$$V_t = H_0(t+1)B_t + \sum_{n=1}^{N} H_n(t+1)S_n(t), t = 1, \ldots, T-1.$$



## 3.1: Trading strategies

Finally, we consider normalized versions of the security prices, selecting the bank account as the divisor, that is, as the numeraire.

**Definition 3.5.**

The discounted price process
$S_n^* = \{S_n^*(t) : n = 0, 1, \ldots, N, t = 0, 1, \ldots, T\}$ is defined by

$$S_n^*(t) = S_n(t)/B_t, t = 0, 1, \ldots, T; n = 1, 2, \ldots, N.$$

The discounted value process $V^* = \{V_t^* : t = 0, 1, \ldots, T\}$ is defined by

$$V_t^* = \begin{cases} H_0(1) + \sum_{n=1}^N H_n(1)S_n^*(0), & t = 0 \\ H_0(t) + \sum_{n=1}^N H_n(t)S_n^*(t), & t \geq 1 \end{cases}$$

The discounted gains process $G^* = \{G_t^* : t = 1, 2, \ldots, T\}$ is defined by

$$G_t^* = \sum^{N} \sum^{t} H_n(u)\Delta S_n^*(u), t \geq 1.$$

## 3.1: Trading strategies

**Proposition 3.6.**

*A trading strategy H is self-financing if and only if*

$$V_t^* = V_0^* + G_t^* \text{ for } t = 1, 2, \dots, T.$$

Proof.

For all $t = 1, 2, \dots, T$

$$G_t^* = G_{t-1}^* + \sum_{n=1}^{N} H_n(t)\Delta S_n^*(t) \tag{3.4}$$

where for convenience we define $G_0^* = 0$.



## 3.1: Trading strategies

Proof.

Assume first that $H$ is self-financing, then using Definitions 3.4 and 3.5 and Equation (3.4) we obtain for all $t = 1, 2, \dots, T$

$$\begin{aligned} V_t^* - G_t^* &= H_0(t) + \sum_{n=1}^{N} H_n(t)S_n^*(t) - \sum_{n=1}^{N} H_n(t)\Delta S_n^*(t) - G_{t-1}^* \\ &= H_0(t) + \sum_{n=1}^{N} H_n(t)(S_n^*(t) - \Delta S_n^*(t)) - G_{t-1}^* \\ &= H_0(t) + \sum_{n=1}^{N} H_n(t)S_n^*(t-1) - G_{t-1}^* \\ &= V_{t-1}^* - G_{t-1}^* \end{aligned}$$

Applying this procedure again we see that $V_t^* - G_t^* = V_{t-2}^* - G_{t-2}^*$ and so on until finally $V_t^* - G_t^* = V_0^*$.

# 3.1: Trading strategies

Proof.

On the other hand if $V_t^* = V_0^* + G_t^*$ for all $t = 1, 2, \ldots, T$, then for all $t = 1, 2, \ldots, T-1$

$$V_t^* - V_{t+1}^* = V_0^* + G_t^* - (V_0^* + G_{t+1}^*) = G_t^* - G_{t+1}^*$$

and therefore using Definition 3.5 and Equation (3.4)

$$\begin{aligned} V_t^* &= V_{t+1}^* - (G_{t+1}^* - G_t^*) \\ &= H_0(t+1) + \sum_{n=1}^{N} H_n(t+1) S_n^*(t+1) - \sum_{n=1}^{N} H_n(t+1) \Delta S_n^*(t+1) \\ &= H_0(t+1) + \sum_{n=1}^{N} H_n(t+1) S_n^*(t). \end{aligned}$$

Thus $H$ is self-financing. □



# 3.2: Arbitrage and martingale measures

We give equivalents of Definition 1.27, Proposition 1.28 and Definition 1.29 for multi-period models.

**Definition 3.7.**

An arbitrage opportunity in the case of a multiperiod model is some trading strategy $H$ such that

(a) $V_0 = 0$,

(b) $V_T \geq 0$,

(c) $\mathbb{E}(V_T) > 0$ and

(d) $H$ is self-financing.

## 3.2: Arbitrage and martingale measures

Arbitrage opportunities can also be characterized by means of the gains process:

**Proposition 3.8.**

*The self-financing strategy $H$ is an arbitrage opportunity if and only if*

(a) $G_T^* \geq 0$,

(b) $\mathbb{E}(G_T^*) > 0$ *and*

(c) $V_0^* = 0$



## 3.2: Arbitrage and martingale measures

**Definition 3.9.**

A martingale measure is a probability measure $\mathbb{Q}$ such that

(a) $\mathbb{Q}(\{\omega\}) > 0$ for all $\omega \in \Omega$ and

(b) The discounted price process $S_n^*$ is a martingale under $\mathbb{Q}$ for every $n = 1, 2, \ldots, N$:

$$\mathbb{E}_Q(S_n^*(t+s)|\mathcal{F}_t) = S_n^*(t) \text{ for all } t, s \geq 0.$$

In other words a martingale measure $\mathbb{Q}$ must satisfy

$$\mathbb{E}_Q(B_t S_n(t+s)/B_{t+s}|\mathcal{F}_t) = S_n(t), \quad t, s \geq 0 \tag{3.5}$$

## 3.2: Arbitrage and martingale measures

There is an extension of the no-arbitrage theorem from the single-period model to the multiperiod model:

**Theorem 3.10.**

*There are no arbitrage opportunities if and only if there exists a martingale measure $\mathbb{Q}$.*

One direction of the proof relies on the following useful result:

**Proposition 3.11.**

*If $\mathbb{Q}$ is a martingale measure and $H$ is a self-financing strategy, then*

1. *the discounted value process $V^*$*
2. *the discounted gains process $G^*$*

*are martingales under $\mathbb{Q}$.*



Proof.

We have to show that $\mathbb{E}_Q(V^*_{t+1}|\mathcal{F}_t) = V^*_t$ for all $t > 0$ which is equivalent to showing that $\mathbb{E}_Q(G^*_{t+1}|\mathcal{F}_t) = G^*_t$ using Proposition (3.6). Using the expression for $G^*_{t+1} - G^*_t$ we derived in Equation (3.4) we conclude that

$$\begin{aligned}\mathbb{E}_Q(G^*_{t+1} - G^*_t|\mathcal{F}_t) &= \mathbb{E}_Q\left(\sum_{n=1}^{N} H_n(t+1)\Delta S^*_n(t+1)|\mathcal{F}_t\right)\\ &= \sum_{n=1}^{N} \mathbb{E}_Q(H_n(t+1)\Delta S^*_n(t+1)|\mathcal{F}_t)\\ &= \sum_{n=1}^{N} H_n(t+1)\mathbb{E}_Q(\Delta S^*_n(t+1)|\mathcal{F}_t)\end{aligned}$$

using that $H_n$ is a trading strategy and thus predictable. Furthermore $\mathbb{E}_Q(\Delta S^*_n(t+1)|\mathcal{F}_t) = 0$ for all $t > 0$ since $S^*_n$ is a martingale under $\mathbb{Q}$. Hence $G^*_t$ and $V^*_t$ are martingales under $\mathbb{Q}$. □

## 3.2: Arbitrage and martingale measures

The result can be used to prove one direction of the no-arbitrage theorem.

**Proof.**

We first show that there can be no arbitrage opportunities provided the existence of a martingale measure $\mathbb{Q}$.
Suppose $H$ is any self-financing trading strategy with

$$V_T^* \geq 0 \text{ and } \mathbb{E}(V_T^*) > 0.$$

This implies

$$\mathbb{E}_Q(V_T^*) > 0.$$

Since $V^*$ is a martingale under $\mathbb{Q}$ (Proposition 3.11), it follows that

$$V_0^* = \mathbb{E}_Q(V_T^*) > 0.$$

Hence $H$ cannot be an arbitrage opportunity, nor can any other trading strategies be arbitrage opportunities, by the arbitrary choice of $H$.



## 3.2: Arbitrage and martingale measures

The other direction could be shown by using again a version of the separating hyperplane theorem, but it is easier to use the results from the single-period model. Since there is no arbitrage for the multiperiod model, there exist for each single-period model a risk neutral probability measure and one can construct a martingale measure $\mathbb{Q}$ from them.

**Proposition 3.12.**

*If the multiperiod model does not have any arbitrage opportunities, then none of the underlying single period models has arbitrage opportunities.*

**Proof.**

For each $t < T$ and for each $A \in \mathcal{P}_t$ there is one underlying SP model.

- Time 0 discounted prices: $S_n^*(t, \omega), n = 1, \ldots, N$ for an arbitrary $\omega \in A$ since $S_n^*(t, \omega)$ are constant on $A$.
- Sample space: one state for each cell $A' \in \mathcal{P}_{t+1}$ such that $A' \subset A$.
- Time 1 discounted prices: $S_n^*(t + 1, \omega), n = 1, \ldots, N$ for $\omega \in A$.

## Proof.

If any underlying single period model has an arbitrage opportunity in the single period sense, then the multiperiod model must have an arbitrage opportunity in the multiperiod sense.
To see this suppose there exists an arbitrage opportunity $\hat{H}$ for the single period model corresponding to some $A \in \mathcal{P}_t$ for $t < T$. This means that the discounted gain

$$\hat{H}_1 \Delta S_1^*(t+1) + \cdots + \hat{H}_N \Delta S_N^*(t+1)$$

is

- non-negative and
- not identical to zero on the event $A$.

We now construct a multiperiod trading strategy $H$ which is an arbitrage opportunity.



## Proof.

$$H_n(s,\omega) = \begin{cases} 0, & s \le t \text{ or } \omega \notin A \\ \hat{H}_n, & s = t+1, \omega \in A, \text{ and } n = 1, \ldots, N \\ -\sum_{i=1}^N \hat{H}_i S_i^*(t), & s = t+1, \omega \in A, \text{ and } n = 0 \\ \sum_{i=1}^N \hat{H}_i \Delta S_i^*(t+1), & s > t+1, \omega \in A, \text{ and } n = 0 \\ 0 & s > t+1 \text{ and } n > 0. \end{cases}$$

- So this strategy starts with zero money,
- does nothing unless the event $A$ occurs at time $t$ in which case at time $t$ the position $\hat{H}_n$ is taken in the $n$th risky security while the position in the bank account is used to self-finance.
- Subsequently, no position is taken in any of the risky securities, any non-zero value of the portfolio is reflected by a position in the bank account.

This trading strategy $H$ is an arbitrage opportunity in the MP model. □

We use this result to finish the proof of the no-arbitrage theorem for the multiperiod model.

## Proof.

We see from Proposition 3.12 that for each $t < T$ and for each $A \in \mathcal{P}_t$ there is a risk neutral probability measure $\mathbb{Q}(t, A)$ for the underlying single period model. This probability measure

- gives positive mass to each cell $A' \in \mathcal{P}_{t+1}$ such that $A' \subset A$,
- it sums to one over such cells,
- and it satisfies
$$\mathbb{E}_{\mathbb{Q}(t,A)}(\Delta S_n^*(t+1)) = 0$$
for $n = 1, \ldots, N$.

Notice that $\mathbb{Q}(t, A)$ puts a probability on each branch in the information tree that emerges from the node corresponding to $(t, A)$.



## Proof.

We calculate a martingale measure $\mathbb{Q}$ for the multiperiod model from these probabilities by

- setting $\mathbb{Q}(\{\omega\})$ equal to the product of the conditional probabilities along the path from the node at $t = 0$ to the node corresponding to $(T, \omega)$.

Then

- $\sum_{\omega \in \Omega} \mathbb{Q}(\{\omega\}) = 1$.
- $\mathbb{Q}(\{\omega\}) > 0$ for every $\omega \in \Omega$ (because all the conditional risk neutral probabilities are strictly positive)
- and $\mathbb{E}_Q(S_n^*(t+1)|\mathcal{F}_t) = S_n^*(t)$ for all $t$ and $n$.

Thus $\mathbb{Q}$ is indeed a martingale measure. □

# 3.2: Arbitrage and martingale measures

**Example 3.13.**

We consider a model with $N = 1$ stock over $T = 2$ periods and $K = 4$ possible states. For simplicity we change the notation of the price process to $S_t$ instead of $S_1(t)$. The process is given by the following table:

| $\omega_k$ | $t = 0$ | $t = 1$ | $t = 2$ |
|---|---|---|---|
| $\omega_1$ | $S_0 = 6$ | $S_1 = 9$ | $S_2 = 10$ |
| $\omega_2$ | $S_0 = 6$ | $S_1 = 9$ | $S_2 = 7$ |
| $\omega_3$ | $S_0 = 6$ | $S_1 = 3$ | $S_2 = 7$ |
| $\omega_4$ | $S_0 = 6$ | $S_1 = 3$ | $S_2 = 1$ |

- We further assume that the interest rate is $r$ in both periods.
- Therefore the bank account process is $B_t = (1 + r)^t$.



# 3.2: Arbitrage and martingale measures

**Example 3.13.**

- So the filtration (collection of $\sigma$-algebras) generated from the price process is

$$\begin{aligned}
\mathcal{F}_0 =& \{\emptyset, \Omega\} \\
\mathcal{F}_1 =& \{\emptyset, \{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \Omega\} \\
\mathcal{F}_2 =& \{\emptyset, \{\omega_1\}, \{\omega_2\}, \{\omega_3\}, \{\omega_4\}, \{\omega_1, \omega_2\}, \{\omega_1, \omega_3\}, \{\omega_1, \omega_4\}, \\
& \{\omega_2, \omega_3\}, \{\omega_2, \omega_4\}, \{\omega_3, \omega_4\}, \{\omega_1, \omega_2, \omega_3\}, \{\omega_1, \omega_2, \omega_4\}, \\
& \{\omega_1, \omega_3, \omega_4\}, \{\omega_2, \omega_3, \omega_4\}, \Omega\}.
\end{aligned}$$

## 3.2: Arbitrage and martingale measures

### Example 3.13.

- Recall that the trading strategy has components $H_n(t)(\omega)$, giving the number of units of asset $n$ held between times $t-1$ and $t$.
- Let $H$ be an arbitrary trading strategy, then for the value process we have $V_0(\omega) = H_0(1)(\omega) + 6H_1(1)(\omega)$.
- $$V_1(\omega) = \begin{cases} (1+r)H_0(1)(\omega) + 9H_1(1)(\omega), & \text{if } \omega = \{\omega_1, \omega_2\} \\ (1+r)H_0(1)(\omega) + 3H_1(1)(\omega), & \text{if } \omega = \{\omega_3, \omega_4\} \end{cases}$$
- $$V_2(\omega) = \begin{cases} (1+r)^2 H_0(2)(\omega) + 10H_1(2)(\omega), & \text{if } \omega = \omega_1 \\ (1+r)^2 H_0(2)(\omega) + 7H_1(2)(\omega), & \text{if } \omega = \omega_2, \omega_3 \\ (1+r)^2 H_0(2)(\omega) + H_1(2)(\omega), & \text{if } \omega = \omega_4. \end{cases}$$
- The trading strategy is predictable so $H_i(1)(\omega)$ is constant for all $\omega$, and $H_i(2)(\omega_1) = H_i(2)(\omega_2) = H_i(\omega_1, \omega_2)$ (say) and $H_i(2)(\omega_3) = H_i(2)(\omega_4) = H_i(\omega_3, \omega_4)$ (say).





### Example 3.13.

- The gains process is given by

$$G_1(\omega) = \begin{cases} rH_0(1) + 3H_1(1), & \text{if } \omega = \omega_1, \omega_2 \\ rH_0(1) - 3H_1(1), & \text{if } \omega = \omega_3, \omega_4 \end{cases}$$

- Writing $H_n(t)$ instead of $H_n(t)(\omega)$ for brevity:

$$G_2(\omega) = \begin{cases} rH_0(1) + 3H_1(1) + r(1+r)H_0(2) + H_1(2), & \text{if } \omega = \omega_1 \\ rH_0(1) + 3H_1(1) + r(1+r)H_0(2) - 2H_1(2), & \text{if } \omega = \omega_2 \\ rH_0(1) - 3H_1(1) + r(1+r)H_0(2) + 4H_1(2), & \text{if } \omega = \omega_3 \\ rH_0(1) - 3H_1(1) + r(1+r)H_0(2) - 2H_1(2), & \text{if } \omega = \omega_4. \end{cases}$$

**Example 3.13.**

- For the trading strategy $H$ to be self-financing, one must have at time 1 in states $\omega_1$ and $\omega_2$

$$V_1 = (1+r)H_0(1) + 9H_1(1) = (1+r)H_0(2) + 9H_1(2)$$

- In states $\omega_3$ and $\omega_4$, require

$$V_1 = (1+r)H_0(1) + 3H_1(1) = (1+r)H_0(2) + 3H_1(2)$$





**Example 3.13.**

We consider possible arbitrage opportunities without considering the existence of martingale measures for the moment. If interest rate is $r = 1/9$, then one possible strategy is:

- Start with an amount of 0
- Do nothing at time point $t = 0$ or at time point $t = 1$ if $S_1 = 3$.
- If $S_1 = 9$ at time $t = 1$ sell short one share of the risky asset, $H_1(2) = -1$, put proceeds on bank account, $H_0(2) = 9/(1+r)$.
- Then at time $t = 2$ the value of the portfolio is

$$V_2 = \begin{cases} (1+r)^2 H_0(2) + 10H_1(2) = 10 - 10 = 0, & \text{if } \omega = \omega_1 \\ (1+r)^2 H_0(2) + 7H_1(2) = 10 - 7 > 0, & \text{if } \omega = \omega_2 \\ 0, & \text{if } \omega = \omega_3, \omega_4. \end{cases}$$

If on the other hand $r = 0$, then there are no possible arbitrage opportunities.

## Example 3.13.

- We want to compute an equivalent martingale measure and use conditional probabilities.
- At $t = 0$ we can obtain $p_1 = \mathbb{Q}(S_1 = 9)$ by solving

$$6(1 + r) = 9p_1 + 3(1 - p_1)$$

  and find that $\mathbb{Q}(S_1 = 9) = (1 + 2r)/2$ and $\mathbb{Q}(S_1 = 3) = (1 - 2r)/2$.
- At $t = 1$ if $\{\omega_1, \omega_2\}$, write $p_2 = \mathbb{Q}(S_2 = 10|S_1 = 9)$ and

$$9(1 + r) = 10p_2 + 7(1 - p_2).$$

  Get $p_2 = \mathbb{Q}(S_2 = 10|S_1 = 9) = (2 + 9r)/3$ and $\mathbb{Q}(S_2 = 7|S_1 = 9) = (1 - 9r)/3$.
- At $t = 1$ if $\{\omega_3, \omega_4\}$, solving

$$3(1 + r) = 7p_3 + (1 - p_3),$$

  get $\mathbb{Q}(S_2 = 7|S_1 = 3) = (2 + 3r)/6$ and $\mathbb{Q}(S_2 = 1|S_1 = 3) = (4 - 3r)/6$.



## Example 3.13.

- By multiplying conditional probabilities along the paths leading to the four states in $\Omega$ we obtain

$$\begin{aligned}
\mathbb{Q}(\omega_1) &= \frac{1}{6}(1 + 2r)(2 + 9r),\\
\mathbb{Q}(\omega_2) &= \frac{1}{6}(1 + 2r)(1 - 9r),\\
\mathbb{Q}(\omega_3) &= \frac{1}{12}(1 - 2r)(2 + 3r),\\
\mathbb{Q}(\omega_4) &= \frac{1}{12}(1 - 2r)(4 - 3r).
\end{aligned}$$

- All probabilities are $> 0$ if $r < 1/9$, so conclude a martingale measure exists iff $0 \leq r < 1/9$.

## 3.3: Valuation of contingent claims

We now give extensions of Definition 1.32 and Proposition 1.33 to the multi-period model.

**Definition 3.14.**

A *contingent* claim in the multiperiod model is a random variable $X$ representing a payoff at time $t = T$. A contingent claim $X$ is said to be *attainable* if there exists some self-financing trading strategy $H$ such that $V_T(\omega) = X(\omega)$ for all $\omega \in \Omega$. In this case one says that $H$ generates $X$ and $H$ is called the replicating portfolio.

What is the time $t$ value of an attainable contingent claim?

**Proposition 3.15 (Risk-neutral valuation principle).**

*The time $t$ value of an attainable contingent claim $X$ is equal to $V_t$, the time $t$ value of the portfolio which replicates $X$. Moreover,*

$$V_t^* = V_t/B_t = \mathbb{E}_Q(X/B_T|\mathcal{F}_t), \ t = 0, 1, \ldots, T$$

*for all risk neutral probability measures $\mathbb{Q}$.*



## 3.3: Valuation of contingent claims

Proof.

Let $P_t$ denote the actual time $t$ price of the contingent claim. Then by the same arguments as in the risk-neutral valuation principle for the single-period model or by Proposition 1.13 $P_t = V_t$ is the only possibility to avoid arbitrage opportunities.
Let now $\mathbb{Q}$ be an arbitrary martingale measure, then for every $t < T$ we have that

$$V_t^* = \mathbb{E}_Q(V_T^*|\mathcal{F}_t)$$

because $V_t^*$ is a martingale by Proposition 3.11. Furthermore since $V_T^*$ is the discounted value of the portfolio that replicates $X$ we have

$$\mathbb{E}_Q(V_T^*|\mathcal{F}_t) = \mathbb{E}_Q(X/B_T|\mathcal{F}_t).$$

Thus $V_t^* = \mathbb{E}_Q(X/B_T|\mathcal{F}_t)$ independent of the choice of martingale measure $\mathbb{Q}$. □

## 3.3: Valuation of contingent claims

The next question is how to compute a generating trading strategy $H$. If we know already the value process $V$ for the replicating portfolio, we need to solve for the trading strategy $H$ using the linear equations in the definition of the value process

$$V_t(\omega_i) = H_0(t)B_t + \sum_{n=1}^{N} H_n(t)S_n(t)(\omega_i) \quad \text{for each } i,$$

keeping in mind that $H$ is predictable.



## 3.3: Valuation of contingent claims

If on the other hand we only know $X$, we need to work backwards in time, deriving $V$ and $H$ simultaneously.

Since $V_T = X$, we must first solve

$$X(\omega_i) = H_0(T)B_T + \sum_{n=1}^{N} H_n(T)S_n(T)(\omega_i).$$

for $H(T)$ as in the first case.

Since $H$ is self-financing, we can calculate $V_{T-1}$:

$$V_{T-1} = H_0(T)B_{T-1} + \sum_{n=1}^{N} H_n(T)S_n(T-1)(\omega_i)$$

Therefore our next step is to solve

$$V_{T-1} = H_0(T-1)B_{T-1} + \sum_{n=1}^{N} H_n(T-1)S_n(T-1)(\omega_i)$$

for $H(T-1)$ as above, and then we compute $V_{T-2}$ and so on until we end up with $V_0$.

# 3.3: Valuation of contingent claims

**Example 3.16.**

- Consider a European call option with exercise price $e = 5$.
- The corresponding claim is $X = (S_2 - 5)_+$ and takes the values $5, 2, 2$ and $0$ in states $\omega_1, \omega_2, \omega_3$, and $\omega_4$ respectively.
- Suppose $r = 0$ then $\mathbb{Q} = (1/3, 1/6, 1/6, 1/3)$.
- Since the martingale measure is unique, Proposition 3.17 will show below that the model is complete and therefore every contingent claim is attainable, including the European option.



# 3.3: Valuation of contingent claims

**Example 3.16.**

- The time $t = 0$ value must be

$$V_0 = \mathbb{E}_{\mathbb{Q}}(X) = \frac{1}{3}(5) + \frac{1}{6}(2) + \frac{1}{6}(2) + \frac{1}{3}(0) = 7/3.$$

- The time $t = 1$ value in states $\omega_1$ and $\omega_2$ is

$$V_1(\omega_1) = V_1(\omega_2) = \mathbb{E}_{\mathbb{Q}}(X|S_1 = 9) = \frac{2}{3}(5) + \frac{1}{3}(2) = 4$$

- The time $t = 1$ value in states $\omega_3$ and $\omega_4$ is

$$V_1(\omega_3) = V_1(\omega_4) = \mathbb{E}_{\mathbb{Q}}(X|S_1 = 3) = \frac{1}{3}(2) + \frac{2}{3}(0) = \frac{2}{3}.$$

## 3.3: Valuation of contingent claims

**Example 3.16.**

- To find a replicating portfolio we start at time $t = 2$ and find that

$$\begin{aligned} V_2(\omega_1) &= 5 = H_0(2)(\omega_1,\omega_2)1 + H_1(2)(\omega_1,\omega_2)10 \\ V_2(\omega_2) &= 2 = H_0(2)(\omega_1,\omega_2)1 + H_1(2)(\omega_1,\omega_2)7, \end{aligned}$$

  to obtain $H_0(2)(\omega_1,\omega_2) = -5$ and $H_1(2)(\omega_1,\omega_2) = 1$.
- Similarly

$$\begin{aligned} V_2(\omega_3) &= 2 = H_0(2)(\omega_3,\omega_4)1 + H_1(2)(\omega_3,\omega_4)7 \\ V_2(\omega_4) &= 0 = H_0(2)(\omega_3,\omega_4)1 + H_1(2)(\omega_3,\omega_4)1. \end{aligned}$$

  to obtain $H_0(2)(\omega_3,\omega_4) = -1/3$ and $H_1(2)(\omega_3,\omega_4) = 1/3$.



## 3.3: Valuation of contingent claims

**Example 3.16.**

- Now there are two ways to proceed.
- **Method 1**: Since we have already calculated the value of the option at time point 1 we can proceed as in the second period.
- Writing $H_0(1)$ for $H_0(1)(\omega_1,\ldots,\omega_4)$ and $H_1(1)$ for $H_1(1)(\omega_1,\ldots,\omega_4)$ We have

$$V_1(\omega) = 4 = H_0(1)1 + H_1(1)9, \quad \omega = \omega_1,\omega_2$$
$$V_1(\omega) = \frac{2}{3} = H_0(1)1 + H_1(1)3, \quad \omega = \omega_3,\omega_4$$

- Solving these equations gives $H_0(1) = -1$ and $H_1(1) = 5/9$.
- Note this confirms our calculation for $V_0$, since this portfolio costs $-1 + 6(5/9) = 7/3$ at time 0.

## 3.3: Valuation of contingent claims

**Example 3.16.**

- **Method 2:** using that $H$ is self-financing we calculate the value of the portfolio at time point 1.
- So we assume that we have already calculated $H_0(2)$ and $H_1(2)$. Then
$$V_1 = H_0(2)(1) + H_1(2)(9) = -5(1) + 1(9) = 4$$
in states $\omega_1$ and $\omega_2$.
- Similarly,
$$V_1 = -1/3(1) + (1/3)(3) = 2/3$$
in states $\omega_3$ and $\omega_4$.
- We proceed as above obtaining
$$H_0(1) = -1 \text{ and } H_1(1) = 5/9.$$



## 3.3: Valuation of contingent claims

**Proposition 3.17.**

*An arbitrage-free multi-period model is complete if and only if there exists a unique martingale measure $\mathbb{Q}$.*

Proof.

- If the MP model is complete, for any claim $X$ we can work backwards in time to compute the trading strategy that generates $X$.
Hence for each underlying SP model the matrix $A$ must have enough independent columns and the model is complete.
- Conversely, if every underlying SP model is complete, then the computational procedure for the MP model succeeds.

Therefore completeness of the MP model is equivalent to completeness of all underlying SP models.
In particular the MP model is complete if and only if each underlying SP model has a unique risk neutral probability measure.

## 3.3: Valuation of contingent claims

**Proof.**

On the other hand uniqueness of the martingale measure $\mathbb{Q}$ is equivalent to uniqueness of the risk-neutral probability measures of the underlying single-period models.
Obviously the existence of several risk-neutral probability measures for some SP model leads to several multi-period martingale measures.
On the other hand assume there are two multi-period martingale measures. Then the conditional probabilities must be different for at least one specific single-period model.

□



## 3.4: American claims

**Definition 3.18.**

Let $\{Y_t, t = 0, \ldots, T\}$ be a stochastic process describing the payoff of a financial derivative if exercised at time $t$ and let $\tau$ be a stopping time representing the exercise date in an exercise strategy. Then $Y_\tau$ is called the American claim with respect to $(Y_t)$ and $\tau$.

**Definition 3.19.**

The American claim $Y_\tau$ is said to be attainable if there exists a self-financing trading strategy such that the corresponding portfolio value $V$ satisfies $V_\tau = Y_\tau$. An American payoff process $(Y_t)$ is said to be attainable if for every stopping time $\tau$ the American claim $Y_\tau$ is attainable.

# 3.4: American claims

**Proposition 3.20.**

*If a financial market is complete, then every American claim is attainable.*

Proof.

Let $\{Y_t, t = 0, \dots, T\}$ be a payoff process and $\tau$ be an exercise strategy. We have to find a self-financing trading strategy such that $V_\tau = Y_\tau$. Consider the (European) claim $X = Y_\tau B_T / B_\tau$ that corresponds to someone exercising the American claim $Y$ at time point $\tau$ and then earning interest until time point $T$. Since the model is complete, there must be a replicating trading strategy $H$ such that $V_T = X = Y_\tau B_T / B_\tau$. This portfolio which starts at time $\tau$ with the amount of $Y_\tau$, all of which is put into and kept in the bank account until time $T$, has the same value at time $T$ as $H$. We conclude that $V_\tau = Y_\tau$. □



# 3.4: American claims

**Definition 3.21.**

Let $\{X_t, t = 0, \dots, T\}$ be a stochastic process adapted to some filtration $\mathcal{F}_t$. Then the process $\{Z_t, t = 0, \dots, T\}$ defined by

$$Z_t = \begin{cases} X_T, & \text{if } t = T \\ \max\{X_t, \mathbb{E}(Z_{t+1}|\mathcal{F}_t)\}, & \text{if } t < T \end{cases}$$

is called the Snell envelope of $X$.

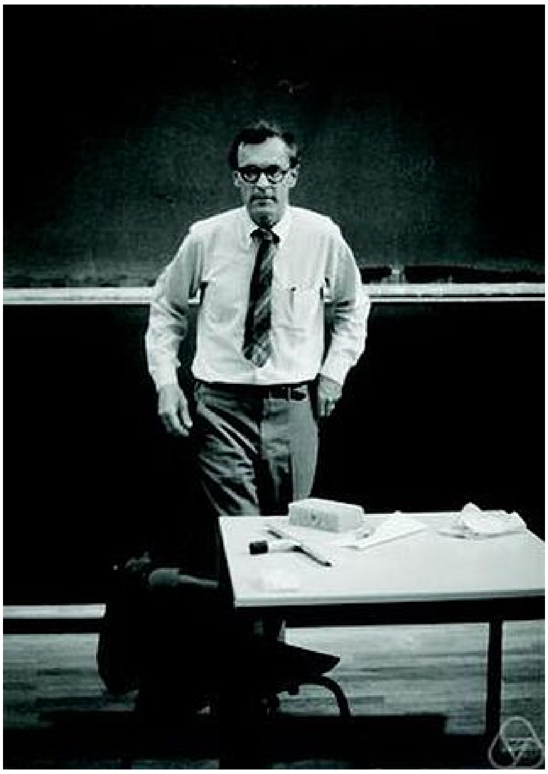
**Proposition 3.22.**

*Consider a financial market with a martingale measure $\mathbb{Q}$ and an attainable American payoff process $(Y_t)$. Then the Snell envelope $Z$ of the discounted payoff process $Y/B$ is the discounted value process for $Y$.*

Proof.

See further down below. □

# 3.4: American claims



James Laurie Snell (1925-2011)



**Example 3.23.**

Consider rolling a six-sided die 3 times. After the $i$th roll, you can either keep the score $X_i$ on the die, or roll again. What should your strategy be?

- We have no choice whether to keep the 3rd score. That is, $Z_3 = X_3$.
- After the second roll, we take

$$Z_2 = \max(X_2, \mathbb{E}(Z_3|\mathcal{F}_2)) = \max(X_2, \mathbb{E}X_3) = \max(X_2, 7/2).$$

  (That is, keep $X_2$ if it is greater than or equal to 4, otherwise reroll.)
- Notice that

$$\mathbb{E}Z_2 = \mathbb{E}(\max(X_2, 7/2)) = \frac{1}{6}(7/2 + 7/2 + 7/2 + 4 + 5 + 6) = \frac{17}{4}.$$

- After the first roll, we take

$$Z_1 = \max(X_1, \mathbb{E}(Z_2|\mathcal{F}_1)) = \max(X_1, \mathbb{E}Z_2) = \max(X_1, 17/4).$$

  (That is, keep $X_1$ if it is greater than or equal to 5, otherwise reroll).

**Theorem 3.24.**

*The Snell envelope $Z$ of $X$ is the smallest supermartingale dominating $X$ (that is, with $Z_t \geq X_t$ for all $t$).*

Proof.

First, $Z_t \geq \mathbb{E}(Z_{t+1}|\mathcal{F}_t)$ and $Z_t \geq X_t$, so $Z$ is a supermartingale and dominates $X$. Next, let $\{U_t, t = 0, \dots, T\}$ be any other supermartingale dominating $X$. Since by definition $Z_T = X_T$ and $U$ dominates $X$, we must have $U_T \geq Z_T$. Assume inductively that $U_t \geq Z_t$. Then

$$U_{t-1} \geq \mathbb{E}(U_t|\mathcal{F}_{t-1}) \geq \mathbb{E}(Z_t|\mathcal{F}_{t-1}) \quad (U \text{ is a supermartingale})$$

and

$$U_{t-1} \geq X_{t-1} \quad (U \text{ dominates } X).$$

Combining,

$$U_{t-1} \geq \max\{X_{t-1}, \mathbb{E}(Z_t|\mathcal{F}_{t-1})\} = Z_{t-1}.$$

By repeating this argument, $U_t \geq Z_t$ for all $t$. □



## 3.4: American claims

The best stopping rule is one based on the Snell envelope.

**Theorem 3.25 (Optimal stopping theorem).**

*Let $\{X_t, t = 0, \dots, T\}$ be a stochastic process adapted to some filtration $\mathcal{F}_t$ and $Z_t$ the Snell envelope of $\{X_t\}$. For any $t = 0, 1, \dots, T$ we define a stopping time by $\tau(t) = \min\{s \geq t : Z_s = X_s\}$, then this stopping rule is optimal:*

$$Z_t = \mathbb{E}(X_{\tau(t)}|\mathcal{F}_t) = \max\{\mathbb{E}(X_\tau|\mathcal{F}_t) : \textit{for all stopping times } t \leq \tau \leq T\} \tag{3.6}$$

*for all $t = 0, 1, \dots, T$. In particular*

$$Z_0 = \mathbb{E}(X_{\tau(0)}) = \max\{\mathbb{E}(X_\tau) : \textit{for all stopping times } \tau \leq T\}.$$

Proof.

The proof is again based on induction backwards in time. Note first that (3.6) is clearly true for $t = T$ because by definition $Z_T = X_T$ and therefore the stopping time $\tau(T)$ stops at $T$.

# 3.4: American claims

**Proof.**

So assume now that we already know that (3.6) is satisfied for some $t$ and let $\tau$ be an arbitrary stopping time between $t-1$ and $T$. Define another stopping time by $\tau' = \max\{\tau, t\}$. Then (since $\tau \geq t$ implies $\tau = \tau'$)

$$\begin{aligned}
\mathbb{E}(X_\tau|\mathcal{F}_{t-1}) &= \mathbb{E}(1_{\{\tau=t-1\}}X_{t-1} + 1_{\{\tau>t-1\}}X_\tau|\mathcal{F}_{t-1}) \\
&= \mathbb{E}(1_{\{\tau=t-1\}}X_{t-1} + 1_{\{\tau>t-1\}}X_{\tau'}|\mathcal{F}_{t-1}) \\
&= 1_{\{\tau=t-1\}}X_{t-1} + 1_{\{\tau>t-1\}}\mathbb{E}(X_{\tau'}|\mathcal{F}_{t-1}) \\
&= 1_{\{\tau=t-1\}}X_{t-1} + 1_{\{\tau>t-1\}}\mathbb{E}(\mathbb{E}(X_{\tau'}|\mathcal{F}_t)|\mathcal{F}_{t-1})
\end{aligned}$$

Since $\tau'$ is a stopping time such that $t \leq \tau' \leq T$, we find that $\mathbb{E}(X_{\tau'}|\mathcal{F}_t) \leq Z_t$ because we have already proved (3.6) for $t$. Using the definition of $Z_{t-1}$ we see that

$$\mathbb{E}(X_\tau|\mathcal{F}_{t-1}) \leq 1_{\{\tau=t-1\}}X_{t-1} + 1_{\{\tau>t-1\}}\mathbb{E}(Z_t|\mathcal{F}_{t-1}) \leq Z_{t-1}.$$



# 3.4: American claims

**Proof.**

In the special case where $\tau = \tau(t-1)$ we find that $\tau(t-1)$ stops in $t-1$ if and only if $Z_{t-1} = X_{t-1}$. Otherwise $Z_{t-1} > X_{t-1}$ and $\tau(t-1) = \tau(t)$ and hence

$$\begin{aligned}
\mathbb{E}(X_{\tau(t-1)}|\mathcal{F}_{t-1}) &= 1_{\{Z_{t-1}=X_{t-1}\}}X_{t-1} + 1_{\{Z_{t-1}>X_{t-1}\}}\mathbb{E}(\mathbb{E}(X_{\tau(t)}|\mathcal{F}_t)|\mathcal{F}_{t-1}) \\
&= 1_{\{Z_{t-1}=X_{t-1}\}}X_{t-1} + 1_{\{Z_{t-1}>X_{t-1}\}}\mathbb{E}(Z_t|\mathcal{F}_{t-1}) \\
&= Z_{t-1}.
\end{aligned}$$

□

# 3.4: American claims

We prove Prop. 3.22 using the Optimal Stopping thm. (non-examinable).

## Proof.

Let $p$ denote the time $t$ price of the American claim with respect to the payoff process $(Y_t/B_t)$. Suppose first that $p < Z_t$, then:

- We buy the option for $p$.
- If $\tau(t) = t$, then $Z_t = Y_t/B_t$ and so we can exercise the option immediately for $Y_t/B_t > p$ to make riskless profit.
- If $\tau(t) > t$ we undertake the negative of the trading strategy that replicates $Y_{\tau(t)}/B_{\tau(t)}$. The price of the replicating strategy is $\mathbb{E}_Q(Y_{\tau(t)}/B_{\tau(t)}|\mathcal{F}_t)$ by the risk-neutral valuation principle, but by the Optimal Stopping Theorem this is equal to $Z_t$ and so we can invest the difference $Z_t - p$ in the bank account.
- Later at time $\tau(t)$ we exercise the option and liquidate the replicating portfolio at the same time. The amount we collect from the option seller is exactly equal to our liability on the portfolio. Meanwhile we have $(Z_t - p)B_{\tau(t)}/B_t > 0$ in the bank account.



# 3.4: American claims

## Proof.

Consider now the case where $p > Z_t$, then:

- We sell the option for $p$.

Let us first consider in detail the case where $\tau(t) > t$. Then

- We undertake the trading strategy that replicates $Y_{\tau(t)}/B_{\tau(t)}$. Again we find using risk-neutral valuation and the Optimal Stopping Theorem that the price of building up the portfolio is

$$Z_t = \mathbb{E}_Q(Y_{\tau(t)}/B_{\tau(t)}|\mathcal{F}_t).$$

- Therefore there is a profit of $p - Z_t$ which we put into a bank account.

How we proceed will depend on when the buyer exercises the option...

## 3.4: American claims

**Proof.**

If the buyer exercises the option at a time point $s \le \tau(t)$, then

- We pay the buyer the payoff $Y_s/B_s$.
- We liquidate our portfolio and using risk-neutral valuation the value of our portfolio at time $s$ is

$$\mathbb{E}_Q(Y_{\tau(t)}/B_{\tau(t)}|\mathcal{F}_s).$$

  Since $t \le s \le \tau(t)$ we see from the definition of $\tau(t)$ that $\tau(t) = \tau(s)$.
  Using this and the optimal stopping theorem we can deduce that the value of our portfolio at time $s$ is

$$\mathbb{E}_Q(Y_{\tau(t)}/B_{\tau(t)}|\mathcal{F}_s) = \mathbb{E}_Q(Y_{\tau(s)}/B_{\tau(s)}|\mathcal{F}_s) = Z_s \ge Y_s/B_s.$$

- Therefore all transactions at time point $s$ will only add to our portfolio and our total profit is strictly positive.



## 3.4: American claims

**Proof.**

On the other hand, suppose the option buyer does not exercise by time $s = \tau(t)$ where $\tau(t) < T$, then:

- We repeat the process, undertaking the trading strategy that replicates $Y_{\tau(s+1)}/B_{\tau(s+1)}$.
- The value of the portfolio to be built up is equal to

$$\mathbb{E}_Q(Y_{\tau(s+1)}/B_{\tau(s+1)}|\mathcal{F}_s) \le \mathbb{E}_Q(Y_{\tau(s)}/B_{\tau(s)}|\mathcal{F}_s) = Z_s.$$

  Therefore the change of the portfolio will only pay us some money which we put in the bank account.
- As before, if the option buyer exercises at some time $u \le \tau(s+1)$, then the value of the portfolio will be enough to cover the payoff $Y_u$.

If the buyer still has not exercised by time $\tau(s+1)$, then we repeat this process again, and so forth. There will always be enough money in our portfolio to cover the payoff, our overall profit will be at least $p - Z_t$.

# 3.4: American claims

**Proof.**

Finally, we consider the case where $\tau(t) = t$. Then the optimal strategy would be to exercise the option immediately.

- If the buyer indeed exercises at time $t$, then we pay them $Y_t/B_t = Z_t < p$ and make riskless profit.
- If not, then we proceed as in the previous case where the buyer does not exercise by the optimal stopping time and undertake the trading strategy that replicates $Y_{\tau(t+1)}/B_{\tau(t+1)}$ and so forth making again profit of at least $p - Z_t$.

□



# 3.4: American claims

## 3.4: American claims

**Example 3.26.**

- Return to extended example 3.13 where the price process is given by

| $\omega_k$ | $t=0$ | $t=1$ | $t=2$ |
|---|---|---|---|
| $\omega_1$ | $S_0=6$ | $S_1=9$ | $S_2=10$ |
| $\omega_2$ | $S_0=6$ | $S_1=9$ | $S_2=7$ |
| $\omega_3$ | $S_0=6$ | $S_1=3$ | $S_2=7$ |
| $\omega_4$ | $S_0=6$ | $S_1=3$ | $S_2=1$ |

- In the case $r=0$, recall that $\mathbb{Q}(\omega_1)=1/3$, $\mathbb{Q}(\omega_2)=1/6$, $\mathbb{Q}(\omega_3)=1/6$, $\mathbb{Q}(\omega_4)=1/3$.



## 3.4: American claims

**Example 3.26.**

- Consider an American call option with exercise price 5. Payoff process

| $\omega_k$ | $t=0$ | $t=1$ | $t=2$ |
|---|---|---|---|
| $\omega_1$ | $Y_0=1$ | $Y_1=4$ | $Y_2=5$ |
| $\omega_2$ | $Y_0=1$ | $Y_1=4$ | $Y_2=2$ |
| $\omega_3$ | $Y_0=1$ | $Y_1=0$ | $Y_2=2$ |
| $\omega_4$ | $Y_0=1$ | $Y_1=0$ | $Y_2=0$ |

- In states $\omega_3$ and $\omega_4$ the price $Z_1=\max(Y_1,\mathbb{E}_{\mathbb{Q}}(Z_2|\mathcal{F}_1))=\max(0,\mathbb{E}_{\mathbb{Q}}(Z_2|\mathcal{F}_1))=\max(0,2/3)=2/3$. This is equal to $V_1(\omega_3,\omega_4)$ for the European option above.
- In states $\omega_1$ and $\omega_2$ the price $Z_1=\max(4,\mathbb{E}_{\mathbb{Q}}(Z_2|\mathcal{F}_1))=\max(4,5(2/3)+2(1/3))=\max(4,4)=4$, again equalling $V_1(\omega_1,\omega_2)$.
- So $Z_0=\max(1,1/2(4)+1/2(2/3))=7/3$.
- Time 0 price equal to European option, cf Proposition 1.15.

## Example 3.27.

We consider a modified pay-off process by changing $Y_1(\omega_1) = Y_1(\omega_2) = 5$.

| $\omega_k$ | $t = 0$ | $t = 1$ | $t = 2$ |
|---|---|---|---|
| $\omega_1$ | $Y_0 = 1$ | $Y_1 = 5$ | $Y_2 = 5$ |
| $\omega_2$ | $Y_0 = 1$ | $Y_1 = 5$ | $Y_2 = 2$ |
| $\omega_3$ | $Y_0 = 1$ | $Y_1 = 0$ | $Y_2 = 2$ |
| $\omega_4$ | $Y_0 = 1$ | $Y_1 = 0$ | $Y_2 = 0$ |

- At time point 2 obviously again $Z_2 = Y_2$. Furthermore $\mathbb{E}_{\mathbb{Q}}(Z_2|\mathcal{F}_1) = \mathbb{E}_{\mathbb{Q}}(Y_2|\mathcal{F}_1)$ and the latter has already been calculated above because $Z_2$ is equal to the payoff of the European option above.
- So $\mathbb{E}_{\mathbb{Q}}(Z_2|\mathcal{F}_1) = \frac{2}{3}5 + \frac{1}{3}(2) = 4$ in states $\omega_1$ and $\omega_2$ and $\mathbb{E}_{\mathbb{Q}}(Z_2|\mathcal{F}_1) = \frac{1}{3}2 + \frac{2}{3}0 = 2/3$ in states $\omega_3$ and $\omega_4$.
- Hence $Z_1 = \max\{Y_1, \mathbb{E}_{\mathbb{Q}}(Z_2|\mathcal{F}_1)\}$ equals $\max\{5, 4\} = 5$ and $\max\{0, 2/3\} = 2/3$ respectively.





## Example 3.27.

- For time 0 we have

$$\mathbb{E}_{\mathbb{Q}}(Z_1|\mathcal{F}_0) = \mathbb{E}_{\mathbb{Q}}(Z_1) = \frac{1}{2}5 + \frac{1}{2}\frac{2}{3} = \frac{17}{6}$$

  in which case $Z_0 = \max\{Y_0, \mathbb{E}_{\mathbb{Q}}(Z_1)\} = \max\{1, 17/6\} = 17/6$.
- The optimal exercise strategies are

$$\tau(0)(\omega) = \tau(1)(\omega) = \begin{cases} 1, & \text{if } \omega = \omega_1, \omega_2 \\ 2, & \text{if } \omega = \omega_3, \omega_4 \end{cases}.$$

  and, of course, $\tau(2)(\omega) = 2$ for all $\omega \in \Omega$.

**Example 3.27.**

- Finally we compute a trading strategy that replicates $\tau(0)$.
- In states $\omega_3$ and $\omega_4$ the stopping time $\tau(0)$ stops in time 2, so the portfolio in the second period must satisfy

$$H_0(2) + 7H_1(2) = Y_2(\omega_3) = 2$$
$$H_0(2) + 1H_1(2) = Y_2(\omega_4) = 0,$$

writing $H_i(2)$ for $H_i(2)(\omega_3, \omega_4)$.

- The solution to this is $H_0(2) = -1/3$ and $H_1(2) = 1/3$ in states $\omega_3$ and $\omega_4$.
- The value of that portfolio at time 1 is equal to $-1/3 + 3(1/3) = 2/3$ which equals $V_1$ because $H$ is self-financing.
- The values of $H_0(2)$ and $H_1(2)$ in states $\omega_1$ and $\omega_2$ do not matter.





**Example 3.27.**

- The portfolio in the first period must satisfy

$$H_0(1) + 9H_1(1) = Z_1(\omega_1) = Z_1(\omega_2) = 5$$
$$H_0(1) + 3H_1(1) = Z_1(\omega_3) = Z_1(\omega_4) = 2/3$$

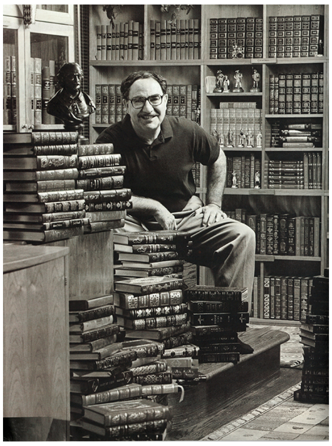
with solution $H_0(1) = -3/2$ and $H_1(1) = 13/18$.

- The American option is attainable with time 0 value is $-3/2 + 6(13/18) = 17/6$ (confirming our previous calculations).

# 3.5: The Cox-Ross-Rubinstein model

John Carrington Cox
(1943-)

Stephen Ross
(1944-2017)



Mark Rubinstein
(1944-2019)



# 3.5: The Cox-Ross-Rubinstein model

**Definition 3.28.**

The stochastic process $\{X_t : t = 1, 2, \ldots\}$ is said to be a Bernoulli process with parameter $p$ if the random variables $X_1, X_2, \ldots$ are independent and

$$\mathbb{P}(X_t = 1) = 1 - \mathbb{P}(X_t = 0) = p \quad \text{for all } t.$$

The random walk process $\{N_t : t = 1, 2, \ldots\}$ is defined in terms of the Bernoulli process by setting

$$N_t = X_1 + \cdots + X_t. \tag{3.7}$$

# 3.5: The Cox-Ross-Rubinstein model

**Definition 3.29.**

A random variable $B_{t,p}$ is binomially distributed with parameters $t$ and $p$ if for all $k = 0, \dots, t$

$$\mathbb{P}(B_{t,p} = k) = \binom{t}{k} p^k (1-p)^{t-k}.$$

**Lemma 3.30.**

*For any $t$, the random walk process $N_t$ is binomially distributed with parameters $t$ and $p$, and hence*

$$\mathbb{E}(N_t) = tp \quad \text{and} \quad \mathrm{Var}\,(N_t) = tp(1-p).$$





- Simplify the general multi-period model.
- Assume price always moves up or down by a fixed ratio.
- Assume moves take place with fixed probability.

**Definition 3.31.**

- The Cox-Ross-Rubinstein model (or Binomial model) consists of a bank account with constant interest rate $r \geq 0$ and one risky asset with price process $S_t$ defined by

$$S_t = S_0 u^{N_t} d^{t-N_t}, \quad t = 1, 2, \dots, T.$$

- Here $N_t = X_1 + \dots + X_t$ is the random walk process associated with a Bernoulli process $X_t$ with parameter $p$, where $0 < p < 1$.
- The up and down ratios $u, d$ and start price $S_0$ are real numbers with $S_0 > 0$ and $0 < d < 1 < u$.

## 3.5: The Cox-Ross-Rubinstein model

- In each period of this multi-period model there are two possibilities:
  - price goes up by factor $u$, which happens with probability $p$
  - price goes down by factor $d$ which happens with probability $1 - p$.
- The presence of these multiplicative factors means that we often think in terms of the logarithm of the price.
- In particular, the logarithm of the process $S_t$ behaves like an (asymmetric) random walk (with drift).



## 3.5: The Cox-Ross-Rubinstein model

**Proposition 3.32.**

*The Cox-Ross-Rubinstein model has no arbitrage opportunities if and only if $d < 1 + r < u$. In that case it is also complete and the unique martingale measure $\mathbb{Q}$ is defined by*

$$\mathbb{Q}(\{\omega\}) = q^n(1-q)^{T-n}$$

*where $\omega \in \Omega$ is any state corresponding to $n$ 'ups' and $T - n$ 'downs' and*

$$q = \frac{1+r-d}{u-d}.$$

*In particular, adding all contributions from states with exactly $n$ 'ups'*

$$\mathbb{Q}(S_t = S_0 u^n d^{t-n}) = \binom{t}{n} q^n (1-q)^{t-n}, n = 0, 1, \ldots, t.$$

**Proposition 3.32.**

*Moreover, if $X$ is a contingent claim of the form $X = g(S_T)$ for some real-valued function $g$, then the time $t = 0$ value of $X$ is given by*

$$V_0 = (1+r)^{-T} \sum_{n=0}^{T} \binom{T}{n} q^n (1-q)^{T-n} g(S_0 u^n d^{T-n}).$$

*More generally, the time $t$ value of $X$ is given by*

$$\Pi(t) = \frac{1}{(1+r)^{T-t}} \sum_{n=0}^{T-t} \binom{T-t}{n} q^n (1-q)^{T-t-n} g(S_t u^n d^{T-t-n}).$$



Proof.

- The no-arbitrage theorem shows the Cox-Ross-Rubinstein model is free of arbitrage if and only if there is a martingale measure $\mathbb{Q}$, with

$$S_t = \frac{1}{1+r} \mathbb{E}_{\mathbb{Q}}(S_{t+1} | \mathcal{F}_t) \quad \text{for all } t < T. \tag{3.8}$$

- Now let $q = \mathbb{Q}(X_{t+1} = 1) = \mathbb{Q}(S_{t+1}/S_t = u)$, which means that $\mathbb{Q}(S_{t+1}/S_t = d) = 1 - q$.
- $S_t$ is known at time $t$, so we put it inside the conditional expectation with respect to $\mathcal{F}_t$, ie divide by $S_t$ and rewrite Equation (3.8)

$$1 = \frac{1}{1+r} \mathbb{E}_{\mathbb{Q}} \left( \frac{S_{t+1}}{S_t} \middle| \mathcal{F}_t \right) = \frac{1}{1+r} (qu + (1-q)d)\,.$$

- This means that the no-arbitrage property is satisfied if and only if $qu + (1-q)d = 1 + r$. Rearranging, we see that

$$q = \frac{1+r-d}{u-d}.$$

## 3.5: The Cox-Ross-Rubinstein model

**Proof.**

- By definition $\mathbb{Q}$ must be positive everywhere, so that $0 < q < 1$, or equivalently $d < 1 + r < u$ must be satisfied.
- The form of martingale measure $\mathbb{Q}$ can now be obtained (as in the proof of the no-arbitrage theorem) by multiplying the conditional probabilities along the paths that lead to each state $\omega_i$.
- By the uniqueness of $q$ we see that the Cox-Ross-Rubinstein model is complete.
- The expression for $\mathbb{Q}(S_t = S_0 u^n d^{t-n})$ is straightforward.
- We can calculate the time 0 price as the expectation with respect to $\mathbb{Q}$ (risk-neutral valuation formula).
- We think of $\Pi(t)$ as a time-shifted version of the formula for the time 0 price, replacing $T$ by $T - t$ and $S_0$ by $S_t$.

□



## 3.5: The Cox-Ross-Rubinstein model

**Example 3.33.**

- Consider a Cox-Ross-Rubinstein model with $T = 3$ periods, start price $S_0 = 8$, interest rate $r = 0.1$ and parameters $u = 2$ and $d = 1/2$.
- We can calculate $q = (1 + r - d)/(u - d) = (1.1 - 0.5)/(1.5) = 0.4$.
- After 3 periods, the possible values of $S_T$ are 64, 16, 4, 1.
- In each case, the value of a European call with exercise price 4 is $60, 12, 0, 0$.
- The time 0 price of the European call option is therefore

$$\begin{aligned}\Pi_K(0) &= \frac{1}{(1+r)^T} \sum_{n=0}^{T} \binom{T}{n} q^n (1-q)^{T-n} (S_0 u^n d^{T-n} - K)_+ \\ &= (60q^3 + 36q^2(1-q))/(1+r)^3 = 5.4816.\end{aligned}$$

## 3.5: The Cox-Ross-Rubinstein model

**Proposition 3.34.**

*Consider a Cox-Ross-Rubinstein model with $T$ periods, start price $S_0$, interest rate $r$ and parameters $d$ and $u$. Then the time $t$ price of a European call option with exercise price $K$ is*

$$\Pi_K(t) = (1+r)^{-T+t} \sum_{n=0}^{T-t} \binom{T-t}{n} q^n (1-q)^{T-t-n} (S_t u^n d^{T-t-n} - K)^+$$

*where as usual*

$$q = \frac{1+r-d}{u-d}.$$



## 3.6: The CRR model and the Black-Scholes formula

Suppose we observe two assets in a continuous time interval $[0, U]$,

- a riskless bond
- and a stock.

We try to approximate them by a Cox-Ross-Rubinstein model with $T$ periods where trading is only possible at the $T+1$ time points

$$0, U/T, 2U/T, \dots, U.$$

Since we assume continuous compounding we define the interest rate for the Cox-Ross-Rubinstein model as $r_T = e^{rU/T} - 1$ because at time point $j \cdot U/T$ of the discrete model the bank account process then has value $(1+r_T)^j = e^{jrU/T}$.

In the special case where $u_T = e^{\sigma\sqrt{U/T}} = 1/d_T$ we obtain the Black-Scholes-Formula as the limit of the prices of a European Call option in our Cox-Ross-Rubinstein models if $T \to \infty$.

## 3.6: The CRR model and the Black-Scholes formula

**Proposition 3.35 (Black-Scholes-Formula).**

*Consider a European call option with exercise price $K$ maturing at time $U$. Let $\Pi_K^{(T)}(0)$ denote its time 0 price in a Cox-Ross-Rubinstein model with $T+1$ time points $0, \frac{U}{T}, 2\frac{U}{T}, \ldots, U$, constant interest rate $r_T = e^{rU/T} - 1$, and $u_T = e^{\sigma\sqrt{U/T}} = 1/d_T$. Then*

$$\lim_{T\to\infty} \Pi_K^{(T)}(0) = \Pi_K^{BS}(0)$$

*with $\Pi_K^{BS}(0)$ given by the Black-Scholes-Formula*

$$\Pi_K^{BS}(0) = S_0\Phi(d_1(S_0, U)) - Ke^{-rU}\Phi(d_2(S_0, U)).$$



## 3.6: The CRR model and the Black-Scholes formula

**Proposition 3.35 (Black-Scholes-Formula).**

*The functions $d_1(s, u)$ and $d_2(s, u)$ are given by*

$$d_1(s, u) = \frac{\log(s/K) + (r + \sigma^2/2)u}{\sigma\sqrt{u}}$$

$$d_2(s, u) = \frac{\log(s/K) + (r - \sigma^2/2)u}{\sigma\sqrt{u}}$$

*and $\Phi$ is the standard normal cumulative distribution function.*

## 3.6: The CRR model and the Black-Scholes formula

We will need a special version of the central limit theorem for binomially distributed random variables that we use without proof:

**Lemma 3.36.**

*Let $Y_n$ be binomially distributed with parameters $n$ and $\pi_n$. Then*

$$\tilde{Y}_n = \frac{Y_n - n\pi_n}{\sqrt{n\pi_n(1-\pi_n)}}$$

*converges in distribution to the standard normal distribution.*



## 3.6: The CRR model and the Black-Scholes formula

Proof (of Black-Scholes-Formula).

Let

$$\alpha_T = \min\{n : S_0 u_T^n d_T^{T-n} > K\}.$$

We rewrite the time 0 price $\Pi_K^{(T)}(0)$ in the $T$-th CRR model:

$$\begin{aligned}
\Pi_K^{(T)}(0) &= (1+r)^{-T} \sum_{n=\alpha_T}^{T} \binom{T}{n} q_T^n (1-q_T)^{T-n} (S_0 u_T^n d_T^{T-n} - K) \\
&= S_0 \left( \sum_{n=\alpha_T}^{T} \binom{T}{n} \left(\frac{q_T u_T}{1+r_T}\right)^n \left(\frac{(1-q_T)d_T}{1+r_T}\right)^{T-n} \right) \qquad (3.9) \\
&\quad - (1+r_T)^{-T} K \left( \sum_{n=\alpha_T}^{T} \binom{T}{n} q_T^n (1-q_T)^{T-n} \right).
\end{aligned}$$

We immediately identify the terms involved in the second sum as the density of a Binomial distribution with parameters $T$ and $q_T$.

## Proof.

For the first sum we also define

$$\hat{q}_T = \frac{q_T u_T}{1 + r_T}.$$

This is a probability because

$$0 < \hat{q}_T = \frac{q_T u_T}{1 + r_T} = \frac{\frac{1+r_T-d_T}{u_T-d_T} u_T}{1 + r_T} = \frac{u_T - \frac{d_T u_T}{1+r_T}}{u_T - d_T} < \frac{u_T - \frac{d_T u_T}{u_T}}{u_T - d_T} = 1.$$

Moreover we see from the definition of $q_T$ that

$$1 - \hat{q}_T = \frac{1 + r_T - q_T u_T}{1 + r_T} = \frac{1 + r_T - (1 + r_T - d_T) - q_T d_T}{1 + r_T}$$
$$= \frac{(1 - q_T) d_T}{1 + r_T}.$$

Thus we identify the first sum in (3.9) as the density of a Binomial distribution with parameters $T$ and $\hat{q}_T$.



## Proof.

Let now $Y_T$ and $\hat{Y}_T$ be binomially distributed random variables, $Y_T$ with parameters $T$ and $q_T$, $\hat{Y}_T$ with parameters $T$ and $\hat{q}_T$. We rewrite (3.9)

$$\Pi_K^{(T)}(0) = S_0 \mathbb{P}(\hat{Y}_T > \alpha_T - 1) - K(1 + r_T)^{-T} \mathbb{P}(Y_T > \alpha_T - 1).$$

Thus we need to show that

(i) $\lim_{T\to\infty} \mathbb{P}(\hat{Y}_T > \alpha_T - 1) = \Phi(d_1(S_0, U))$

(ii) $\lim_{T\to\infty} \mathbb{P}(Y_T > \alpha_T - 1) = \Phi(d_2(S_0, U))$.

Starting with (ii)

$$\mathbb{P}(Y_T > \alpha_T - 1) = \mathbb{P}\left(\frac{Y_T - Tq_T}{\sqrt{Tq_T(1 - q_T)}} > \frac{\alpha_T - 1 - Tq_T}{\sqrt{Tq_T(1 - q_T)}}\right).$$

We want to use Lemma 3.36 and therefore determine the convergence of the term $\dfrac{\alpha_T - 1 - Tq_T}{\sqrt{Tq_T(1 - q_T)}}$.

## Proof.

Using a Taylor decomposition we see that

$$q_T = \frac{e^{rU/T} - e^{-\sigma\sqrt{U/T}}}{e^{\sigma\sqrt{U/T}} - e^{-\sigma\sqrt{U/T}}}$$

$$= \frac{1 + rU/T + o(1/T) - 1 + \sigma\sqrt{U/T} - \frac{1}{2}\sigma^2\frac{U}{T} - o(1/T)}{1 + \sigma\sqrt{U/T} + \frac{1}{2}\sigma^2\frac{U}{T} + o(1/T) - 1 + \sigma\sqrt{U/T} - \frac{1}{2}\sigma^2\frac{U}{T} - o(1/T)}$$

$$= \frac{rU/T + o(1/T) + \sigma\sqrt{U/T} - \frac{1}{2}\sigma^2\frac{U}{T}}{2\sigma\sqrt{U/T} + o(1/T)}$$

$$= \frac{1}{2}\left(\left(\frac{r}{\sigma} - \frac{\sigma}{2}\right)\frac{\sqrt{U}}{\sqrt{T}} + 1 + o\left(\frac{1}{\sqrt{T}}\right)\right).$$

Thus we obtain the limiting relations

$$\lim_{T\to\infty} q_T = \frac{1}{2} \text{ and } \lim_{T\to\infty}(1 - 2q_T)\sqrt{UT} = -U\left(\frac{r}{\sigma} - \frac{\sigma}{2}\right).$$



## Proof.

Finally we see from the definition of $\alpha_T$ that for some $|\gamma_T| < 1$

$$\alpha_T = \frac{\log(K/S_0 d_T^T)}{\log(u_T/d_T)} + \gamma_T = \frac{\log(K/S_0) - T\log(d_T)}{\log(u_T^2)} + \gamma_T$$

$$= \frac{\log(K/S_0) + T\sigma\sqrt{U/T}}{2\sigma\sqrt{U/T}} + \gamma_T.$$

Hence

$$\lim_{T\to\infty}\frac{\alpha_T - 1 - Tq_T}{\sqrt{Tq_T(1-q_T)}} = \lim_{T\to\infty}\frac{\frac{\log(K/S_0) + T\sigma\sqrt{U/T}}{2\sigma\sqrt{U/T}} + \gamma_T - 1 - Tq_T}{\sqrt{Tq_T(1-q_T)}}$$

$$= \lim_{T\to\infty}\frac{\log(K/S_0) + \sigma\sqrt{UT}(1 - 2q_T)}{2\sigma\sqrt{Uq_T(1-q_T)}}$$

$$= \frac{\log(K/S_0) - (r - \sigma^2/2)U}{\sigma\sqrt{U}} = -d_2(S_0, U).$$

## Proof.

Using Lemma 3.36 we conclude that

$$\lim_{T\to\infty} \mathbb{P}(Y_T > \alpha_T - 1) = 1 - \Phi(-d_2(S_0, U)) = \Phi(d_2(S_0, U)),$$

completing the proof of (ii).
To prove (i) we can argue in very much the same way. Using the limiting relations

$$\lim_{T\to\infty} \hat{q}_T = \frac{1}{2} \text{ and } \lim_{T\to\infty} (1 - 2\hat{q}_T)\sqrt{UT} = -U\left(\frac{r}{\sigma} + \frac{\sigma}{2}\right)$$

we can conclude that

$$\lim_{T\to\infty} \frac{\alpha_T - 1 - T\hat{q}_T}{\sqrt{T\hat{q}_T(1-\hat{q}_T)}} = \frac{\log(K/S_0) - (r + \sigma^2/2)U}{\sigma\sqrt{U}} = -d_1(S_0, U).$$

Hence

$$\lim_{T\to\infty} \mathbb{P}(\hat{Y}_T > \alpha_T - 1) = 1 - \Phi(-d_1(S_0, U)) = \Phi(d_1(S_0, U)).$$

# 4. Stochastic Processes in Continuous Time





# 4.1: The Brownian motion

**Definition 4.1.**

A stochastic process $X = (X_t)_{t\geq 0}$ is a family of random variables $X_t : \Omega \to \mathbb{R}$.
For a given $\omega \in \Omega$, $t \longmapsto X(t, \omega)$ is called the trajectory of the process $X$.
We say $X$ is adapted to a filtration $(\mathcal{F}_t)_{t\geq 0}$, if $X_t$ is $\mathcal{F}_t$-measurable for each $t$.

**Definition 4.2.**

A stochastic process $X = (X_t)_{t\geq 0}$ is a martingale with respect to the filtration $\mathcal{F} = (\mathcal{F}_t)_{t\geq 0}$ if

(i) if $X$ is adapted to the filtration $\mathcal{F}_t$,

(ii) $\mathbb{E}(|X_t|) < \infty$; for all $t \geq 0$,

(iii) $\mathbb{E}(X_t|\mathcal{F}_s) = X_s$ for all $0 \leq s \leq t$.

There are similar definitions for sub- and supermartingales.

# 4.1: The Brownian motion

We introduce a process known as Brownian motion, from which we can construct a good model of prices.

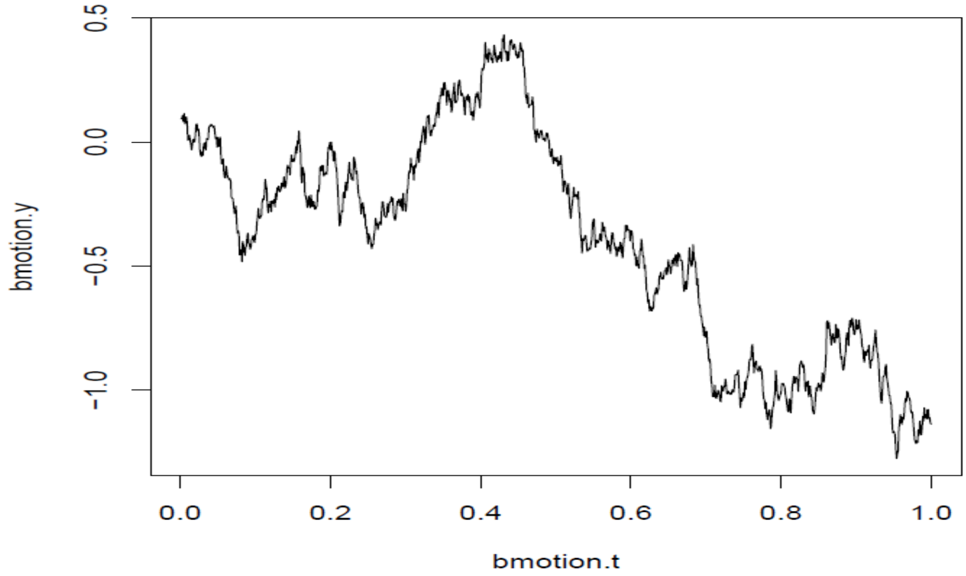



Figure: Sample of Brownian motion



# 4.1: The Brownian motion

## Year's worth of FTSE

# 4.1: The Brownian motion

Morning's worth of GBP vs USD




# 4.1: The Brownian motion

- Graphs illustrate a characteristic 'wigglyness', coupled with a slow drift.
- Graphs look much the same, even though they represent very different time periods.
- Introduce mathematical model which shares these properties.

**Definition 4.3.**

- Let $(S_j)_{j \in \mathbb{N}}$ be a random walk defined by $S_0 = 0$ and $S_n = S_{n-1} + X_n$ where $X_1, X_2, \ldots$ are independent, identically distributed random variables with mean 0 and variance 1.
- Define a process $(S^n_t)_{t \in [0,1]}$ on $[0, 1]$ by setting $S^n_t = S_j/\sqrt{n}$ for every $t = j/n$ and linear interpolation in between.

**Lemma 4.4.**

*Notice that $S_t^n$ has the following properties:*

1. $S_0^n = S_0/\sqrt{n} = 0,$
2. *Taking $t = j/n$ and $u = k/n$,* $S_{t+u}^n - S_t^n = (S_{j+k} - S_j)/\sqrt{n} = (X_{j+1} + \cdots + X_{j+k})/\sqrt{n}$ *is independent of $X_1, \ldots, X_j$.*
3. *Taking $t = j/n$ and $u = k/n$, for $n$ large, by the central limit theorem,*

$$S_{t+u}^n - S_t^n = \frac{X_{j+1} + \ldots + X_{j+k}}{\sqrt{n}} = \frac{\sqrt{k}}{\sqrt{n}} \left( \sum_{i=j+1}^{j+k} \frac{X_i}{\sqrt{k}} \right)$$

   *tends to a normal distribution with mean 0 and variance $k/n = u$.*
4. *$S_t^n(\omega)$ is continuous as a function of $t$ for any $n$ and for any $\omega \in \Omega$.*





(non-examinable) In fact, as $n$ tends to infinity, $S_t^n$ tends to a process $W_t$ known as Brownian motion. The properties of $S_t^n$ described above carry over to give corresponding properties of $W_t$, as listed in Definition 4.5. This can be made more precise by the theorem of Donsker which is also known as the functional central limit theorem. It states that for every bounded and continuous function $h : C(0,1) \to \mathbb{R}$

$$\mathbb{E}(h((S_t^n)_{t\in[0,1]})) \to \mathbb{E}(h((W_t)_{t\in[0,1]})).$$

This theorem has a lot of applications. For example we can conclude that

$$\lim_{n\to\infty} \mathbb{P}\left( \sup_{k\le n} \frac{S_k}{\sqrt{n}} \ge a \right) = \mathbb{P}\left( \sup_{t\in[0,1]} W_t \ge a \right).$$

# 4.1: The Brownian motion

**Definition 4.5.**

Let $\mathcal{F}_t$ be a filtration. An adapted stochastic process $W = (W_t)_{t \geq 0}$ is a standard Brownian motion or Wiener process if

(i) $W_0 = 0$ a.s.

(ii) $W$ has independent increments: $W_{t+u} - W_t$ is independent of $\mathcal{F}_t$ for each $u, t \geq 0$

(iii) $W$ has stationary Gaussian increments: $W_{t+u} - W_t$ is normally distributed with mean 0 and variance $u$.

(iv) $W$ has continuous paths: $W_t(\omega)$ is a continuous function of $t$ for all $\omega \in \Omega$.



# 4.1: The Brownian motion

Existence of Brownian motion was proved by Wiener in 1923.

**Theorem 4.6 (Wiener).**

*Brownian motion exists.*

Proof.

We omit the proof here. □

# 4.1: The Brownian motion

Some properties of Brownian motion are as follows:

**Lemma 4.7.**

*If $W_t$ is standard Brownian motion:*

(a) *For all $t$, $\mathbb{E}W_t = 0$ and $\mathrm{Var}\ W_t = t$.*

(b) *For all $s \le t$, $\mathrm{Cov}\ (W_s, W_t) = s$.*

(c) *$-W_t$ is itself a standard Brownian motion*

(d) *For fixed $t$, $X_s = W_{s+t} - W_t$ is a standard Brownian motion.*

(e) *For any $\alpha > 0$, $Y_s = W_{\alpha s}/\sqrt{\alpha}$ is a standard Brownian motion.*



# 4.1: The Brownian motion

Proof.

(a) Follows from property 3. of Definition 4.5.

(b) Using the previous result we conclude that

$$\begin{aligned}\mathrm{Cov}\ (W_s, W_t) &= \mathbb{E}W_sW_t = \mathbb{E}W_s(W_t - W_s) + \mathbb{E}W_s^2 \\ &= \mathbb{E}W_s\mathbb{E}(W_t - W_s) + s = s.\end{aligned}$$

(c) Follows from the symmetry of the normal distribution.

(d) In the case $t = 0$ we find that $X_0 = W_t - W_t = 0$. The other properties in Definition 4.5 are shift-invariant and therefore follow as well.

(e) The main part to check here is property (iii) of Definition 4.5. Indeed

$$Y_{t+s} - Y_t = (W_{\alpha(t+s)} - W_{\alpha t})/\sqrt{\alpha} \sim N(0, s).$$

□

# 4.1: The Brownian motion

Some further properties, partly the continuous equivalent of Example 2.24.

**Lemma 4.8.**

*Let $(W_t, t \geq 0)$ be a standard Brownian motion with respect to a filtration $(\mathcal{F}_t)$. Then*

(a) $(W_t)_{t\geq 0}$ *is a martingale.*

(b) $(W_t^2 - t)_{t\geq 0}$ *is a martingale.*

(c) *For all $\lambda \in \mathbb{R}$, the process $(Z_t)_{t\geq 0}$ defined by*

$$Z_t = e^{\lambda W_t - \frac{\lambda^2}{2} t}$$

*is a martingale. More generally, the geometric Brownian motion with volatility $\sigma > 0$ and drift $\mu \in \mathbb{R}$ can be defined as*

$$\tilde{Z}_t = e^{\sigma W_t + (\mu - \sigma^2/2)t} \tag{4.10}$$

*and is a martingale if and only if $\mu = 0$.*



**Proof.**

(a) We can write

$$W_t = (W_t - W_s) + W_s.$$

As $(W_t - W_s)$ is independent of $\mathcal{F}_s$ and has mean zero we conclude that

$$\mathbb{E}(W_t|\mathcal{F}_s) = \mathbb{E}(W_t - W_s|\mathcal{F}_s) + \mathbb{E}(W_s|\mathcal{F}_s) = \mathbb{E}(W_t - W_s) + W_s = W_s.$$

(b) Similarly we write

$$W_t^2 = (W_t - W_s)^2 + 2W_s(W_t - W_s) + W_s^2.$$

Then

$$\begin{aligned}
\mathbb{E}(W_t^2 - t|\mathcal{F}_s) &= \mathbb{E}((W_t - W_s)^2|\mathcal{F}_s) + 2W_s\mathbb{E}(W_t - W_s|\mathcal{F}_s) \\
&\quad + W_s^2 - t \\
&= (t - s) + 0 + W_s^2 - t \\
&= W_s^2 - s.
\end{aligned}$$

# 4.1: The Brownian motion

Proof.

(c) We find that

$$\begin{aligned}\mathbb{E}(\tilde{Z}_t|\mathcal{F}_s) = \mathbb{E}(e^{\sigma W_t + at}|\mathcal{F}_s) &= e^{\sigma W_s + at} \cdot \mathbb{E}(e^{\sigma(W_t - W_s)}|\mathcal{F}_s)\\ &= e^{\sigma W_s + at} \cdot \mathbb{E}(e^{\sigma N})\\ &= e^{\sigma W_s + as} e^{a(t-s)} \mathbb{E}(e^{\sigma N})\end{aligned}$$

where $N$ is a normal random variable with mean 0 and variance $t - s$. Its moment-generating function is $\mathbb{E}(e^{\sigma N}) = e^{(t-s)\sigma^2/2}$, therefore

$$\begin{aligned}\mathbb{E}(\tilde{Z}_t|\mathcal{F}_s) &= \tilde{Z}_s \cdot e^{a(t-s)} \cdot (e^{(t-s)\sigma^2/2})\\ &= \tilde{Z}_s e^{(t-s)(a+\sigma^2/2)}.\end{aligned}$$

We conclude that $\tilde{Z}_t$ is a martingale if and only if $a = -\sigma^2/2$.

□



# 4.2: Stochastic integration

Consider gains process in multiperiod model:

$$G_T = \sum_{u=1}^{T} H_0(u)\Delta B_u + \sum_{n=1}^{N}\sum_{u=1}^{T} H_n(u)\Delta S_n(u)$$

where $T$ is the number of periods. In moving to a model in continuous time, we want to make differences between time points smaller and smaller, ie for a time point $t$ in the continuous model we consider grid points $0, t/T, 2t/T, \ldots, t$ and consider what happens if $T \to \infty$.

# 4.2: Stochastic integration

For the bank account the corresponding sum converges like

$$\sum_{u=1}^{T} H_0(u)\Delta B_u \to \int_0^t H_0(u)dB_u \quad (T \to \infty)$$

If we further assume that $B_u$ is deterministic, eg $B(u) = e^{ru}$, then this integral becomes

$$\int_0^t H_0(u)B(u)'du = \int_0^t H_0(u)re^{ru}du.$$



# 4.2: Stochastic integration

Things are more difficult for stock prices as $S_n(u)$ will be modelled by Geometric Brownian motion and then $\Delta S_n$ will be differences of GMB. We want to define something like $\int_0^t H(u)dW(u)$. Here $H(u)$ portfolio at time $u$ and assume for now that $W(u)$ is the stock price (later this will be GMB). Problem is that $W(u)$ is not differentiable. The solution is the Ito integral that we look at in Section 4.2.

# 4.2: Stochastic integration

Starting point is the indicator function, ie for $0 < a < b$,

$$H(u) = \begin{cases} 1, & u \in [a, b] \\ 0, & \text{otherwise} \end{cases}$$

This corresponds to the strategy where be buy some stock at $a$ and sell at $b$.
Then

$$\int_0^t H(u)dW(u) = \int_0^t 1_{[a,b]}(u)dW(u) = \begin{cases} 0, & t < a \\ W(t) - W(a), & a \leq t < b \\ W(b) - W(a), & t \geq b \end{cases}$$



# 4.2: Stochastic integration

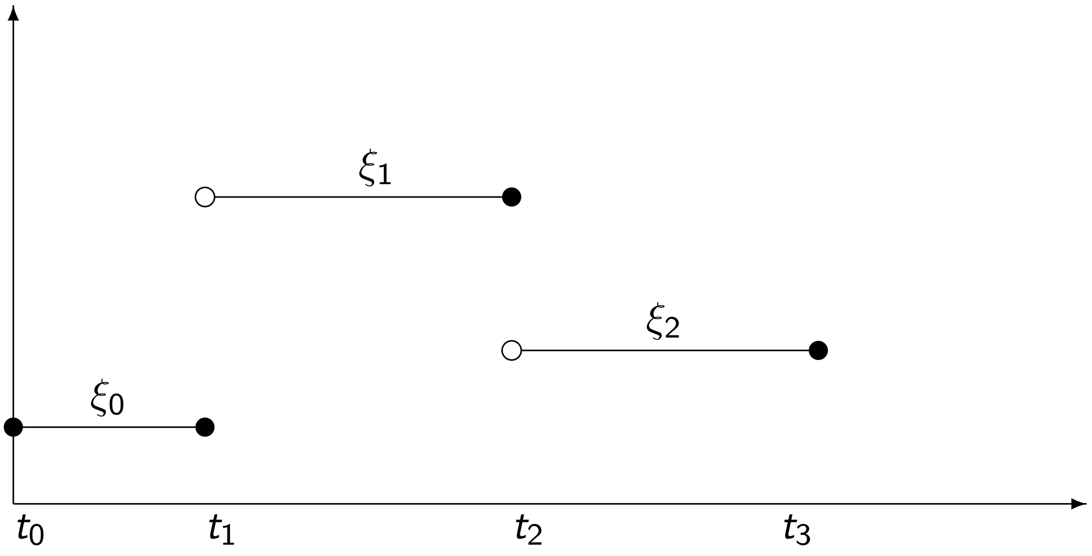
It is possible to extend the notion of integrals and to construct integrals with respect to a stochastic process. First we look at simple processes.

**Definition 4.9.**

A stochastic process $X = (X_t)_{0 \leq t \leq T}$ is called *simple* if there is a partition $0 = t_0 < t_1 < \cdots < t_n = T$ and $\mathcal{F}_{t_k}$-measurable random variables $\xi_k$ satisfying $\mathbb{E}(|\xi_i|) < \infty$ such that $X_t(\omega)$ can be written in the form

$$X_t(\omega) = \xi_0(\omega)1_0(t) + \sum_{i=0}^{n-1} \xi_i(\omega)1_{(t_i,t_{i+1}]}(t) \; (0 \leq t \leq T, \omega \in \Omega).$$

# 4.2: Stochastic integration






# 4.2: Stochastic integration

**Definition 4.10.**

For a simple stochastic process $X = (X_t)_{0 \leq t \leq T}$ we define its stochastic integral $I_t(X)$ by

$$I_t(X) := \int_0^t X dW = \sum_{i=0}^{n-1} \xi_i (W_{t \wedge t_{i+1}} - W_{t \wedge t_i})$$

where $s \wedge t := \min(s, t)$ and $W_t$ is Brownian motion.

In other words, if $t_k \leq t < t_{k+1}$, then

$$I_t(X) = \sum_{i=0}^{k-1} \xi_i (W_{t_{i+1}} - W_{t_i}) + \xi_k (W_t - W_{t_k}).$$

Notice that for each $t$, $I_t(X)$ is a random variable. Therefore $(I_t(X))_t$ is itself a stochastic process.

# 4.2: Stochastic integration

**Example 4.11.**

1. Consider the simplest possible $X_t$; that is $X_t = c$ for all $t$ (so $X_t$ is not really random and $n = 1$). Then
$$I_t(X) = cW_t.$$
2. Even if $X_t = X$ is random but fixed for all $t$ (so $n = 1$), then
$$I_t(X) = XW_t.$$
3. Suppose $X_t = c$ for $t \leq 1/2$ and $X_t = d$ for $t > 1/2$. Then
$$I_t(X) = \begin{cases} cW_t & \text{for } t < 1/2 \\ cW_{1/2} + d(W_t - W_{1/2}) & \text{for } t \geq 1/2. \end{cases}$$



**Lemma 4.12.**

*Let $X = (X_t)_{0 \leq t \leq T}$ be a simple stochastic process, then:*

(a) *The expectation of the stochastic integral is zero:* $\mathbb{E}(I_t(X)) = 0$.

(b) $I_t(X)$ *satisfies Itô's isometry:*
$$\mathbb{E}((I_t(X))^2) = \int_0^t \mathbb{E}(X_s^2)ds \tag{4.11}$$

(c) $I_t(X)$ *is a (continuous) martingale with respect to the natural Brownian motion filtration $\mathcal{F}_t$: For all $0 \leq s < t \leq T$*
$$\mathbb{E}(I_t(X)|\mathcal{F}_s) = I_s(X).$$

(d) *Linearity: If $Y$ is another simple stochastic process and $a$ and $b$ are arbitrary real numbers, then*
$$I_t(aX + bY) = aI_t(X) + bI_t(y).$$

(e) *The stochastic process $I_t(X)$ has continuous sample paths.*

# 4.2: Stochastic integration

**Proof.**

(a) Since $\xi_i$ and $W_{t_{i+1}} - W_{t_i}$ are independent for each $i$ and since $W_{t_{i+1}} - W_{t_i}$ has mean zero, we obtain $\mathbb{E}(\xi_i(W_{t_{i+1}} - W_{t_i})) = 0$ and thus $\mathbb{E}(I_t(X)) = 0$.

(b) Consider a partition of $[0, t]$, so that $0 = t_0 < t_1 < \cdots < t_{k-1} < t_k = t$. Then

$$\mathbb{E}((I_t(X))^2) = \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} \mathbb{E}(\xi_i(W_{t_{i+1}} - W_{t_i})\xi_j(W_{t_{j+1}} - W_{t_j})).$$

If $i > j$, then $W_{t_{i+1}} - W_{t_i}$ is independent of all the other factors and has expectation 0. Thus

$$\begin{aligned}
&\mathbb{E}(\xi_i(W_{t_{i+1}} - W_{t_i})\xi_j(W_{t_{j+1}} - W_{t_j})) \\
&\qquad = \mathbb{E}(W_{t_{i+1}} - W_{t_i})\mathbb{E}(\xi_i\xi_j(W_{t_{j+1}} - W_{t_j})) = 0.
\end{aligned}$$



# 4.2: Stochastic integration

**Proof.**

So all terms with $i > j$ and, similarly, $i < j$ vanish and we conclude that

$$\begin{aligned}
\mathbb{E}((I_t(X))^2) &= \sum_{i=0}^{k-1} \mathbb{E}(\xi_i(W_{t_{i+1}} - W_{t_i})\xi_i(W_{t_{i+1}} - W_{t_i})) \\
&= \sum_{i=0}^{k-1} \mathbb{E}(\xi_i^2)\mathbb{E}((W_{t_{i+1}} - W_{t_i})^2) \\
&= \sum_{i=0}^{k-1} \mathbb{E}(\xi_i^2)(t_{i+1} - t_i).
\end{aligned}$$

The right-hand side is the usual Riemann integral $\int_0^T f(s)ds$ of the step function $f(s) = \mathbb{E}(X_s^2)$ which coincides with $\mathbb{E}(\xi_i^2)$ for $t_i \leq s < t_{i+1}$

**Proof.**

(c) Adaptedness of $I(X)$ follows since at time $t$ all $\xi_i$ and $W_{t_{i+1}} - W_{t_i}$ contributing to $I_t(X)$ are functions of Brownian motion up to time $t$. The condition $\mathbb{E}(|I_t(X)|) < \infty$ follows from the isometry property (b). It remains to show $\mathbb{E}(I_t(X)|\mathcal{F}_s) = I_s(X)$ for all $s < t$. First assume that $s < t$ and $s, t \in [t_k, t_{k+1}]$. Notice that

$$I_t(X) = I_{t_k}(X) + \xi_k(W_t - W_{t_k})$$
$$I_s(X) = I_{t_k}(X) + \xi_k(W_s - W_{t_k})$$

Hence, $I_t(X) = I_s(X) + \xi_k(W_t - W_s)$, where $I_s(X)$ and $\xi_k$ are known at time $s$ and $W_t - W_s$ is independent of $\mathcal{F}_s$ and has mean 0. Hence

$$\mathbb{E}(I_t(X)|\mathcal{F}_s) = I_s(X) + \xi_k \mathbb{E}(W_t - W_s) = I_s(X).$$

The case where $s < t_k \leq t$ can be handled analogously.



**Proof.**

(d) Assume that the simple process $X$ is defined using the partition

$$0 = t_0 < t_1 < \cdots < t_n = T$$

and that $Y$ uses the partition $0 = s_0 < s_1 < \cdots < s_m = T$. Then consider the partition

$$0 = u_0 < u_1 < \cdots < u_l = T$$

which combines all the points from the other two partitions. The values of $I_t(X)$ and $I_t(Y)$ with regard to the finer partition remain the same, the linearity follows from the linearity of the underlying sums.

(e) This follows from the definition of $I_t(X)$ since

$$I_t(X) = I_{t_{k-1}}(X) + \xi_k(W_t - W_{t_{k-1}})$$

for all $t_{k-1} \leq t \leq t_k$ and the sample paths of Brownian motion are continuous.

# 4.2: Stochastic integration

For the general case the following theorem states that we can approximate any stochastic process of interest by a simple process.

**Theorem 4.13.**

*Let $X = (X_t)_{0\le t\le T}$ be a stochastic process which is adapted to Brownian motion, i.e. $X_t$ is a function of $W_s, s \le t$ and which satisfies that $\int_0^T \mathbb{E}(X_t^2) < \infty$.*
*Then one can find a sequence $X^{(n)}$ of simple processes such that*

$$\int_0^T \mathbb{E}\left((X_s - X_s^{(n)})^2\right) ds \to 0 \quad (n \to \infty).$$

*Moreover there is some process $I(X)$ on $[0, T]$ such that*

$$\mathbb{E}\left(\sup_{0\le t\le T} (I_t(X) - I_t(X^{(n)}))^2\right) \to 0 \quad (n \to \infty).$$



**Definition 4.14.**

The mean square limit $I(X)$ in the preceding theorem is called the Itô stochastic integral of $X$. It is denoted by

$$I_t(X) = \int_0^t X_s dW_s, \quad t \in [0, T].$$

For practical purposes the following rule of thumb is useful:

**Remark 4.15.**

*The Itô stochastic integrals $I_t(X), t \in [0, T]$ constitute a stochastic process. For a given partition $0 = t_0 < t_1 < \cdots < t_n = T$ and $t \in [t_k, t_{k+1}]$ the random variable $I_t(X)$ is approximately*

$$I_t(X) \approx \sum_{i=0}^{k-1} X_{t_i}(W_{t_{i+1}} - W_{t_i}) + X_{t_k}(W_t - W_{t_k})$$

*and this approximation is the closer to the value of $I_t(X)$ the more dense the partition in $[0, T]$.*

## 4.2: Stochastic integration

Kiyosi Itô (1915-2008)



## 4.2: Stochastic integration

All the properties described above for the stochastic integral of a simple process can be carried over to the general case.

**Theorem 4.16.**

*Let $X = (X_t)_{0\leq t\leq T}$ be a stochastic process which satisfies the requirements of Theorem 4.13. Then the stochastic integral $I_t(X)$*

(a) *has expectation zero,*

(b) *satisfies the Itô isometry property (4.11),*

(c) *is a martingale with respect to the natural Brownian filtration,*

(d) *is linear and*

(e) *has continuous sample paths.*

# 4.2: Stochastic integration

**Example 4.17.**

We calculate
$$I_t(W) = \int_0^t W_s dW_s.$$
We start by approximating the integrand by a sequence of simple functions
$$X_u^{(n)} = \begin{cases} W_0 = 0, & \text{if } 0 \le u \le t_1, \\ W_{t_1}, & \text{if } t_1 < u \le t_2, \\ \dots & \dots \\ W_{t_{n-1}}, & \text{if } t_{n-1} < u \le t, \end{cases}$$
where $t_k = tk/n$ for $k = 0, 1, \dots, n$.



**Example 4.17.**

Then
$$\begin{aligned} I_t(X^{(n)}) &= \sum_{k=0}^{n-1} W_{t_k}(W_{t_{k+1}} - W_{t_k}) \\ &= \sum_{k=0}^{n-1} W_{t_{k+1}} W_{t_k} - \frac{1}{2}\sum_{k=0}^{n-1}\left(W_{t_{k+1}}^2 + W_{t_k}^2\right) + \frac{1}{2}W_{t_n}^2 \\ &= \frac{1}{2}W_t^2 - \frac{1}{2}\sum_{k=0}^{n-1}(W_{t_{k+1}} - W_{t_k})^2 \end{aligned}$$
Since the differences $(W_{t_{k+1}} - W_{t_k})^2$ are independent and identically distributed with mean $t/n$ the sum on the right-hand side converges to $t$ if $n \to \infty$. Thus we obtain
$$I_t(W) = \int_0^t W_s dW_s = \frac{1}{2}W_t^2 - \frac{1}{2}t.$$

## 4.2 Stochastic integration

The argument used in the previous example leads to a general principle. If we write

$$\Delta W_{t_k} = W_{t_{k+1}} - W_{t_k} \text{ and } \Delta t_k = t_{k+1} - t_k,$$

then we can derive the following mean square limit properties:

$$\sum_{k=0}^{n-1} (\Delta W_{t_k})^2 \to t, \sum_{k=0}^{n-1} (\Delta W_{t_k}) \Delta t_k \to 0, \text{ and } \sum_{k=0}^{n-1} (\Delta t_k)^2 \to 0 \quad (n \to \infty).$$

These are often written as the formal multiplication rules

$$dW \cdot dW = dt,$$
$$dt \cdot dW = 0, \text{ and}$$
$$dt \cdot dt = 0.$$



## 4.3: Itô's lemma

We start with a special case of Itô's lemma:

**Theorem 4.18.**

*Let $f(x)$ be a twice continuously differentiable function. Then for any $t > 0$*

$$f(W_t) - f(W_0) = \int_0^t f'(W_u) dW_u + \frac{1}{2} \int_0^t f''(W_u) du,$$

*or in differential form*

$$df(W_t) = f'(W_t) dW_t + \frac{1}{2} f''(W_t) dt.$$

The second term $\frac{1}{2} f''(W_t) dt$ is known as the Itô correction term.

# 4.3: Itô's lemma

## Proof.

Given a partition of $[0, t]$, i.e.

$$0 = t_0 < t_1 < \cdots < t_n = t,$$

we can use Taylor's formula to obtain

$$\begin{aligned} f(W_t) - f(W_0) &= \sum_{i=0}^{n-1} f(W_{t_{i+1}}) - f(W_{t_i}) \\ &= \sum_{i=0}^{n-1} f'(W_{t_i})(W_{t_{i+1}} - W_{t_i}) \\ &+ \frac{1}{2}\sum_{i=0}^{n-1} f''(W_{t_i} + \theta_i(W_{t_{i+1}} - W_{t_i}))(W_{t_{i+1}} - W_{t_i})^2 \end{aligned}$$

with $0 < \theta_i < 1$.



# 4.3: Itô's lemma

## Proof.

The first sum is an approximating sequence of a stochastic integral; indeed, we find

$$\sum_{i=0}^{n-1} f'(W_{t_i})(W_{t_{i+1}} - W_{t_i}) \to \int_0^t f'(W_u)dW_u.$$

We also know that

$$\lim_{n\to\infty} \sum_{i=0}^{n-1}(W_{t_{i+1}} - W_{t_i})^2 = t,$$

and with a little more effort one can prove that

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} f''(W_{t_i} + \theta_i(W_{t_{i+1}} - W_{t_i}))(W_{t_{i+1}} - W_{t_i})^2 = \int_0^t f''(W_u)du.$$

□

# 4.3: Itô's lemma

**Example 4.19.**

Consider the function $f(x) = x^2$. Then $f'(x) = 2x$ and $f''(x) = 2$ and Theorem 4.18 gives

$$W_t^2 = \int_0^t 2W_u dW_u + \frac{1}{2}\int_0^t 2du = \int_0^t 2W_u dW_u + \frac{1}{2}2t$$

and hence

$$\frac{1}{2}(W_t^2 - t) = \int_0^t W_u dW_u$$

as in Example 4.17.



# 4.3: Itô's lemma

The next theorem is a more general version of the preceding result, although not yet Itô's lemma:

**Theorem 4.20.**

*Let $f(t,x)$ be a function which is continuously differentiable, once in its first argument and twice in its second argument. Then*

$$f(t,W_t) - f(0,W_0) = \int_0^t f_t(u,W_u) + \frac{1}{2}f_{xx}(u,W_u)du + \int_0^t f_x(u,W_u)dW_u$$

*or in differential form*

$$df(t,W_t) = (f_t(t,W_t) + \frac{1}{2}f_{xx}(t,W_t))dt + f_x(t,W_t)dW_t$$

**Definition 4.21.**

We use subscripts to denote partial derivatives of a function of two variables $f(t,x)$, e.g. $f_t = \partial f/\partial t$, $f_{tx} = \partial^2 f/\partial t \partial x$.

**Proof.**

By the Taylor expansion of a smooth function of several variables we get for $t$ close to $t_0$ that

$$\begin{aligned} f(t, W_t) = & f(t_0, W_{t_0}) + (t - t_0) f_t(t_0, W_{t_0}) + (W_t - W_{t_0}) f_x(t_0, W_{t_0}) \\ & + \frac{1}{2}(t - t_0)^2 f_{tt}(t_0, W_{t_0}) + \frac{1}{2}(W_t - W_{t_0})^2 f_{xx}(t_0, W_{t_0}) \\ & + (t - t_0)(W_t - W_{t_0}) f_{tx}(t_0, W_{t_0}) + \cdots \end{aligned}$$

which may be written symbolically as

$$df = f_t dt + f_x dW + \frac{1}{2} f_{tt} (dt)^2 + f_{tx} dt dW + \frac{1}{2} f_{xx} (dW)^2 + \cdots$$

Now using the formal multiplication rules $dt \cdot dt = 0$, $dt \cdot dW = 0$ and $dW \cdot dW = dt$ we get

$$df = f_t dt + f_x dW + \frac{1}{2} f_{xx} dt.$$

□



## 4.3: Itô's lemma

**Example 4.22 (Itô exponential).**

Consider the function

$$f(t, x) = \exp(x - t/2).$$

Then $f_t(t, x) = -f(t, x)/2$, $f_x(t, x) = f(t, x)$ and $f_{xx}(t, x) = f(t, x)$. Thus by Theorem 4.20, since $f(0, W_0) = 1$,

$$\begin{aligned} f(t, W_t) - 1 &= \int_0^t -\frac{1}{2} f(u, W_u) + \frac{1}{2} f(u, W_u) du + \int_0^t f(u, W_u) dW_u \\ &= \int_0^t f(u, W_u) dW_u. \end{aligned}$$

Hence $f$ satisfies the differential equation $df = f dW$ in the Itô sense and is therefore called the *Itô exponential*.

**Example 4.23 (Geometric Brownian Motion).**

Consider the function $f(t,x) = \exp(\sigma x + (\mu - \sigma^2/2)t)$. Then $f_t(t,x) = (\mu - \sigma^2/2)f(t,x)$, $f_x(t,x) = \sigma f(t,x)$ and $f_{xx}(t,x) = \sigma^2 f(t,x)$. Thus $f_t + f_{xx}/2 = \mu f$ so by Theorem 4.20:

$$f(t,W_t) - f(0,W_0) = \int_0^t \mu f(u,W_u)du + \int_0^t \sigma f(u,W_u)dW_u.$$

The process $X_t = f(t,W_t)$ satisfies the linear stochastic differential equation

$$X_t - X_0 = \mu \int_0^t X_s ds + \sigma \int_0^t X_s dW_s$$

which is in differential notation

$$dX = \mu X ds + \sigma X dW = X(\mu ds + \sigma dW).$$





**Definition 4.24.**

Let $X = (X_t)_{0\le t\le T}$ be a stochastic process such that

$$X_t = X_0 + \int_0^t b_u du + \int_0^t \sigma_u dW_u$$

where both, $b$ and $\sigma$, are adapted to Brownian motion. Then $X$ is called an Itô process. We also say that the process $X$ has a stochastic differential

$$dX_t = b_t dt + \sigma_t dW_t.$$

# 4.3: Itô's lemma

**Theorem 4.25 (Itô's Lemma).**

*Let $X$ be an Itô process and $f(t,x)$ be a function whose second order partial derivatives are continuous. Then for any $t > 0$*

$$f(t,X_t) - f(0,X_0)$$
$$= \int_0^t f_t(u,X_u) + b_u f_x(u,X_u) + \frac{1}{2}\sigma_u^2 f_{xx}(u,X_u)du + \int_0^t \sigma_u f_x(u,X_u)dW_u$$

*or in differential form*

$$df = \left(f_t + b_t f_x + \frac{1}{2}\sigma_t^2 f_{xx}\right) dt + \sigma_t f_x dW_t.$$



**Proof.**

We proceed as in the preceding version of Itô's formula and consider a Taylor expansion of $f(t,X_t)$ which is in differential notation

$$df = f_t dt + f_x dX + \frac{1}{2} f_{tt}(dt)^2 + f_{tx} dt dX + \frac{1}{2} f_{xx}(dX)^2 + \cdots$$

Now we substitute $dX = bdt + \sigma dW$ and obtain

$$df = f_t dt + f_x(bdt + \sigma dW) + \frac{1}{2} f_{tt}(dt)^2 + f_{tx} dt(bdt + \sigma dW)$$
$$+ \frac{1}{2} f_{xx}(bdt + \sigma dW)^2 + \cdots$$

Again neglecting all $(dt)^2$- and $dtdW$-terms as well as the higher-order terms we obtain

$$df = f_t dt + f_x(bdt + \sigma dW) + \frac{1}{2} f_{xx}(\sigma dW)^2$$
$$= (f_t + f_x b + \frac{1}{2}\sigma^2 f_{xx})dt + f_x \sigma dW.$$

# 4.3: Itô's lemma

Theorem 4.25 gives us the product rule for stochastic calculus:

**Lemma 4.26.**

*Suppose two processes $X_t$ and $Y_t$ are adapted to the same Brownian motion:*

$$\begin{aligned} dX_t &= \sigma_t dW_t + \mu_t dt \\ dY_t &= \rho_t dW_t + \nu_t dt. \end{aligned}$$

*Then*

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + \sigma_t \rho_t dt.$$



# 4.3: Itô's lemma

Proof.

Since $d(X_t + Y_t) = (\sigma_t + \rho_t)dW_t + (\mu_t + \nu_t)dt$, using Theorem 4.25 applied to $f(t,x) = x^2$ we obtain

$$\begin{aligned} d(X_t^2) &= (2\mu_t X_t + \sigma_t^2)dt + 2\sigma_t X_t dW_t \\ d(Y_t^2) &= (2\nu_t Y_t + \rho_t^2)dt + 2\rho_t Y_t dW_t \\ d((X_t + Y_t)^2) &= (2(\mu_t + \nu_t)(X_t + Y_t) + (\sigma_t + \rho_t)^2)dt \\ &\quad + 2(\sigma_t + \rho_t)(X_t + Y_t)dW_t \end{aligned}$$

Subtracting, the result follows since

$$\begin{aligned} 2d(X_t Y_t) &= d((X_t + Y_t)^2 - X_t^2 - Y_t^2) \\ &= 2(\mu_t Y_t + \nu_t X_t)dt + 2\sigma_t \rho_t dt + 2(\sigma_t Y_t + \rho_t X_t)dW_t \\ &= 2Y_t dX_t + 2X_t dY_t + 2\sigma_t \rho_t dt, \end{aligned}$$

□

**Example 4.27 (Geometric Brownian motion).**

We want to solve the stochastic differential equation

$$X_0 = x_0, \quad dX_t = aX_t dt + bX_t dW_t.$$

Let $f(t,x) = \ln x$. Then we have $f_t = 0, f_x = 1/x, f_{xx} = -1/x^2$. By Itô's Lemma (with $b_t = aX_t$ and $\sigma_t = bX_t$), we have

$$\begin{aligned} df(t,X_t) &= (a - \frac{1}{2}b^2)dt + bdW_t, \\ f(t,X_t) &= \ln X_t = \ln X_0 + \int_0^t (a - \frac{1}{2}b^2)du + b\int_0^t dW_u. \end{aligned}$$

Consequently

$$X_t = x_0 e^{(a-\frac{1}{2}b^2)t + bW_t}.$$





**Example 4.28 (Ornstein-Uhlenbeck process).**

We want to solve the stochastic differential equation

$$X_0 = x_0, \quad dX_t = (b - aX_t)dt + cdW_t.$$

Let $f(t,x) = e^{at}x$, then we have $f_t = af, f_x = e^{at}, f_{xx} = 0$. Using Itô's Lemma with $b_t = b - aX_t$ and $\sigma_t = c$ we find that

$$\begin{aligned} df(t,X_t) &= (ae^{at}X_t + (b - aX_t)e^{at})dt + ce^{at}dW_t \\ &= be^{at}dt + ce^{at}dW_t \end{aligned}$$

$$f(t,X_t) - f(0,X_0) = e^{at}X_t - x_0 = \int_0^t be^{au}du + \int_0^t ce^{au}dW_u.$$

Consequently

$$X_t = x_0 e^{-at} + \frac{b}{a}(1 - e^{-at}) + c\int_0^t e^{-a(t-u)}dW_u.$$

# 5. Financial market models in continuous time





# 5.1: The financial market model in continuous time

**Definition 5.1.**

The financial market model in continuous time has the following elements:

- A trading interval $[0, T]$ where trading is possible at each time point $t \in [0, T]$.
- A probability space $(\Omega, \mathcal{F}, \mathbb{P})$
- A filtration $\mathcal{F} = \{\mathcal{F}_t : t \in [0, T]\}$
- A bank account process $B = \{B_t : t \in [0, T]\}$ where $B$ is a stochastic process with $B_0 = 1$ and with $\mathbb{P}(B_t > 0) = 1$ for all $t \in [0, T]$. Here again $B_t$ should be thought of as the time $t$ value of a savings account if 1 unit of a currency is deposited at time $t = 0$.
- A price process $S = \{S(t) : t \in [0, T]\}$ where $S(t) = (S_1(t), \dots, S_N(t))$ and $S_n$ is a non-negative stochastic process for each $n = 1, 2, \dots, N$. Here $S_n(t)$ should be thought of as the time $t$ price of the $n$th risky security.

# 5.2: Trading strategies

**Definition 5.2.**

A trading strategy $H = (H_0, H_1, \dots, H_N)$ is a vector of predictable stochastic processes $H_n = \{H_n(t) : t = 0, \dots, T\}, n = 0, 1, \dots, N.$

- $H_n(t)$ : number of shares of asset $n$ held in the portfolio at time $t$.
- Predictability: We do not give a precise definition, a predictable process $H_n(t)$ can be determined on the basis of information available before time $t$. If an adapted process is left-continuous, then it is predictable.



# 5.2: Trading strategies

**Definition 5.3.**

The value process $V = \{V_t : t \in [0, T]\}$ is a stochastic process defined by

$$V_t = H_0(t)B_t + \sum_{n=1}^{N} H_n(t)S_n(t)$$

which is the value of the portfolio $H$ at time $t$.

**Definition 5.4.**

The gains process of a trading strategy $H$ is given by

$$G_t = \int_0^t H_0(u)dB_u + \sum_{n=1}^{N} \int_0^t H_n(u)dS_n(u).$$

## 5.2: Trading strategies

We consider again normalized versions of the above:

**Definition 5.5.**

The discounted price process $S_n^* = \{S_n^*(t) : n = 0, 1, \ldots, N, t \in [0, T]\}$ is defined by

$$S_n^*(t) = S_n(t)/B_t; n = 1, 2, \ldots, N, t \in [0, T].$$

The discounted value process $V^* = \{V_t^* : t \in [0, T]\}$ is defined by

$$V_t^* = \frac{V_t}{B_t} = H_0(t) + \sum_{n=1}^{N} H_n(t) S_n^*(t).$$

The discounted gains process $G^* = \{G_t^* : t \in [0, T]\}$ is defined by

$$G_t^* = \sum_{n=1}^{N} \int_0^t H_n(u) dS_n^*(u).$$



## 5.2: Trading strategies

Again we will only consider trading strategies which do not bring in or consume any wealth at any time:

**Definition 5.6.**

The trading strategy $H$ is self-financing if its value process satisfies

$$V_t = V_0 + G(t) \text{ for all } t \in [0, T]. \tag{5.12}$$

**Proposition 5.7.**

*Let $H$ be a trading strategy. Then $H$ if self-financing if and only if*

$$V_t^* = V_0^* + G_t^* \text{ for all } t \in [0, T]. \tag{5.13}$$

We leave out the proof which requires an application of the multiplication rule to $dV_t^*$.

# 5.3: Arbitrage in the continuous time model

Arbitrage opportunities in the continuous time model are defined exactly as in the multiperiod model:

**Definition 5.8.**

An arbitrage opportunity in the continuous time model is some trading strategy $H$ such that

(a) $V_0 = 0$,

(b) $V_T \geq 0$,

(c) $\mathbb{E}(V_T) > 0$ and

(d) $H$ is self-financing.



# 5.3: Arbitrage in the continuous time model

In the multiperiod model the discounted gains process is always a martingale. This is no longer the case in the continuous model, so we define a subclass of trading strategies.

**Definition 5.9.**

An equivalent martingale measure is a probability measure $\mathbb{Q}$ such that

(a) $\mathbb{Q}$ is equivalent to $\mathbb{P}$: $\mathbb{P}(A) = 0$ if and only if $\mathbb{Q}(A) = 0$.

(b) The discounted price process $S_n^*$ is a martingale under $\mathbb{Q}$ for every $n = 1, 2, \ldots, N$.

**Definition 5.10.**

A self-financing trading strategy $H$ is called admissible with respect to an equivalent martingale measure $\mathbb{Q}$ if the discounted gains process $G^*(t)$ is a martingale under $\mathbb{Q}$.

**Proposition 5.11.**

*For each admissible strategy and for each equivalent martingale measure $\mathbb{Q}$ the discounted value process $V^*(t)$ is a martingale with respect to $\mathbb{Q}$.*

Proof.

This follows immediately from Proposition 5.7. □





**Theorem 5.12.**

*Assume there exists an equivalent martingale measure $\mathbb{Q}$. Then the market model contains no admissible arbitrage opportunities.*

Proof.

Assume that $H$ is an admissible arbitrage opportunity. Then

$$V_0^* = 0,\ V_T^* \geq 0 \text{ and } \mathbb{E}(V_T^*) > 0.$$

Since $\mathbb{Q}$ is equivalent to $\mathbb{P}$, we also have $\mathbb{E}_{\mathbb{Q}}(V_T^*) > 0$.
On the other hand $V_t^*$ is a martingale under $\mathbb{Q}$ using Proposition 5.7.
That is,

$$\mathbb{E}_{\mathbb{Q}}(V_t^* | \mathcal{F}_u) = V_u^* \quad \text{for all } u \leq t \leq T.$$

which implies

$$\mathbb{E}_{\mathbb{Q}}(V_T^*) = V_0^* = 0.$$

Thus there cannot be an admissible arbitrage opportunity. □

# 5.3: Arbitrage in the continuous time model

**Definition 5.13.**

A contingent claim $X$ is called attainable if there exists at least one admissible trading strategy such that $V_T = X$. We call such a trading strategy $H$ a replicating strategy for $X$. The financial market model is said to be complete if any contingent claim is attainable.

**Proposition 5.14 (Risk-neutral valuation principle).**

*The time $t$ value of an attainable contingent claim $X$ is equal to $V_t$, the time $t$ value of any portfolio which replicates $X$. Moreover,*

$$V_t^* = \frac{V_t}{B_t} = \mathbb{E}_Q\left(\frac{X}{B_T}|\mathcal{F}_t\right), \; t \in [0, \dots, T]$$

*for all equivalent martingale measures $\mathbb{Q}$. In particular, the time 0 value is*

$$\pi_0 = V_0 = \mathbb{E}_Q\left(\frac{X}{B_T}\right).$$



# 5.3: Arbitrage in the continuous time model

Proof.

Let $H$ be an arbitrary replicating strategy for $X$ and $P_t$ the time $t$ price of the contingent claim $X$. If $P_t$ does not match the value of the strategy $H$ at time $t$, then there exist arbitrage opportunities as described in the proof of the Risk-neutral valuation principle in the multiperiod model.

Moreover by Proposition 5.11 the discounted value process $V_t^*$ is a martingale under $\mathbb{Q}$ and thus

$$V_t^* = \mathbb{E}_Q(V_T^*|\mathcal{F}_t) = \mathbb{E}_Q(V_T/B_T|\mathcal{F}_t) = \mathbb{E}_Q(X/B_T|\mathcal{F}_t).$$

Note that the last expression is independent of the particular replicating strategy. □

# 5.4: The Black-Scholes model

## Definition 5.15.

The Black-Scholes model is a financial market model in continuous time with two assets:

- A riskless bond with constant interest rate $r$, and corresponding bank account process $B_t = e^{rt}$.
- A risky asset with price process $S_t$ modelled as

$$S_t = S_0 e^{\sigma W_t + (\mu - \sigma^2/2)t}.$$

where $W_t$ is a Brownian motion, $\mu$ is the drift and $\sigma$ is the volatility.

Brownian motion has the characteristic 'randomness' of share prices, and geometric Brownian motion ensures prices do not become negative.



# 5.4: The Black-Scholes model

## Proposition 5.16.

*The bond price satisfies the differential equation $dB_t = rB_t dt$ while the stock price satisfies the stochastic differential equation*

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

### Proof.

We have already proved this in Proposition 4.23: The function $f(t, W_t) = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$ has partial derivatives

$$f_t(t, W_t) = (\mu - \sigma^2/2)S_t, \quad f_x(t, W_t) = \sigma S_t \quad f_{xx}(t, W_t) = \sigma^2 S_t.$$

We can now apply Itô's Lemma to obtain

$$\begin{aligned} dS_t &= (f_t(t, W_t) + f_{xx}(t, W_t)/2)\, dt + f_x(t, W_t) dW_t \\ &= \left((\mu - \sigma^2/2) + \sigma^2/2\right) S_t dt + \sigma S_t dW_t \\ &= \mu S_t dt + \sigma S_t dW_t. \end{aligned}$$

## 5.5: Equivalent martingale measures in the BS model

We need a special version of Girsanov's theorem:

**Theorem 5.17 (Girsanov's Theorem for BS model).**

*Let $(X_t)_{t\geq 0}$ be a Brownian motion with drift $b$ and volatility $\sigma$. Let $a \in \mathbb{R}$ and $T > 0$ and define*

$$L_T = \exp\left(\frac{a-b}{\sigma^2} X_T - \frac{a^2-b^2}{2\sigma^2} T\right).$$

*Then by setting*

$$\mathbb{Q}(A) = \mathbb{E}(L_T 1_A), \text{ for all } A \in \mathcal{F}_T$$

*we can define a probability measure $\mathbb{Q}$ such that*

- *$\mathbb{Q}$ is equivalent to $\mathbb{P}$ and*
- *$(X_t)_{t\geq 0}$ is a Brownian motion with volatility $\sigma$ and drift $a$ under $\mathbb{Q}$.*



**Proof.**

We write $X_T = \sigma W_T + bT$ and obtain

$$\begin{aligned}
\mathbb{Q}(A) &= \mathbb{E}\left(1_A \exp\left(\frac{a-b}{\sigma} W_T + \frac{a-b}{\sigma^2} bT - \frac{a^2-b^2}{2\sigma^2} T\right)\right) \\
&= \mathbb{E}\left(1_A \exp\left(\frac{a-b}{\sigma} W_T + \frac{T}{2\sigma^2}(a-b)(2b-(a+b))\right)\right). \\
&= \mathbb{E}\left(1_A \exp\left(\theta W_T - T\theta^2/2\right)\right)
\end{aligned}$$

where $\theta = (a-b)/\sigma$. Since $W_T$ has a $\mathcal{N}(0,T)$ distribution we obtain in particular for $A = \Omega$ that

$$\begin{aligned}
\mathbb{Q}(\Omega) &= \int_{-\infty}^{\infty} \exp\left(\theta z - T\theta^2/2\right) \frac{1}{\sqrt{2\pi T}} \exp\left(-\frac{z^2}{2T}\right) dz \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi T}} \exp\left(-\frac{z^2 - 2T\theta z + T^2\theta^2}{2T}\right) dz \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi T}} \exp\left(-\frac{(z-T\theta)^2}{2T}\right) dz = \int_{-\infty}^{\infty} g(z) dz.
\end{aligned}$$

# 5.5: Equivalent martingale measures in the BS model

**Proof.**

We identify the integrand $g(z)$ as the density of a $\mathcal{N}(T\theta, T)$ distribution. Therefore

$$\mathbb{Q}(\Omega) = \int_{-\infty}^{\infty} g(z)dz = 1.$$

Hence $\mathbb{Q}$ is a probability measure and, since $L_T > 0$, equivalent to $\mathbb{P}$. Similarly we obtain for every $s \in \mathbb{R}$ that

$$\begin{aligned}\mathbb{Q}(X_T \leq s) = \mathbb{Q}\left(W_T \leq \frac{s - bT}{\sigma}\right) &= \int_{-\infty}^{\frac{s-bT}{\sigma}} g(z)dz \\ &= \int_{-\infty}^{s} \frac{1}{\sigma} g\left(\frac{u - Tb}{\sigma}\right) du\end{aligned}$$

using the transformation $z = (u - Tb)/\sigma$.



# 5.5: Equivalent martingale measures in the BS model

**Proof.**

We can rewrite the integrand as

$$\begin{aligned}\frac{1}{\sigma} g\left(\frac{u - Tb}{\sigma}\right) &= \frac{1}{\sqrt{2\pi T}\sigma} \exp\left(-\frac{(\frac{u-Tb}{\sigma} - T\theta)^2}{2T}\right) \\ &= \frac{1}{\sqrt{2\pi T}\sigma} \exp\left(-\frac{(u - T(b + \theta\sigma))^2}{2T\sigma^2}\right)\end{aligned}$$

and note that $b + \theta\sigma = a$. Therefore $X_T$ is under $\mathbb{Q}$ normally distributed with mean $aT$ and variance $T\sigma^2$.

It remains to show that for $t < T$ the distribution of $X_t$ is again a normal distribution with mean $at$ and variance $t\sigma^2$ and that the increments of $X_t$ are independent. $\square$

# 5.5: Equivalent martingale measures in the BS model

**Proposition 5.18.**

*Consider the Black-Scholes model as defined in Definition 5.15. An equivalent martingale measure $\mathbb{Q}$ is defined by setting*

$$\mathbb{Q}(A) = \mathbb{E}(L_T 1_A)$$

*where*

$$L_T = \exp\left(\frac{r-\mu}{\sigma}W_T - \frac{(r-\mu)^2}{2\sigma^2}T\right). \tag{5.14}$$

**Proof.**

The discounted price process in the Black-Scholes model is given by

$$S_t^* = e^{-rt}S_t = S_0 e^{\sigma W_t + (\mu - r - \sigma^2/2)t} = S_0 e^{X_t}.$$

where $X_t$ is a Brownian motion with volatility $\sigma$ and drift $\mu - r - \sigma^2/2$.



**Proof.**

Writing

$$\begin{aligned}
L_T &= \exp\left(\frac{r-\mu}{\sigma^2}(X_T - (\mu - r - \sigma^2/2)T) - \frac{(r-\mu)^2}{2\sigma^2}T\right)\\
&= \exp\left(\frac{r-\mu}{\sigma^2}X_T - \frac{(r-\mu)(\mu - r - \sigma^2/2)T)}{\sigma^2} - \frac{(r-\mu)^2}{2\sigma^2}T\right)\\
&= \exp\left(\frac{r-\mu}{\sigma^2}X_T - \frac{(r-\mu)(2\mu - 2r - \sigma^2)}{2\sigma^2}T - \frac{(r-\mu)^2}{2\sigma^2}T\right)\\
&= \exp\left(\frac{r-\mu}{\sigma^2}X_T - \frac{-2(r-\mu)^2 - (r-\mu)\sigma^2)}{2\sigma^2}T - \frac{(r-\mu)^2}{2\sigma^2}T\right)\\
&= \exp\left(\frac{r-\mu}{\sigma^2}X_T - \frac{-(r-\mu)^2 - (r-\mu)\sigma^2}{2\sigma^2}T\right)
\end{aligned}$$

and applying Girsanov's theorem with $a = -\sigma^2/2$ and $b = \mu - r - \sigma^2/2$ we see that $X_T$ is under $\mathbb{Q}$ a Brownian motion with volatility $\sigma$ and drift $-\sigma^2/2$. Using Lemma 4.8 we see that $S_t^*$ is a martingale under $\mathbb{Q}$. □

## 5.5: Equivalent martingale measures in the BS model

**Remark 5.19.**

*The stock price process is given by*

$$S_t = S_0 e^{\sigma W_t + (\mu - \sigma^2/2)t}$$

*under $\mathbb{P}$ and*

$$S_t = S_0 e^{\sigma \overline{W_t} + (r - \sigma^2/2)t}$$

*under $\mathbb{Q}$ where*

$$\overline{W_t} = W_t - \frac{r-\mu}{\sigma} t$$

*is a standard Brownian motion with respect to $\mathbb{Q}$. This shows that computations using the risk-neutral measure can be made by replacing the trend $\mu$ simply by the constant interest rate $r$.*



## 5.6: Pricing in the BS model

We price in the Black-Scholes model using the risk-neutral valuation principle.

**Proposition 5.20.**

*Consider the Black-Scholes model as defined in 5.15 and let $X$ be a contingent claim. Then the price of $X$ at time $t$ is given by*

$$P_t = e^{-r(T-t)} \mathbb{E}_Q(X | \mathcal{F}_t)$$

*where $\mathbb{E}_Q$ is given via the Girsanov density (5.14).*

## 5.6: Pricing in the BS model

**Example 5.21.**

For $S_t$ satisfying the Black-Scholes model we calculate the time $t$ price of a European option with payoff $S_T^3$ at time $T$. Since $S_t = S_0 e^{(r-\sigma^2/2)t+\sigma \overline{W}_t}$ under $\mathbb{Q}$, the time $t$ value is using Proposition 5.20:

$$\begin{aligned}
e^{-r(T-t)}\mathbb{E}_Q(S_T^3|\mathcal{F}_t) &= e^{-r(T-t)}S_t^3\mathbb{E}_Q(S_T^3/S_t^3|\mathcal{F}_t)\\
&= e^{-r(T-t)}S_t^3\mathbb{E}_Q(e^{3\sigma(\overline{W}_T-\overline{W}_t)+3(r-\sigma^2/2)(T-t)}|\mathcal{F}_t)\\
&= S_t^3 e^{-r(T-t)+3(r-\sigma^2/2)(T-t)}\mathbb{E}_Q(e^{3\sigma(\overline{W}_T-\overline{W}_t)}|\mathcal{F}_t)\\
&= S_t^3 e^{(2r-3\sigma^2/2)(T-t)}e^{9\sigma^2(T-t)/2}\\
&= S_t^3 e^{(2r+3\sigma^2)(T-t)}
\end{aligned}$$

where we used that $\overline{W}_T - \overline{W}_t$ has a $\mathcal{N}(0, T-t)$ distribution and therefore its moment generating function is

$$\mathbb{E}_Q(e^{\theta(\overline{W}_T-\overline{W}_t)}|\mathcal{F}_t) = e^{\theta^2(T-t)/2}.$$



## 5.6: Pricing in the BS model

We use Proposition 5.20 now to derive the Black-Scholes-Formula that we have already seen in Proposition 3.35 First we need a technical lemma:

**Lemma 5.22.**

*For $Z \sim \mathcal{N}(a, \gamma^2)$, for any $b, c > 0$:*

$$\mathbb{E}((be^Z - c)_+) = be^{a+\frac{\gamma^2}{2}}\Phi\left(\frac{\log(b/c)+a+\gamma^2}{\gamma}\right) - c\Phi\left(\frac{\log(b/c)+a}{\gamma}\right)$$

*where $\Phi$ is the cumulative distribution function of a standard normal random variable.*

**Proof.**

We recall that for a standard normal random variable $X$

$$\mathbb{P}(X > x) = 1 - \Phi(x) = \Phi(-x) \quad \text{for all } x \in \mathbb{R}.$$

Therefore we can calculate $\mathbb{E}((be^Z - c)_+)$ as follows:

Proof.

$$
\begin{aligned}
\mathbb{E}((be^Z - c)_+) &= \int_{-\infty}^{\infty} (be^x - c)_+ \frac{1}{\sqrt{2\pi\gamma^2}} e^{-\frac{(x-a)^2}{2\gamma^2}}\, dx \\
&= \int_{\log(c/b)}^{\infty} (be^x - c) \frac{1}{\sqrt{2\pi\gamma^2}} e^{-\frac{(x-a)^2}{2\gamma^2}}\, dx \\
&= b \int_{\log(c/b)}^{\infty} \frac{1}{\sqrt{2\pi\gamma^2}} e^{-\frac{-2x\gamma^2+(x-a)^2}{2\gamma^2}}\, dx - c \int_{\log(c/b)}^{\infty} \frac{1}{\sqrt{2\pi\gamma^2}} e^{-\frac{(x-a)^2}{2\gamma^2}}\, dx \\
&= be^{\frac{2a\gamma^2+\gamma^4}{2\gamma^2}} \int_{\log(c/b)}^{\infty} \frac{1}{\sqrt{2\pi\gamma^2}} e^{-\frac{(x-a-\gamma^2)^2}{2\gamma^2}}\, dx - c\mathbb{P}(Z > \log(c/b)) \\
&= be^{a+\frac{\gamma^2}{2}} \mathbb{P}(Z + \gamma^2 > \log(c/b)) - c\mathbb{P}(Z > \log(c/b)) \\
&= be^{a+\frac{\gamma^2}{2}} \mathbb{P}\left(\frac{Z-a}{\gamma} > \frac{\log(c/b)-\gamma^2-a}{\gamma}\right) - c\mathbb{P}\left(\frac{Z-a}{\gamma} > \frac{\log(c/b)-a}{\gamma}\right) \\
&= be^{a+\frac{\gamma^2}{2}} \Phi\left(\frac{\log(b/c)+a+\gamma^2}{\gamma}\right) - c\Phi\left(\frac{\log(b/c)+a}{\gamma}\right).
\end{aligned}
$$

□



## 5.6: Pricing in the BS model

**Proposition 5.23 (Black-Scholes-Formula).**

*Consider a European call option with exercise price $K$ maturing at time $T$ which corresponds to the contingent claim $X = (S_T - K)_+$. Then the Black-Scholes price process of that European call is given by*

$$\Pi_t = S_t\Phi(d_1(S_t, T-t)) - Ke^{-r(T-t)}\Phi(d_2(S_t, T-t)). \qquad (5.15)$$

*The functions $d_1(u,v)$ and $d_2(u,v)$ are given by*

$$d_1(u,v) = \frac{\log(u/K) + (r + \sigma^2/2)v}{\sigma\sqrt{v}}$$

$$d_2(u,v) = \frac{\log(u/K) + (r - \sigma^2/2)v}{\sigma\sqrt{v}}$$

*and $\Phi$ is the standard normal cumulative distribution function.*

# 5.6: Pricing in the BS model

**Proof.**

Let $\overline{W_t}$ be a standard Brownian motion under $\mathbb{Q}$. The price $\Pi_t$ is given by

$$\begin{aligned}
\Pi_t &= e^{-r(T-t)}\mathbb{E}_Q((S_T-K)_+|\mathcal{F}_t) \\
&= e^{-r(T-t)}\mathbb{E}_Q\left((S_t e^{\sigma(\overline{W_T}-\overline{W_t})+(r-\frac{\sigma^2}{2})(T-t)}-K)_+|\mathcal{F}_t\right) \\
&= e^{-r(T-t)}\mathbb{E}_Q((S_t e^{Z}-K)_+|\mathcal{F}_t)
\end{aligned}$$

where $Z$ has a $\mathcal{N}(a,\gamma^2)$ distribution under $\mathbb{Q}$ with mean $a=(r-\frac{\sigma^2}{2})(T-t)$ and standard deviation $\gamma=\sigma\sqrt{T-t}$ and is independent of $\mathcal{F}_t$. We obtain the Black-Scholes formula by applying Lemma 5.22 with $b=S_t, c=K$:

$$\begin{aligned}
\Pi_t &= e^{-r(T-t)}S_t e^{a+\frac{\sigma^2(T-t)}{2}}\Phi\left(\frac{\log(S_t/K)+a+\sigma^2(T-t)}{\sigma\sqrt{T-t}}\right) \\
&\quad - e^{-r(T-t)}K\Phi\left(\frac{\log(S_t/K)+a}{\sigma\sqrt{T-t}}\right)
\end{aligned}$$



□

# 5.6: Pricing in the BS model

**Remark 5.24.**

*The key assumptions in the Black-Scholes framework are:*

- *No-arbitrage.*
- *No transaction costs, unlimited shortselling at no costs.*
- *Constant risk-free interest rate.*
- *The underlying price follows a geometric Brownian motion.*
- *Constant volatility.*

## 5.6: Pricing in the BS model

**Remark 5.25.**

*Some discussion of those assumptions:*

- *No arbitrage is a standard assumption.*
- *The assumptions of zero transaction costs and constant risk-free rate can be relaxed easily, for example, stochastic interest rates can be introduced.*
- *In reality, prices can jump, so there are extensions of the Black-Scholes model with some jump components (eg. Merton (1976) who introduced a compound Poisson process to the dynamics of $S$).*
- *Volatility is neither known nor constant.*
- *Empirical evidence shows that asset returns exhibit fat-tailed behaviour, unlike the log-normal distributions assumed by the Black-Scholes model.*



## 5.6: Pricing in the BS model

A big problem is the estimation of the unobservable parameter $\sigma$. There are mainly 2 approaches:

1. Statistical approach: We can estimate $\sigma$ by looking at historical prices $S$ and deduce the historical volatility. However, there is no reason for the volatility over the past to be a good estimator for the future volatility.
2. Market-based approach: We extract information of volatility from traded instruments. Volatility obtained in this way is the market view of the volatility, called implied volatility. The Black-Scholes formula is numerically inverted to find what volatility would give the quoted price.

Furthermore, $\sigma$ is constant in the Black-Scholes model, but in practice, there is a volatility smile.

## 5.6: Pricing in the BS model

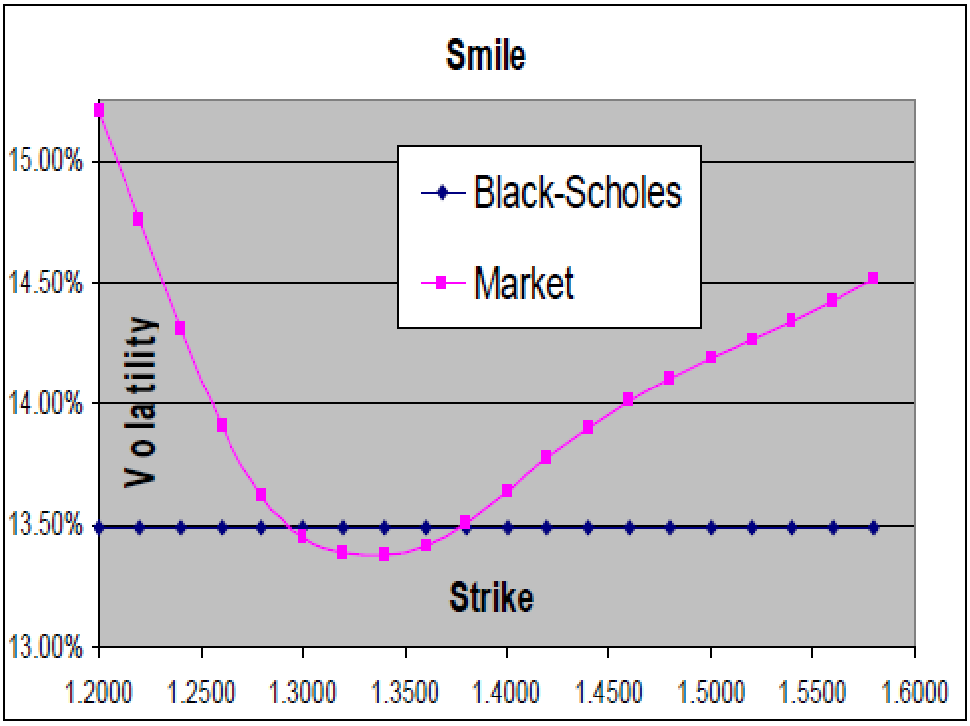



Figure: Implied volatility of a European call option with different strikes



## 5.6: Pricing in the BS model

**Example 5.26.**

Consider the Black-Scholes model with interest rate $r = 0.16$. Suppose a European call option on the stock with exercise price 40 and a maturity date of six months is trading at a value of 4.25 when the current stock price is 38. Compute the implied volatility of the call option.

Using the Black-Scholes formula, for values of $\sigma = (0.3, 0.35, 0.4)$, the price of the call option is $(3.73, 4.25, 4.77)$. Hence, the implied volatility is $\sigma = 0.35$.

## 5.6: Pricing in the BS model

The put-call parity is useful to derive the value of put options. We recall that

$$S_t + P_t - C_t = Ke^{-r(T-t)}.$$

where $P_t$ and $C_t$ are the prices of European put and call options on the same stock.

**Example 5.27.**

Assuming the implied volatility is the true value of $\sigma$, find the time zero value of a European put option on the same stock, with maturity 1 year and exercise price 50.

Plugging in the Black-Scholes formula, the price of the European call is 3.59. Using the put-call parity, we get the price of the corresponding put is 8.20.



## 5.7: Replicating strategies and the BSM equation

- The risk neutral pricing principle allows us to compute the no-arbitrage price of the option. However, it is somewhat unsatisfactory because it does not provide an explicit expression for the replicating portfolio.
- Next, we show that our techniques can price any attainable contingent claim, and deduce the remarkable result that all price functions satisfy the same partial differential equation.
- The boundary conditions (value of the price function at time $T$) determine the differences between different types of options.
- Solving the Black-Scholes-Merton equation gives an alternative way of pricing a contingent claim, once we choose boundary conditions corresponding to that contingent claim.
- This method also gives us the replicating portfolio.

## 5.7: Replicating strategies and the BSM equation

**Theorem 5.28.**

*Let $X = g(S_T)$ be an attainable contingent claim in the Black-Scholes model with drift $\mu$ and volatility $\sigma$. Assume that the price $\Pi_t$ at time $t$ satisfies $\Pi_t = f(t, S_t)$, for some function $f$ with two continuous derivatives in $x$ and one in $t$.*
*Then $f(t,x)$ satisfies the Black-Scholes-Merton equation*

$$f_t + rS_t f_x + \frac{1}{2}\sigma^2 S_t^2 f_{xx} - rf = 0, \quad f(T, S_T) = g(S_T). \tag{5.16}$$

*Further, the replicating strategy is given by*

$$H_0(t) = \frac{f(t,S_t) - f_x(t,S_t)S_t}{B_t} = \frac{f_t(t,S_t) + \frac{1}{2}f_{xx}(t,S_t)S_t^2\sigma^2}{rB_t}$$
$$H_1(t) = f_x(t, S_t).$$



## 5.7: Replicating strategies and the BSM equation

Proof.

The price process $S_t$ of the stock under $\mathbb{Q}$ is

$$S_t = S_0 e^{\sigma \overline{W}_t + (r - \sigma^2/2)t}.$$

An application of Itô's lemma shows that it has dynamics

$$dS_t = S_t(rdt + \sigma d\overline{W}_t). \tag{5.17}$$

Similarly the discounted price process $S_t^* = S_0 e^{\sigma \overline{W}_t - \sigma^2/2t}$ has dynamics

$$dS_t^* = S_t^* \sigma d\overline{W}_t. \tag{5.18}$$

Let now

$$h(t, S_t) = f(t, S_t)/B_t = e^{-rt} f(t, S_t)$$

be the discounted fair price of the contingent claim $X$ at time $t$.

## Proof.

Since Equation (5.17) means that $S_t$ is an Itô process with $b_t = rS_t$ and $\sigma_t = \sigma S_t$, using Itô's lemma we find that

$$dh = (h_t + rS_t h_x + \frac{1}{2}\sigma^2 S_t^2 h_{xx})dt + \sigma S_t h_x d\overline{W}_t. \tag{5.19}$$

We know from the risk-neutral valuation principle that

$$h(t, S_t) = \mathbb{E}_{\mathbb{Q}}\left(\frac{X}{B_T}|\mathcal{F}_t\right)$$

and in particular that $h$ is a martingale with respect to $\mathbb{Q}$. Therefore there is no drift meaning that the $dt$ term in (5.19) is zero, ie

$$h_t(t, S_t) + rS_t h_x(t, S_t) + \frac{1}{2}\sigma^2 S_t^2 h_{xx}(t, S_t) = 0. \tag{5.20}$$

Moreover the dynamics in Equation (5.19) reduce to

$$dh(t, S_t) = \sigma S_t h_x(t, S_t) d\overline{W}_t. \tag{5.21}$$



# 5.7: Replicating strategies and the BSM equation

## Proof.

**Black-Scholes-Merton equation:** The ratio between $h$ and $f$ is a function of $t$, not $x$ and therefore $h_x = f_x/B_t$, $h_{xx} = f_{xx}/B_t$. We also know that $h_t = f_t/B_t - rf/B_t$. Therefore Equation (5.20) can be rewritten as

$$0 = h_t + rS_t h_x + \frac{1}{2}\sigma^2 S_t^2 h_{xx} = \frac{1}{B_t}\left(f_t - rf + rS_t f_x + \frac{1}{2}\sigma^2 S_t^2 f_{xx}\right).$$

Multiplying by $B_t$ we see that $f$ satisfies the Black-Scholes-Merton Equation.

**Replicating portfolio:** For all $t$ the value of the replicating portfolio $(H_0(t), H_1(t))$ equals the price process $f(t, S_t)$ of the contingent claim $X$ and the replicating strategy must be self-financing. Therefore (5.18) implies

$$dh(t, S_t) = H_1(t) dS_t^* = H_1(t) S_t^* \sigma d\overline{W}_t = H_1(t)\frac{S_t}{B_t}\sigma d\overline{W}_t. \tag{5.22}$$

## 5.7: Replicating strategies and the BSM equation

**Proof.**

Equating the differentials (5.21) and (5.22) we obtain

$$H_1(t)S_t^*\sigma = \sigma S_t h_x(t, S_t)$$

and therefore

$$H_1(t) = B_t h_x(t, S_t) = f_x(t, S_t).$$

Finally the replicating portfolio has to satisfy

$$f(t, S_t) = H_0(t)B_t + H_1(t)S_t = H_0(t)B_t + f_x(t, S_t)S_t$$

and from that we obtain

$$H_0(t) = \frac{f(t, S_t) - f_x(t, S_t)S_t}{B_t},$$

with the alternative formulation following from the Black-Scholes-Merton equation. □



## 5.7: Replicating strategies and the BSM equation

Solving the Black-Scholes-Merton equation is an alternative way of calculating the price of a contingent claim. The boundary condition has to be chosen accordingly to the contingent claim we want to analyze. In particular for $f(T, S_T) = (S_T - K)^+$ we obtain yet another way of deriving the Black-Scholes formula as we show now.

**Example 5.29.**

For $f(t, S_t) = \Pi_t$ let us recall Equation (5.15):

$$\Pi_t = S_t\Phi(d_1(S_t, T-t)) - Ke^{-r(T-t)}\Phi(d_2(S_t, T-t)).$$

First of all we observe that

$$\frac{\partial d_1}{\partial S}(S_t, T-t) = \frac{K}{K\sigma S_t\sqrt{T-t}} = \frac{1}{\sigma S_t\sqrt{T-t}} = \frac{\partial d_2}{\partial S}(S_t, T-t)$$

and

$$\begin{aligned}\phi(d_2)/\phi(d_1) &= \exp(-\frac{1}{2}(d_2^2 - d_1^2)) \\ &= \exp\left(-\frac{1}{2}\frac{-2(\log(S_t/K) + r(T-t))(\sigma^2(T-t))}{\sigma^2(T-t)}\right) \\ &= \exp\left(\log(S_t/K) + r(T-t)\right) \\ &= (S_t/K)\exp(r(T-t)).\end{aligned}$$



5.7: Replicating strategies and the BSM equation

**Example 5.29.**

We use this to derive

$$\begin{aligned}f_x &= \Phi(d_1) + S_t\phi(d_1)\frac{\partial d_1}{\partial S} - Ke^{-r(T-t)}\phi(d_2)\frac{\partial d_2}{\partial S} \\ &= \Phi(d_1) + S_t\phi(d_1)\left(\frac{\partial d_1}{\partial S} - \frac{\partial d_2}{\partial S}\right) = \Phi(d_1) \\ f_{xx} &= \phi(d_1)\frac{\partial d_1}{\partial S} = \frac{\phi(d_1)}{S_t\sigma\sqrt{T-t}}.\end{aligned}$$

Furthermore

$$\begin{aligned}&\frac{\partial d_1}{\partial t}(S_t, T-t) - \frac{\partial d_2}{\partial t}(S_t, T-t) \\ &= \frac{\partial(d_1 - d_2)}{\partial t}(S_t, T-t) = \frac{\partial}{\partial t}(\sigma\sqrt{T-t}) = \frac{-\sigma}{2\sqrt{T-t}}.\end{aligned}$$

# 5.7: Replicating strategies and the BSM equation

**Example 5.29.**

We can now calculate $f_t$ as

$$\begin{aligned} f_t &= S_t\phi(d_1)\frac{\partial d_1}{\partial t} - Ke^{-r(T-t)}\phi(d_2)\frac{\partial d_2}{\partial t} - Kre^{-r(T-t)}\Phi(d_2) \\ &= S_t\phi(d_1)\left(\frac{\partial d_1}{\partial t} - \frac{\partial d_2}{\partial t}\right) - Kre^{-r(T-t)}\Phi(d_2) \\ &= -S_t\phi(d_1)\frac{\sigma}{2\sqrt{T-t}} - Kre^{-r(T-t)}\Phi(d_2) \end{aligned}$$

and substitute in the BSM equation to obtain

$$\begin{aligned} f_t + rS_tf_x + \frac{1}{2}\sigma^2S_t^2f_{xx} - rf &= -S_t\phi(d_1)\frac{\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}\Phi(d_2) \\ &\quad + rS_t\Phi(d_1) + \frac{1}{2}\sigma^2S_t^2\frac{\phi(d_1)}{S_t\sigma\sqrt{T-t}} - rf \\ &= r(S_t\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2) - f) = 0. \end{aligned}$$